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Analytical theory of SDEs via Rough Paths

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ABSTRACT

The study of stochastic differential equations has long been a prominent field of research, and in the last twenty-five years the theory of rough paths has become an important tool in this area. This approach gives in particular an elegant framework to treat SDEs as controlled ordinary differential equations with an irregular driving path. This thesis first focuses on an approach to controlled ODEs with a driving path of Hölder regularity greater than $1/3$, based on the Sewing Bound and appropriate a priori estimates; the Sewing Bound is a general estimate for functions of two real variables that is contained in the notorious Sewing Lemma but can be stated on its own. Then we take the first steps towards the extension of the theory to paths with arbitrarily low Hölder regularity. To do this, we prove some interesting identities closely related to the algebraic structure of geometric rough paths, particularly shedding light on the relationship between rough paths and controlled ODEs. In the last part of the thesis, we use these results to prove the uniqueness of the solution to the rough difference equation in the general case.

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CHAPTER 1

INTRODUCTION

In recent years, the field of Stochastic (Partial) Differential Equations has experienced an impressive growth and has emerged as a very popular research area (see [22] for a detailed historical report). This interest can also be attributed to the large number of applications that this class of equations has found, notably in Economics, starting from the Black-Scholes model in [2],[17], and in Physics, with, for example, the famous KPZ equation introduced in [15], see also [9],[18],[1]. The specific contribution of the pathwise approach based on rough paths ideas has been fundamental in the construction of a robust theory for these equations

The pathwise method based on rough paths was pioneered by Terry Lyons [16]; later Massimiliano Gubinelli introduced a new approach to *rough integration* for stochastic ODEs, based on the celebrated Sewing Lemma; Martin Hairer understood then how to extend these ideas to SPDEs and developed the theory of *regularity structures* (see [12],[13],[14]), for which he was awarded a Fields Medal in 2014.

This thesis will focus on Davie's notion of solutions to a rough differential equations [10], and on the Sewing method to prove existence, uniqueness, a priori estimates and continuity of the Itô-Lyons map (which associates the solution to the driving rough path). As an illustration, one of the byproducts of the theory is the construction of integrals of the form

$$I_t = \int_0^t Y dX,$$

when X and Y are functions that are continuous but not continuously differentiable. The interest of this problem becomes evident when we think of X as a Brownian motion, which is only (almost) $1/2$ -Hölder continuous. A first answer to this challenge was given in 1936 in [21] by the so called *Young integral*, which allows to consider functions $Y \in \mathcal{C}^\alpha$, $X \in \mathcal{C}^\beta$ Hölder continuous with exponents $\alpha, \beta \in (0, 1]$ such that $\alpha + \beta > 1$. Subsequently the theory of rough path and the *Sewing Lemma* enabled the extension of the integral for $\alpha + \beta < 1$ (see [16],[11]). This result and the fundamental notion of *controlled paths* in [12] are sufficient to build a coherent *rough integration* theory which permits to prove, by the usual contraction arguments, the existence of solution for controlled ODEs with an irregular driving path.

This thesis follows the approach for well-posedness and continuity of the Itô-Lyons map proposed by F. Caravenna, M. Gubinelli and L. Zambotti in their pedagogical exposition [5] and takes the first steps to extend this work to the case of a driving path with an arbitrarily low Hölder exponent. The results are based on analytic bounds and on the algebraic properties of rough paths. Therefore the main focus of the thesis is on the definition, the structure of rough paths and their relation with the ODEs of interest.

CHAPTER 2

PRELIMINARIES

We will start by introducing some tools that will be used repeatedly in the following. The main theorem of this section is the Sewing Bound, which is a simple bound for functions of two real variables that can be stated on its own. This theorem is fundamental for presenting the results of [5].

2.1. FUNCTION SPACES

Let us define for $n \geq 1$ the simplex $[0, T]_{\leq}^n := \{(t_1, \dots, t_n): 0 \leq t_1 \leq \dots \leq t_n \leq T\}$.

On the space $C([0, T]_{\leq}, \mathbb{R}^k)$ of continuous functions $f: [0, T] \rightarrow \mathbb{R}^k$ we consider the usual Hölder space. Defining $(\delta f)_{st} := f_t - f_s$, for $0 \leq s \leq t \leq T$, we define, for $\alpha \in (0, 1]$, the Hölder space $\mathcal{C}^\alpha$ as

$$\mathcal{C}^\alpha := \left\{ f: [0, T] \rightarrow \mathbb{R}^k: \quad \|\delta f\|_\alpha = \sup_{0 \leq s < t \leq T} \frac{|f_t - f_s|}{(t - s)^\alpha} < \infty \right\}.$$

We can consider a similar structure also for $C([0, T]_{\leq}^2, \mathbb{R}^k)$ and $C([0, T]_{\leq}^3, \mathbb{R}^k)$. Given any $\eta \in (0, \infty)$, we define for $F \in C([0, T]_{\leq}^2, \mathbb{R}^k)$ and $G \in C([0, T]_{\leq}^3, \mathbb{R}^k)$

$$\|F\|_\eta := \sup_{0 \leq s < t \leq T} \frac{|F_{st}|}{(t - s)^\eta}, \quad \|G\|_\eta := \sup_{\substack{0 \leq s \leq u \leq t \leq T \\ s < t}} \frac{|G_{sut}|}{(t - s)^\eta},$$

and the function spaces

$$C_2^\eta := \{F \in C([0, T]_{\leq}^2, \mathbb{R}^k): \quad \|F\|_\eta < \infty\}, \quad C_3^\eta := \{G \in C([0, T]_{\leq}^3, \mathbb{R}^k): \quad \|G\|_\eta < \infty\}.$$

2.2. THE SEWING BOUND

LEMMA 2.1. (RIEMANN SUMS) *Let us take any $R \in C([0, T]_{\leq}^2, \mathbb{R}^k)$ such that $R_{st} = o(t - s)$. For any $0 \leq a < b \leq T$ let us take any sequence of partitions $(\mathcal{P}_n)_{n \geq 0}$ of $[a, b]$ with vanishing mesh, i.e. $\lim_{n \rightarrow \infty} |\mathcal{P}_n| = 0$. If we define $I_{\mathcal{P}_n}(R) := \sum_{i=1}^{\#\mathcal{P}} R_{t_{i-1}t_i}$ we have*

$$\lim_{n \rightarrow \infty} I_{\mathcal{P}_n}(R) = 0.$$

Moreover, if $\mathcal{P}_0 = \{a, b\}$, we can write

$$R_{ab} = \sum_{n=0}^{\infty} (I_{\mathcal{P}_n}(R) - I_{\mathcal{P}_{n+1}}(R)), \quad 0 \leq a < b \leq T. \quad (2.1)$$

Proof. For $\mathcal{P}_n = \{a = t_0^n < \dots < t_{\#\mathcal{P}_n}^n = b\}$, we can write

$$|I_{\mathcal{P}_n}(R)| \leq \sum_{i=1}^{\#\mathcal{P}_n} |R_{t_{i-1}^n t_i^n}| \leq \left\{ \max_{j=1, \dots, \#\mathcal{P}_n} \frac{|R_{t_{j-1}^n t_j^n}|}{(t_j^n - t_{j-1}^n)} \right\} \sum_{j=1}^{\#\mathcal{P}_n} (t_j^n - t_{j-1}^n),$$

which implies $|I_{\mathcal{P}_n}(R)| \rightarrow 0$ as $n \rightarrow \infty$, since the sum is finite, $R_{st} = o(t - s)$ and $\max_{j=1, \dots, \#\mathcal{P}_n} (t_j^n - t_{j-1}^n) = |\mathcal{P}_n|$ tends to 0.

Finally, since $\lim_{N \rightarrow \infty} I_{\mathcal{P}_N}(R) = 0$ and $I_{\mathcal{P}_0}(R) = R_{ab}$ for $\mathcal{P}_0 = \{a, b\}$, we have

$$\begin{aligned} R_{ab} &= \lim_{N \rightarrow \infty} (I_{\mathcal{P}_0}(R) - I_{\mathcal{P}_N}(R)) \\ &= \lim_{N \rightarrow \infty} \sum_{n=0}^{N-1} (I_{\mathcal{P}_n}(R) - I_{\mathcal{P}_{n+1}}(R)) \\ &= \sum_{n=0}^{\infty} (I_{\mathcal{P}_n}(R) - I_{\mathcal{P}_{n+1}}(R)), \end{aligned}$$

which proves (2.1). □

THEOREM 2.2. (SEWING BOUND) Let us take any $R \in C([0, T]_{\leq}^2, \mathbb{R}^k)$ such that $R_{st} = o(t - s)$. Then, for any $\eta \in (1, \infty)$,

$$\|R\|_{\eta} \leq K_{\eta} \|\delta R\|_{\eta} \quad \text{where} \quad K_{\eta} := (1 - 2^{1-\eta})^{-1}. \quad (2.2)$$

Proof. Assume $\|\delta R\|_{\eta} < \infty$ for some $\eta > 1$, otherwise we are done.

Fix $0 \leq a < b \leq T$. For $n \geq 0$ we call $\mathcal{P}_n := \{t_i^n := a + \frac{i}{2^n}(b-a) : 0 \leq i \leq 2^n\}$ the dyadic partitions of $[a, b]$. Note that $\mathcal{P}_0$ is the trivial partition. By (2.1) we can write

$$|R_{ab}| \leq \sum_{n=0}^{\infty} |I_{\mathcal{P}_n}(R) - I_{\mathcal{P}_{n+1}}(R)|. \quad (2.3)$$

Now note that if we remove a single point t_i from a partition $\mathcal{P} = \{t_0 < t_1 < \dots < t_m\}$ we obtain a new partition $\mathcal{P}'$ and we can write

$$I_{\mathcal{P}'}(R) - I_{\mathcal{P}}(R) = \delta R_{t_{i-1} t_i t_{i+1}}. \quad (2.4)$$

Observe that removing all the points t_{2j+1}^{n+1} , with $0 \leq j \leq 2^n - 1$, from $\mathcal{P}_{n+1}$ we obtain $\mathcal{P}_n$, so that by (2.4) we have

$$\begin{aligned} |I_{\mathcal{P}_{n+1}}(R) - I_{\mathcal{P}_n}(R)| &\leq \sum_{j=0}^{2^n-1} |\delta R_{t_{2j}^{n+1} t_{2j+1}^{n+1} t_{2j+2}^{n+1}}| \\ &\leq 2^n \|\delta R\|_{\eta} \left(\frac{2(b-a)}{2^{n+1}} \right)^{\eta} \\ &= 2^{-(\eta-1)n} \|\delta R\|_{\eta} (b-a)^{\eta}. \end{aligned} \quad (2.5)$$

Substituting in (2.3) we get, for $0 \leq a < b \leq T$,

$$\begin{aligned} |R_{ab}| &\leq \sum_{n=0}^{\infty} 2^{-(\eta-1)n} \|\delta R\|_{\eta} (b-a)^{\eta} \\ &= (1 - 2^{1-\eta})^{-1} \|\delta R\|_{\eta} (b-a)^{\eta} \\ &= K_{\eta} \|\delta R\|_{\eta} (b-a)^{\eta}, \end{aligned}$$

which proves (2.2). $\square$

2.3. WEIGHTED NORMS

We now define a new norm modifying the usual norm $\|\cdot\|_{\eta}$, the idea is to introduce a weight τ that will be useful when we want to transform local results in global. In particular, the idea is to allow the time interval $[0, T]$ to be large, but to take instead $\tau > 0$ small.

DEFINITION 2.3. (WEIGHTED NORMS) Given $F \in C([0, T]_{\leq}^2, \mathbb{R}^k)$ and $G \in C([0, T]_{\leq}^3, \mathbb{R}^k)$, for $\eta, \tau \in (0, \infty)$ we define

$$\|F\|_{\eta, \tau} := \sup_{0 \leq s \leq t \leq T} \mathbb{1}_{\{0 < t-s \leq \tau\}} e^{-\frac{t}{\tau}} \frac{|F_{st}|}{(t-s)^{\eta}}, \quad (2.6)$$

$$\|G\|_{\eta, \tau} := \sup_{0 \leq s \leq u \leq t \leq T} \mathbb{1}_{\{0 < t-s \leq \tau\}} e^{-\frac{t}{\tau}} \frac{|G_{sut}|}{(t-s)^{\eta}}. \quad (2.7)$$

Remark 2.4. Note that when we let τ tend to infinity we recover the η norm:

$$\lim_{\tau \rightarrow \infty} \|\cdot\|_{\eta, \tau} = \|\cdot\|_{\eta}.$$

Remark 2.5. Note that $\|\cdot\|_{\eta, \tau}$ is, in general, not a norm but a seminorm. In case we have $F = \delta f$, then the norm $\|\cdot\|_{\eta, \tau}$ is equivalent to $\|\cdot\|_{\eta}$.

We also define an analogous for the supremum norm.

DEFINITION 2.6. (WEIGHTED SUPREMUM NORM) For $f \in C([0, T]_{\leq}, \mathbb{R}^k)$ we define

$$\|f\|_{\infty, \tau} := \sup_{0 \leq t \leq T} e^{-\frac{t}{\tau}} |f_t|.$$

Remark 2.7. Note that $\|\cdot\|_{\infty, \tau}$ is equivalent to $\|\cdot\|_{\infty}$ since

$$\|\cdot\|_{\infty, \tau} \leq \|\cdot\|_{\infty} \leq e^{\frac{T}{\tau}} \|\cdot\|_{\infty, \tau}.$$

2.4. USEFUL BOUNDS

We first need an estimate on the difference of increments of a function. For $\psi: \mathbb{R}^k \rightarrow \mathbb{R}^d$, we define

$$C_{\psi, R} := \sup \{ |\psi(x)| : x \in \mathbb{R}^k, |x| \leq R \}. \quad (2.8)$$

LEMMA 2.8. (DIFFERENCE OF INCREMENTS) *If $\psi: \mathbb{R}^k \rightarrow \mathbb{R}^d$ is C^1 with locally Lipschitz gradient, then for all $x, x', y, y' \in \mathbb{R}^k$ with $\max\{|x|, |x'|, |y|, |y'|\} \leq R$ we have*

$$\begin{aligned} & |[\psi(x) - \psi(y)] - [\psi(x') - \psi(y')]| \\ & \leq C_{\nabla\psi, R} |(x - y) - (x' - y')| + C_{\nabla^2\psi, R} \{|x - x'| + |y - y'|\} |x - y|. \end{aligned} \quad (2.9)$$

Proof. For the technical proof refer to [5]. $\square$

We now introduce some useful bounds on weighted norms. First we show an elementary bound that lets us bound $\|\cdot\|_{\eta, \tau}$ with $\|\cdot\|_{\eta+\eta', \tau}$.

LEMMA 2.9. *Let us fix $F \in C([0, T]_{\leq}, \mathbb{R}^k)$. For all $\eta, \eta' > 0$ we have*

$$\|F\|_{\eta, \tau} \leq (\tau \wedge T)^{\eta'} \|F\|_{\eta+\eta', \tau}. \quad (2.10)$$

Proof. We have

$$\begin{aligned} \|F\|_{\eta, \tau} &= \sup_{0 \leq s \leq t \leq T} \mathbb{1}_{\{0 < t-s \leq \tau\}} e^{-\frac{t}{\tau}} \frac{|F_{st}|}{(t-s)^\eta} \\ &\leq \sup_{0 \leq s \leq t \leq T} \mathbb{1}_{\{0 < t-s \leq \tau\}} e^{-\frac{t}{\tau}(\tau \wedge T)^{\eta'}} \frac{|F_{st}|}{(t-s)^{\eta+\eta'}} \\ &= (\tau \wedge T)^{\eta'} \|F\|_{\eta+\eta', \tau}. \end{aligned}$$

$\square$

Now we show a bound that connects the supremum norms to the Hölder norms.

LEMMA 2.10. (SUPREMUM-HÖLDER BOUND) *For $f \in C([0, T]_{\leq}, \mathbb{R}^k)$, $\eta, \tau \in (0, \infty)$, we have*

$$\|f\|_\infty \leq |f_0| + T^\eta \|\delta f\|_\eta, \quad (2.11)$$

$$\|f\|_{\infty, \tau} \leq |f_0| + 3(\tau \wedge T)^\eta \|\delta f\|_{\eta, \tau}. \quad (2.12)$$

Proof. For any $f \in C([0, T]_{\leq}, \mathbb{R}^k)$ and for $t \in (0, T]$ we can write

$$|f_t| \leq |f_0| + |f_t - f_0| = |f_0| + t^\eta \frac{|f_t - f_0|}{t^\eta} \leq |f_0| + T^\eta \|\delta f\|_\eta.$$

We proved (2.11). For (2.12) we consider two cases, $t \in (0, \tau \wedge T]$ and $\tau < t \leq T$. In the first we have

$$e^{-\frac{t}{\tau}} |f_t| \leq |f_0| + t^\eta e^{-\frac{t}{\tau}} \frac{|f_t - f_0|}{t^\eta} \leq |f_0| + (\tau \wedge T)^\eta \|\delta f\|_{\eta, \tau},$$

which implies (2.12). For the second case we define $N := \min\{n \in \mathbb{N} : n\tau \geq t\} \geq 2$, so that $\frac{t}{N} \leq \tau$. For $i \geq 0$ set $t_i = i\frac{t}{N}$, so that $t_N = t$. We get

$$\begin{aligned} e^{-\frac{t}{\tau}} |f_t| &\leq |f_0| + \sum_{i=1}^N (t_i - t_{i-1})^\eta e^{-\frac{t-t_i}{\tau}} \left[e^{-\frac{t_i}{\tau}} \frac{|f_{t_i} - f_{t_{i-1}}|}{(t_i - t_{i-1})^\eta} \right] \\ &\leq |f_0| + (\tau \wedge T)^\eta \|\delta f\|_{\eta, \tau} \sum_{i=1}^N e^{-\frac{t-t_i}{\tau}} \\ &= |f_0| + (\tau \wedge T)^\eta \|\delta f\|_{\eta, \tau} \sum_{i=1}^N e^{-(N-i)\frac{t}{N\tau}}. \end{aligned}$$

Since $(N-1)\tau < t$ and $\tau < t$, we get $N\tau < 2t$ which implies $\frac{t}{N\tau} \geq \frac{1}{2}$, so that

$$\sum_{i=1}^N e^{-(N-i)\frac{t}{N\tau}} = \sum_{(N-i)=0}^{N-1} e^{-(N-i)\frac{t}{N\tau}} = \frac{1 - e^{-\frac{t}{\tau}}}{1 - e^{-\frac{t}{N\tau}}} \leq \frac{1}{1 - e^{-\frac{1}{2}}} \leq 3.$$

We completed the proof. □

We now show some bounds for functions that are products of two factors.

LEMMA 2.11. (WEIGHTED BOUNDS) *Let $\eta, \tau \in (0, \infty)$. We have*

$$\text{for } F_{st} = g_s H_{st} \text{ or } F_{st} = g_t H_{st} \quad \text{we have } \|F\|_{\eta, \tau} \leq \|g\|_{\infty, \tau} \|H\|_{\eta}, \quad (2.13)$$

$$\text{for } G_{sut} = F_{su} H_{ut} \quad \text{we have } \|G\|_{2\eta, \tau} \leq \|F\|_{\eta, \tau} \|H\|_{\eta}. \quad (2.14)$$

Proof. If $F_{st} = g_t H_{st}$ we write

$$\begin{aligned} \|F\|_{\eta, \tau} &= \sup_{0 \leq s \leq t \leq T} \mathbb{1}_{\{0 < t-s \leq \tau\}} e^{-\frac{t}{\tau}} \frac{|F_{st}|}{(t-s)^\eta} \\ &= \sup_{0 \leq s \leq t \leq T} \mathbb{1}_{\{0 < t-s \leq \tau\}} e^{-\frac{t}{\tau}} \frac{|g_t H_{st}|}{(t-s)^\eta} \\ &\leq \|g\|_{\infty, \tau} \sup_{0 \leq s \leq t \leq T} \frac{|H_{st}|}{(t-s)^\eta} \\ &= \|g\|_{\infty, \tau} \|H\|_{\eta}. \end{aligned}$$

For $F_{st} = g_s H_{st}$ we observe that we can do the same using the bound $e^{-t/\tau} \leq e^{-s/\tau}$ for $s \leq t$.

If $G_{sut} = F_{su} H_{ut}$ we write

$$\begin{aligned} \|G\|_{2\eta, \tau} &= \sup_{0 \leq s \leq u \leq t \leq T} \mathbb{1}_{\{0 < t-s \leq \tau\}} e^{-\frac{t}{\tau}} \frac{|F_{su} H_{ut}|}{(t-s)^{2\eta}} \\ &\leq \sup_{0 \leq s \leq u \leq t \leq T} \mathbb{1}_{\{0 < t-s \leq \tau\}} e^{-\frac{s}{\tau}} \frac{|F_{su}| |H_{ut}|}{(u-s)^\eta (t-u)^\eta} \\ &\leq \|F\|_{\eta, \tau} \|H\|_{\eta}. \end{aligned}$$

□

Remark 2.12. There is a natural generalization of (2.14) which follows immediately. Let us take $G_{sut} = F_{su} H_{ut}$, $\eta, \tau \in (0, \infty)$ and $\eta_1, \eta_2 \in (0, \infty)$ such that $\eta_1 + \eta_2 = \eta$. Then we have

$$\|G\|_{\eta, \tau} \leq \|F\|_{\eta_1, \tau} \|H\|_{\eta_2}. \quad (2.15)$$

Note that in every bound of this kind only one factor takes the weighted norm.

LEMMA 2.13. (EXTRA WEIGHT) *Let $\eta, \bar{\tau} \in (0, \infty)$ and $0 < \tau \leq \bar{\tau}$. Then if $F_{st} = g_s H_{st}$ or $F_{st} = g_t H_{st}$ we have*

$$\|F\|_{\eta, \tau} \leq \|g\|_{\infty, \tau} e^{\frac{\tau}{\bar{\tau}}} \|H\|_{\eta, \bar{\tau}}. \quad (2.16)$$

Proof. For $F_{st} = g_t H_{st}$ we have

$$\begin{aligned} \|F\|_{\eta, \tau} &= \sup_{0 \leq s \leq t \leq T} \mathbb{1}_{\{0 < t-s \leq \tau\}} e^{-\frac{t}{\tau}} \frac{|g_t H_{st}|}{(t-s)^\eta} \\ &\leq \|g\|_{\infty, \tau} \sup_{0 \leq s \leq t \leq T} \mathbb{1}_{\{0 < t-s \leq \tau \leq \bar{\tau}\}} e^{\frac{\tau}{\bar{\tau}}} e^{-\frac{t}{\bar{\tau}}} \frac{|H_{st}|}{(t-s)^\eta} \\ &\leq \|g\|_{\infty, \tau} e^{\frac{\tau}{\bar{\tau}}} \|H\|_{\eta, \bar{\tau}}. \end{aligned}$$

We can do the same for $F_{st} = g_s H_{st}$ since $e^{-t/\tau} \leq e^{-s/\tau}$ for $s \leq t$.

□

2.5. WEIGHTED SEWING BOUND

We show that the Sewing Bound still holds for the weighted norm.

THEOREM 2.14. (WEIGHTED SEWING BOUND) *Let us take $R \in C([0, T]_{\leq}^2, \mathbb{R}^k)$ such that $R_{st} = o(t-s)$. Then for any $\eta \in (1, \infty)$ and $\tau > 0$ we have*

$$\|R\|_{\eta, \tau} \leq K_\eta \|\delta R\|_{\eta, \tau} \quad \text{where} \quad K_\eta := (1 - 2^{1-\eta})^{-1} \quad (2.17)$$

Proof. For $0 \leq a \leq b \leq T$ we define

$$\|\delta R\|_{\eta, [a, b]} := \sup_{\substack{s, u, t \in [a, b] \\ s \leq u \leq t, s < t}} \frac{|\delta R_{sut}|}{(t-s)^\eta}.$$

Now we can follow the proof of (2.2) to obtain $|R_{ab}| \leq K_\eta \|\delta R\|_{\eta, [a, b]} (b-a)^\eta$. Now we write

$$e^{-\frac{b}{\tau}} \frac{|R_{ab}|}{(b-a)^\eta} \leq e^{-\frac{b}{\tau}} K_\eta \|\delta R\|_{\eta, [a, b]} \leq K_\eta \|\delta R\|_{\eta, \tau},$$

which permits us to conclude the proof by taking the supremum over $0 \leq a \leq b \leq T$, $b-a \leq \tau$. □

2.6. DISCRETE SEWING BOUND

We want to give a version of the Sewing Bound for functions defined on a finite set of points. This will be useful when we will use discretization techniques to prove existence of solutions. As in the original sewing bound, we will take a function $R = (R_{st})_{s < t}$, where $s, t \in \mathbb{T} = \{t_0 < \dots < t_{|\mathbb{T}|}\} \subseteq \mathbb{R}$ such that $R_{t_i t_{i+1}} = 0$, i.e. R vanishes on consecutive points of $\mathbb{T}$. We will denote with $\|\cdot\|_\eta^\mathbb{T}$ the norm $\|\cdot\|_\eta$ restricted to $\mathbb{T}$. We have the following result.

THEOREM 2.15. (DISCRETE SEWING BOUND) *Let us take $R = (R_{st})_{s < t}$, where $s, t \in \mathbb{T} = \{t_0 < \dots < t_{|\mathbb{T}|}\} \subseteq \mathbb{R}$ such that $R_{t_i t_{i+1}} = 0$. Then, for any $\eta \in (1, \infty)$, we have*

$$\|R\|_\eta^\mathbb{T} \leq C_\eta \|\delta R\|_\eta^\mathbb{T} \quad \text{where} \quad C_\eta := 2^\eta \sum_{n \geq 1} \frac{1}{n^\eta}. \quad (2.18)$$

Proof. We want to show that

$$|R_{st}| \leq C_\eta \|\delta R\|_\eta^\mathbb{T} (t-s)^\eta,$$

for $s, t \in \mathbb{T}$ such that $s < t$. We may write $s = t_k$, $t = t_{k+m}$ and we can assume that $m \geq 2$, since otherwise s and t are consecutive and $R_{st} = 0$ by hypothesis.

We consider the partition $\mathcal{P} = \{s = t_k < \dots < t_{k+m} = t\}$. We note that for some index $i \in \{1, \dots, m-1\}$ we must have $t_{i+1} - t_{i-1} \leq \frac{2(t-s)}{m-1}$, otherwise we observe that

$$2(t-s) \geq \sum_{i=1}^{m-1} (t_{i+1} - t_{i-1}) > \sum_{i=1}^{m-1} \frac{2(t-s)}{m-1} = 2(t-s),$$

which is clearly a contradiction.

Similarly to the proof of the Sewing Bound we define $I_{\mathcal{P}}(R) := \sum_{i=0}^{m-1} R_{t_i t_{i+1}}$ and we consider a partition $\mathcal{P}'$ obtained removing t_i from $\mathcal{P}$. As in (2.5) we have

$$|I_{\mathcal{P}}(R) - I_{\mathcal{P}'}(R)| \leq \|\delta R\|_\eta^\mathbb{T} \frac{2^\eta(t-s)}{(m-1)^\eta}.$$

Iterating until the partition $\{s, t\}$, we get

$$|I_{\mathcal{P}}(R) - R_{st}| \leq C_\eta \|\delta R\|_\eta^\mathbb{T} (t-s)^\eta \quad \text{where} \quad C_\eta := 2^\eta \sum_{n \geq 1} \frac{1}{n^\eta} < \infty \quad \text{for} \quad \eta > 1.$$

We can conclude since $I_{\mathcal{P}}(R) = 0$ by the assumption $R_{t_i t_{i+1}} = 0$.

□

CHAPTER 3

FINITE DIFFERENCE EQUATIONS

We will now introduce the so called controlled ordinary differential equations, which are the main focus of this thesis. The goal is to give meaning to controlled ODEs with an irregular driving path. We will follow the approach of [5].

3.1. CONTROLLED DIFFERENTIAL EQUATIONS

Let us first show a classical controlled ODE. Let us fix a time horizon $T > 0$, a path $X: [0, T] \rightarrow \mathbb{R}^d$ and a continuous function $\sigma: \mathbb{R}^k \rightarrow \mathbb{R}^k \otimes (\mathbb{R}^d)^*$. We call a controlled ODE an equation in the form

$$\dot{Z}(t) = \sigma(Z(t))\dot{X}(t), \quad t \in [0, T],$$

for a path $Z: [0, T] \rightarrow \mathbb{R}^d$. To ease the notation we will write

$$\dot{Z}_t = \sigma(Z_t)\dot{X}_t, \quad t \in [0, T]. \quad (3.1)$$

When the driving path X is of class C^1 , the equation (3.1) is well-defined and we can write the solution Z as

$$Z_t = Z_0 + \int_0^t \sigma(Z_s)\dot{X}_s ds, \quad t \in [0, T]. \quad (3.2)$$

In general, if we suppose that $\sigma(\cdot)$ is a Lipschitz function, then equation (3.1) is well-posed and we can find a unique solution Z which is of class C^1 , see for example [8].

We want to consider a similar problem, but this time with a driving path X that is continuous, but not continuously differentiable. In this case equation (3.1) does not make sense, since the derivative of X appears. We will therefore need to reformulate (3.1) in order to be meaningful in this new setting.

Supposing X is of class C^1 , by (3.2) we are able to write that, for $0 \leq s \leq t \leq T$,

$$\begin{aligned} Z_t - Z_s &= Z_0 + \int_0^t \sigma(Z_u)\dot{X}_u du - Z_0 - \int_0^s \sigma(Z_u)\dot{X}_u du \\ &= \int_s^t \sigma(Z_u)\dot{X}_u du \\ &= \sigma(Z_s)(X_t - X_s) + \int_s^t (\sigma(Z_u) - \sigma(Z_s))\dot{X}_u du. \end{aligned} \quad (3.3)$$

Since $u \mapsto \sigma(Z_u)$ and $u \mapsto \dot{X}_u$ are continuous and bounded on $[0, T]$, the last term of (3.3) is $o(t - s)$, so that (3.3) implies the controlled difference equation

$$Z_t - Z_s = \sigma(Z_s)(X_t - X_s) + o(t - s), \quad 0 \leq s \leq t \leq T. \quad (3.4)$$

Note that (3.4) is meaningful also for X not continuously differentiable.

We can separate different cases depending on the regularity of X . In particular, when $X \in \mathcal{C}^\alpha$ is Hölder continuous for $\alpha \in (\frac{1}{2}, 1]$, we fall into the *Young case*. It is well known in the literature that in this case equation (3.4) is well-posed, refer for example to [21].

3.2. ROUGH CASE

Much more interesting than the Young case is the so called *rough case*, which consists in assuming that the driving path $X \in \mathcal{C}^\alpha$ is Hölder continuous but only for $\alpha \leq \frac{1}{2}$.

Note that in this case we could take $X = B$, where B is a path of Brownian motion which is almost $\frac{1}{2}$ -Hölder continuous, so that we find ourselves in the classical setting of Stochastic differential equations (SDE), which is of course of great relevance.

Looking at the difference equation (3.4) we can see that in the rough case existence of a solution is not guaranteed. Indeed, by taking increments on both sides of (3.4) we get

$$\delta(Z_t - Z_s) = \delta(\sigma(Z_s)(X_t - X_s)) + o(t - s),$$

which, since $\delta(\delta Z_{st}) = 0$ and

$$\begin{aligned} \delta(\sigma(Z_s)(X_t - X_s)) &= \delta(\sigma(Z_s)(\delta X_{st})) \\ &= \sigma(Z_s)\delta X_{st} - \sigma(Z_s)\delta X_{su} - \sigma(Z_u)\delta X_{ut} \\ &= \sigma(Z_s)\delta X_{ut} - \sigma(Z_u)\delta X_{ut} \\ &= (\sigma(Z_u) - \sigma(Z_s))\delta X_{su}, \end{aligned}$$

implies

$$(\sigma(Z_u) - \sigma(Z_s))(X_t - X_u) = o(t - s), \quad \text{for } 0 \leq s \leq u \leq t \leq T. \quad (3.5)$$

It is easy to see (for example in [5]) that, when $X \in \mathcal{C}^\alpha$, the solution Z of (3.4) can also be at most of class $\mathcal{C}^\alpha$, so that the left hand side of (3.5) is at most of order $o((u - s)^\alpha(t - u)^\alpha) \lesssim o((t - s)^{2\alpha})$ which is not $o(t - s)$ for $\alpha < \frac{1}{2}$. Therefore (3.4) does not necessarily admit a solution in the rough case.

3.3. ROUGH DIFFERENCE EQUATION

Let us suppose for now that $X \in \mathcal{C}^\alpha$ is Hölder continuous for $\alpha \in (\frac{1}{3}, \frac{1}{2}]$. We saw that in the Young case we started from equation (3.1) supposing $X \in C^1$ and we derived the equation (3.3), which we rewrite:

$$Z_t - Z_s = \sigma(Z_s)(X_t - X_s) + \int_s^t (\sigma(Z_u) - \sigma(Z_s))\dot{X}_u du, \quad \text{for } 0 \leq s \leq t \leq T. \quad (3.6)$$

The idea now is to expand the right hand side in order to make the integral $O((t - s)^3)$. This way we could do something similar to the Young case and impose that the integral is only $o(t - s)$.

Let us assume that σ is differentiable and observe that we can write, for $0 \leq s \leq u \leq T$,

$$\begin{aligned} \sigma(Z_u) - \sigma(Z_s) &= \int_0^u \frac{d}{dr} \sigma(Z_r) dr \\ &= \int_0^u \nabla \sigma(Z_r) \dot{Z}_r dr \\ &= \int_0^u \nabla \sigma(Z_r) \sigma(Z_r) \dot{X}_r dr \\ &= \nabla \sigma(Z_s) \sigma(Z_s) (X_u - X_s) + \int_s^u (\nabla \sigma(Z_r) \sigma(Z_r) - \nabla \sigma(Z_s) \sigma(Z_s)) \dot{X}_r dr, \end{aligned}$$

which, defining the function $\sigma_2: \mathbb{R}^k \rightarrow \mathbb{R}^k \otimes (\mathbb{R}^d)^* \otimes (\mathbb{R}^d)^*$ by

$$\sigma_2(z) := \nabla \sigma(z) \sigma(z), \quad \text{or} \quad [\sigma_2(z)]_{jl}^i := \sum_{a=1}^k \frac{\partial \sigma_j^i}{\partial z_a}(z) \sigma_l^a(z),$$

we can rewrite as

$$\sigma(Z_u) - \sigma(Z_s) = \sigma_2(Z_s)(X_u - X_s) + \int_s^u (\sigma_2(Z_r) - \sigma_2(Z_s)) \dot{X}_r dr. \quad (3.7)$$

If we assume that σ_2 is locally Lipschitz, the integral in (3.7) is $O((t-s)^2)$, so that, plugging (3.7) into (3.6), we can write

$$Z_t - Z_s = \sigma(Z_s)(X_t - X_s) + \sigma_2(Z_s) \int_s^t (X_u - X_s) \otimes \dot{X}_u du + O((t-s)^3). \quad (3.8)$$

Defining for convenience

$$\mathbb{X}_{st}^{(1)} := X_t - X_s, \quad \mathbb{X}_{st}^{(2)} := \int_s^t (X_u - X_s) \otimes \dot{X}_u du, \quad 0 \leq s \leq t \leq T, \quad (3.9)$$

we can rewrite (3.8) as

$$Z_t - Z_s = \sigma(Z_s) \mathbb{X}_{st}^{(1)} + \sigma_2(Z_s) \mathbb{X}_{st}^{(2)} + O((t-s)^3), \quad 0 \leq s \leq t \leq T.$$

Now, substituting $O((t-s)^3)$ with $o(t-s)$ we obtain a rough difference equation for $\alpha \in (\frac{1}{3}, \frac{1}{2}]$:

$$Z_t - Z_s = \sigma(Z_s) \mathbb{X}_{st}^{(1)} + \sigma_2(Z_s) \mathbb{X}_{st}^{(2)} + o((t-s)), \quad 0 \leq s \leq t \leq T. \quad (3.10)$$

Notice that we still have a problem since in the definition of $\mathbb{X}^{(2)}$ the derivative of X appears. We will therefore have to use an approximation of the integral $\int_s^t (X_u - X_s) \otimes \dot{X}_u du$ and this leads to the fundamental notion of rough paths. In practice we will define the integral as a suitable function, with some intuitive properties.

3.4. GENERAL ROUGH DIFFERENCE EQUATION

We want to define a rough difference equation when $X \in \mathcal{C}^\alpha$ is Hölder continuous for $\alpha \in (\frac{1}{n}, \frac{1}{n+1}]$. We can do something similar to the simple case.

Let us suppose for now that is X of class C^1 . Let us also set $E := \mathbb{R}^d$ and for $n \geq 1$

$$E^{\otimes 0} := \mathbb{R}, \quad E^{\otimes n} := \underbrace{E \otimes \cdots \otimes E}_{n \text{ times}}, \quad (E^*)^{\otimes 0} := \mathbb{R}, \quad (E^*)^{\otimes n} := \underbrace{E^* \otimes \cdots \otimes E^*}_{n \text{ times}}.$$

Then we define recursively $\sigma_n: \mathbb{R}^k \rightarrow \mathbb{R}^k \otimes (E^*)^{\otimes n}$ in the following way:

$$\sigma_0 := \text{Id}_{\mathbb{R}^k}, \quad (\sigma_{n+1}(z))_{a_1 \dots a_{n+1}}^i := \sum_{b=1}^k \partial_b (\sigma_n(z))_{a_1 \dots a_n}^i (\sigma)_{a_{n+1}}^b(z), \quad (3.11)$$

and $\mathbb{X}^{(n)}: [0, T]_{\leq}^2 \rightarrow E^{\otimes n}$ such that

$$\mathbb{X}_{st}^{(0)} := 1, \quad \mathbb{X}_{st}^{(i+1)} := \int_s^t \mathbb{X}_{su}^{(i)} \otimes \dot{X}_u \, du.$$

In particular $\mathbb{X}^{(1)} = \delta X$. Note that

$$[\mathbb{X}_{st}^{(n)}]^{a_n \dots a_1} = \int_s^t \int_s^{r_1} \cdots \int_s^{r_{n-1}} \dot{X}_{r_n}^{a_n} \, dr_n \cdots \dot{X}_{r_1}^{a_1} \, dr_1,$$

and that we will use the contraction rule

$$[\sigma_n(z) \mathbb{X}_{st}^{(n)}]^i = \sum_{a_1, \dots, a_n=1}^d [\sigma_n(z)]_{a_1 \dots a_n}^i [\mathbb{X}_{st}^{(n)}]^{a_n \dots a_1}.$$

For fixed $Z_0 \in \mathbb{R}^k$ we consider the unique $Z: [0, T] \rightarrow \mathbb{R}^k$ such that

$$Z_t = Z_0 + \int_0^t \sigma(Z_s) \dot{X}_s \, ds, \quad t \in [0, T]. \quad (3.12)$$

THEOREM 3.1. *We have for $0 \leq s \leq t \leq T$ and $n \geq 0$*

$$Z_t - \sum_{i=0}^n \sigma_i(Z_s) \mathbb{X}_{st}^{(i)} = O(|t - s|^{n+1}). \quad (3.13)$$

Proof. First we prove that

$$\begin{aligned} Z_t - \sum_{i=0}^n \sigma_i(Z_s) \mathbb{X}_{st}^{(i)} &= \\ &= \int_{t > r_1 > \cdots > r_{n+1} > s} \sigma_{n+1}(Z_{r_{n+1}}) \dot{X}_{r_{n+1}} \otimes \cdots \otimes \dot{X}_{r_1} \, dr_1 \cdots dr_{n+1}. \end{aligned} \quad (3.14)$$

We argue by recurrence on $n \geq 0$. For $n = 0$ formula (3.14) reduces to (3.12).

If $f: \mathbb{R}^k \rightarrow \mathbb{R}^m$ then

$$\frac{d}{dt} f(Z_t) = \sum_{a=1}^k \partial_a f(Z_t) \dot{Z}_t^a = \sum_{a=1}^k \sum_{i=1}^d \partial_a f(Z_t) \sigma_i^a(Z_t) \dot{X}_t^i. \quad (3.15)$$

By (3.15) we have, by recurrence on n ,

$$\sigma_n(Z_t) - \sigma_n(Z_s) = \int_s^t \sigma_{n+1}(Z_r) \dot{X}_r \, dr, \quad n \geq 0. \quad (3.16)$$

Now suppose we have proved (3.14) for $n \geq 0$. Then

$$\begin{aligned} Z_t - \sum_{i=0}^{n+1} \sigma_i(Z_s) \mathbb{X}_{st}^{(i)} &= \\ &= \int_{t > r_1 > \dots > r_{n+1} > s} (\sigma_{n+1}(Z_{r_{n+1}}) - \sigma_{n+1}(Z_s)) \dot{X}_{r_{n+1}} \otimes \dots \otimes \dot{X}_{r_1} dr_1 \dots dr_{n+1} \\ &= \int_{t > r_1 > \dots > r_{n+2} > s} \sigma_{n+2}(Z_{r_{n+2}}) \dot{X}_{r_{n+2}} \otimes \dots \otimes \dot{X}_{r_1} dr_1 \dots dr_{n+2} \end{aligned}$$

where we have used (3.16). Now (3.13) follows from (3.14). $\square$

Remark 3.2. Note that we also have

$$\sigma(Z_t) - \sum_{n=0}^N \sigma_{n+1}(Z_s) \mathbb{X}_{st}^{(n)} = O(|t - s|^{n+1}).$$

The case $n = 0$ follows from (3.16) with $n = 1$. By recurrence and by (3.16)

$$\begin{aligned} \sigma(Z_t) - \sum_{i=0}^N \sigma_{i+1}(Z_s) \mathbb{X}_{st}^{(i)} &= \\ &= \int_{t > r_1 > \dots > r_{n+1} > s} \sigma_{n+2}(Z_{r_{n+1}}) \dot{X}_{r_{n+1}} \otimes \dots \otimes \dot{X}_{r_1} dr_1 \dots dr_{n+1}. \end{aligned}$$

As in the simple case we want to define $\mathbb{X}^{(i)}$ for $i \in \{1, \dots, n\}$ as functions that do not depend on $\dot{X}$ and that mimic the integrals $\mathbb{X}_{st}^{(i)} := \int_s^t \mathbb{X}_{su}^{(i-1)} \otimes \dot{X}_u du$.

CHAPTER 4

ROUGH PATHS

4.1. ROUGH PATHS FOR $\alpha \in (1/3, 1/2]$

We first give a definition for the most basic rough path. From the discussion in the last chapter we understood that in order to make sense of (3.10) we need to define $\mathbb{X} = (\mathbb{X}^{(1)}, \mathbb{X}^{(2)})$ which mimics the expressions $X_t - X_s$ and $\int_s^t (X_u - X_s) \otimes \dot{X}_u du$ and is meaningful when X is a path of class $\mathcal{C}^\alpha$ for $\alpha \in (\frac{1}{3}, \frac{1}{2}]$.

We have no problem in defining $\mathbb{X}_{st}^{(1)} := X_t - X_s$, but for $\mathbb{X}_{st}^{(2)}$ the situation is different. Let us suppose for an instant that $X \in C^1$ and that $\mathbb{X}_{st}^{(2)} = \int_s^t (X_u - X_s) \otimes \dot{X}_u du$. It was first shown in [7] that $\mathbb{X}^{(2)}$ satisfies an algebraic identity called *Chen's relation*: for $0 \leq s \leq u \leq t \leq T$

$$\mathbb{X}_{st}^{(2)} - \mathbb{X}_{su}^{(2)} - \mathbb{X}_{ut}^{(2)} = \mathbb{X}_{su}^{(1)} \otimes \mathbb{X}_{ut}^{(1)}$$

and that

$$|\mathbb{X}_{st}^{(1)}| \lesssim |t - s|, \quad |\mathbb{X}_{st}^{(2)}| \lesssim |t - s|^2.$$

Then it feels natural to give the following definition.

DEFINITION 4.1. (ROUGH PATHS $\alpha \in (\frac{1}{3}, \frac{1}{2}]$) Fix $X: [0, T] \rightarrow \mathbb{R}^d$ of class $\mathcal{C}^\alpha$ for $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ and $d \in \mathbb{N}$. We call a d -dimensional α -rough path over X a pair $\mathbb{X} = (\mathbb{X}^{(1)}, \mathbb{X}^{(2)})$ of functions $\mathbb{X}^{(1)}: [0, T]_{\leq}^2 \rightarrow \mathbb{R}^d$ and $\mathbb{X}^{(2)}: [0, T]_{\leq}^2 \rightarrow \mathbb{R}^d \otimes \mathbb{R}^d$ such that, for $0 \leq u \leq t \leq T$,

$$\mathbb{X}_{st}^{(1)} := X_t - X_s, \tag{4.1}$$

$$\delta \mathbb{X}_{sut}^{(2)} := \mathbb{X}_{st}^{(2)} - \mathbb{X}_{su}^{(2)} - \mathbb{X}_{ut}^{(2)} = \mathbb{X}_{su}^{(1)} \otimes \mathbb{X}_{ut}^{(1)}, \quad (\text{Chen's relation}) \tag{4.2}$$

$$|\mathbb{X}_{st}^{(1)}| \lesssim |t - s|^\alpha, \quad |\mathbb{X}_{st}^{(2)}| \lesssim |t - s|^{2\alpha}. \tag{4.3}$$

We call $\mathcal{R}_{\alpha,d}(X)$ the set of d -dimensional α -rough path over X .

DEFINITION 4.2. (GEOMETRIC ROUGH PATHS) We call $\mathcal{R}_{\alpha,d}^g(X)$, the set of geometric rough paths, the closure of $\mathcal{R}_{1,d}(X)$ in $\mathcal{R}_{\alpha,d}(X)$ for $\alpha \in (\frac{1}{3}, 1]$.

Note that for two smooth paths $f, g \in C^1$ the integration by parts formula holds:

$$\int_s^t f(u) dg(u) = f(t)g(t) - f(s)g(s) - \int_s^t g(u) df(u).$$

This implies

$$\int_s^t (f(u) - f(s)) dg(u) + \int_s^t (g(u) - g(s)) df(u) = (f(t) - f(s))(g(t) - g(s)).$$

We want to highlight rough paths that have a similar property. Let us take $f = X_i$ and $g = X_j$, now for a smooth path X , defining $(\mathbb{X}^{(1)}, \mathbb{X}^{(2)})$ as in (3.9), the previous relation reads as

$$(\mathbb{X}_{st}^{(2)})^{ij} + (\mathbb{X}_{st}^{(2)})^{ji} = (\mathbb{X}_{st}^{(1)})^i (\mathbb{X}_{st}^{(1)})^j.$$

This motivates the following definition.

DEFINITION 4.3. (WEAKLY GEOMETRIC ROUGH PATHS) *We call $\mathbb{X} \in \mathcal{R}_{\alpha,d}(X)$ weakly geometric if*

$$(\mathbb{X}_{st}^{(2)})^{ij} + (\mathbb{X}_{st}^{(2)})^{ji} = (\mathbb{X}_{st}^{(1)})^i (\mathbb{X}_{st}^{(1)})^j.$$

We write $\mathbb{X} \in \mathcal{R}_{\alpha,d}^{\text{wg}}(X)$.

LEMMA 4.4. $\mathcal{R}_{\alpha,d}^g(X) \subset \mathcal{R}_{\alpha,d}^{\text{wg}}(X)$.

Proof. Refer to [5] for a proof. □

We want now to extend the definition for $\alpha > 0$ arbitrary small.

4.2. GENERAL ROUGH PATHS

As we saw in section 3.4 we took some functions $\mathbb{X}^{(i)}: [0, T]_{\leq}^2 \rightarrow E^{\otimes i}$ for $i \in \{1, \dots, n\}$, such that

$$\mathbb{X}_{st}^{(0)} := 1 \in \mathbb{R}, \quad \mathbb{X}_{st}^{(i+1)} := \int_s^t \mathbb{X}_{su}^{(i)} \otimes \dot{X}_u \, du.$$

In particular $\mathbb{X}^{(1)} = \delta X$. Note that

$$[\mathbb{X}_{st}^{(i)}]^{a_n \dots a_1} = \int_s^t \int_s^{r_1} \dots \int_s^{r_{i-1}} \dot{X}_{r_i}^{a_i} \, dr_i \dots \dot{X}_{r_1}^{a_1} \, dr_1.$$

As in the simple case we have some analytic bounds on the functions:

$$|\mathbb{X}_{st}^{(i)}| \lesssim |t - s|^i \quad \text{for } i \in \{1, \dots, n\}.$$

We want to analyze the Chen's relation in this setting.

4.2.1. Chen's relation

We define $T((E))$ as the space of formal series

$$T((E)) := \left\{ \sum_{i \geq 0} v^i : v^i \in E^{\otimes i}, \quad \forall n \geq 0 \right\}.$$

Then we set

$$\mathbb{X}_{st} := \sum_{i \geq 0} \mathbb{X}_{st}^{(i)} \in T((E)), \quad 0 \leq s \leq t \leq T. \quad (4.4)$$

We have a canonical associative non-commutative product on $T((E))$ given by the natural extension of the tensor product:

$$\left(\sum_{i \geq 0} u^i\right) \otimes \left(\sum_{i \geq 0} v^i\right) := \sum_{i \geq 0} w^i, \quad w^i := \sum_{j=0}^i u^j \otimes v^{i-j}.$$

We can write the Chen's relation in this setting.

PROPOSITION 4.5. (GENERAL CHEN'S RELATION) *For $\mathbb{X}$ defined in (4.4) we have*

$$\mathbb{X}_{st} = \mathbb{X}_{su} \otimes \mathbb{X}_{ut} \quad 0 \leq s \leq u \leq t \leq T.$$

Componentwise, this means

$$\mathbb{X}_{st}^{(i)} = \sum_{j=0}^i \mathbb{X}_{su}^{(i-j)} \otimes \mathbb{X}_{ut}^{(j)}, \quad \text{for } i \in \{1, \dots, n\}.$$

Proof. This formula is based on the following equality which holds for all $(r_1, \dots, r_n) \in \mathbb{R}^n$:

$$\mathbb{1}_{(s < r_n < \dots < r_1 < t)} = \mathbb{1}_{(s < r_n < \dots < r_1 < t)} \left[\sum_{i=0}^n \mathbb{1}_{(r_{i+1} < u < r_i)} + \sum_{i=1}^n \mathbb{1}_{(u = r_i)} \right],$$

where we set $r_{n+1} := s$ and $r_0 := t$. Since the set $\{(r_1, \dots, r_n) \in \mathbb{R}^n : r_i = u\}$ has a n -dimensional Lebesgue measure equal to 0, we can write

$$\begin{aligned} [\mathbb{X}_{st}^{(n)}]^{a_i \dots a_1} &= \int_{s < r_n < \dots < r_1 < t} \dot{X}_{r_n}^{a_n} \dots \dot{X}_{r_1}^{a_1} dr_1 \dots dr_n \\ &= \sum_{i=0}^n \int_{s < r_n < \dots < r_1 < t} \mathbb{1}_{(r_{i+1} < u < r_i)} \dot{X}_{r_n}^{a_n} \dots \dot{X}_{r_1}^{a_1} dr_1 \dots dr_n \\ &= \sum_{i=0}^n \int_{s < r_n < \dots < r_{i+1} < u} \dot{X}_{r_n}^{a_n} \dots \dot{X}_{r_{i+1}}^{a_{i+1}} dr_{i+1} \dots dr_n \int_{u < r_i < \dots < r_1 < t} \dot{X}_{r_i}^{a_i} \dots \dot{X}_{r_1}^{a_1} dr_1 \dots dr_i \\ &= \sum_{i=0}^n [\mathbb{X}_{st}^{(n-i)}]^{a_n \dots a_{i+1}} [\mathbb{X}_{st}^{(i)}]^{a_i \dots a_1} = \sum_{i=0}^n [\mathbb{X}_{su}^{(n-i)} \otimes \mathbb{X}_{ut}^{(i)}]^{a_n \dots a_1}. \end{aligned}$$

The proof is complete. $\square$

Remark 4.6. Note that the space

$$G := \left\{ 1 + \sum_{n \geq 1} v^n : v^n \in V^n \right\}$$

is a group for the tensor product:

- $(1 + \sum_{n \geq 1} v^n) \otimes (1 + \sum_{n \geq 1} w^n) = 1 + \sum_{n \geq 1} \sum_{i=0}^n v^{n-i} \otimes w^i$
- $(1 + \sum_{n \geq 1} v^n)^{-1} = 1 + \sum_{n \geq 1} u^n$ where $(u^n)_{n \geq 1}$ is obtained recursively by

$$u^n = -v^n - \sum_{i=1}^{n-1} v^{n-i} \otimes u^i.$$

4.2.2. General Rough Paths

Motivated by the previous discussion, it is natural to give the following definition of a general rough path.

DEFINITION 4.7. (ROUGH PATHS $\alpha \in (\frac{1}{n+1}, \frac{1}{n}]$) Fix $X: [0, T] \rightarrow \mathbb{R}^d$ of class $\mathcal{C}^\alpha$ for $\alpha \in (\frac{1}{n+1}, \frac{1}{n}]$ and $d \in \mathbb{N}$. We call a d -dimensional α -rough path over X a family $\mathbb{X} = (\mathbb{X}^{(1)}, \dots, \mathbb{X}^{(n)})$ of functions $\mathbb{X}^{(i)}: [0, T]_{\leq}^2 \rightarrow E^{\otimes i}$ such that, for $0 \leq u \leq t \leq T$ and $i \in \{1, \dots, n\}$,

$$\mathbb{X}_{st}^{(1)} := X_t - X_s, \quad (4.5)$$

$$\mathbb{X}_{st}^{(i)} = \sum_{j=0}^i \mathbb{X}_{su}^{(i-j)} \otimes \mathbb{X}_{ut}^{(j)}, \quad \text{for } i \in \{1, \dots, n\}, \quad (4.6)$$

$$|\mathbb{X}_{st}^{(i)}| \lesssim |t - s|^{i\alpha}. \quad (4.7)$$

We call $\mathcal{R}_{\alpha,d}(X)$ the set of d -dimensional α -rough path over X .

4.2.3. (Weakly) Geometric Rough Paths

Now we want to give a notion of chain rule for general rough paths. We know that in the simple case geometric rough paths satisfy the chain rule and we want to extend it to the general case.

We define the space $T((\mathbb{R}^d)^*)$ as

$$T(E^*) := \bigoplus_{n \geq 0} (E^*)^{\otimes n} = \bigoplus_{n \geq 0} \underbrace{(\mathbb{R}^d)^* \otimes \dots \otimes (\mathbb{R}^d)^*}_{n \text{ times}}.$$

Then $T((E))$ and $T(E^*)$ are naturally in duality:

$$\left\langle \sum_{n \geq 0} v^n, u^m \right\rangle_{(E^*)^{\otimes n}} := \langle v^m, u^m \rangle_{E^{\otimes m}}, \quad u_m \in E^{\otimes m}.$$

We use the following notation: for $i_1, \dots, i_n \in \{1, \dots, d\}$

$$i_1 \cdots i_n := e_{i_1} \otimes \dots \otimes e_{i_n} \in E^{\otimes n}.$$

The element $e_{i_1} \otimes \dots \otimes e_{i_n}$ is interpreted as a *word* in the alphabet $\{1, \dots, d\}$ and the tensor product becomes a *concatenation* of words:

$$(e_{i_1} \otimes \dots \otimes e_{i_n}) \otimes (e_{j_1} \otimes \dots \otimes e_{j_m}) = (i_1 \cdots i_n)(j_1 \cdots j_m) = i_1 \cdots i_n j_1 \cdots j_m.$$

We still denote by 1 the empty word. On $T(E^*)$ we define the *shuffle product*

$$i_1 \cdots i_n \sqcup i_{n+1} \cdots i_{n+m} := \sum_{\alpha^{-1} \in \text{Sh}(n, m)} i_{\alpha(1)} \cdots i_{\alpha(n+m)},$$

where $\text{Sh}(n, m)$ is the set of permutations $\beta \in S_{n+m}$ of $\{1, \dots, n+m\}$ such that

$$\beta(1) < \dots < \beta(n), \quad \beta(n+1) < \dots < \beta(n+m).$$

The shuffle product can also be defined recursively: for all $v, w \in T(E^*)$ and $i, j \in \{1, \dots, d\}$

$$1 \sqcup v = v \sqcup 1 = v, \quad (va) \sqcup (wb) = (v \sqcup wb)a + (va \sqcup w)b.$$

We can now introduce a subset $H \subset G$ given by

$$H := \{\mathbb{X} \in G: \langle \mathbb{X}, v \sqcup w \rangle = \langle \mathbb{X}, v \rangle \langle \mathbb{X}, w \rangle \ \forall v, w \in T((\mathbb{R}^d)^*)\}. \quad (4.8)$$

H is the set of weakly geometric rough paths in this new setting. See [19] for details on the algebraic properties of shuffles.

In the general setting we will always work with (weakly) geometric rough paths, and (4.8) will be one of the main tools.

CHAPTER 5

ROUGH EQUATION FOR $\alpha \in (1/3, 1/2]$

The goal now is to give results on the well-posedness of a rough difference equation for the simplest case. We will follow closely [5]. Let us fix a time horizon $T > 0$ and the dimensions $d, k \in \mathbb{N}$. We also fix a path $X: [0, T] \rightarrow \mathbb{R}^d$ such that $X \in \mathcal{C}^\alpha$ for $\alpha \in (\frac{1}{3}, \frac{1}{2}]$, an α -rough path $\mathbb{X} = (\mathbb{X}^{(1)}, \mathbb{X}^{(2)})$ over X and a differentiable function $\sigma: \mathbb{R}^k \rightarrow \mathbb{R}^k \otimes (\mathbb{R}^d)^*$.

Since σ is differentiable we are able to define

$$\sigma_2: \mathbb{R}^k \rightarrow \mathbb{R}^k \otimes (\mathbb{R}^d)^* \otimes (\mathbb{R}^d)^* \quad \text{with} \quad \sigma_2(x) := \nabla \sigma(x) \sigma(x). \quad (5.1)$$

We consider the rough difference equation

$$\delta Z_{st} = \sigma(Z_s) \mathbb{X}_{st}^{(1)} + \sigma_2(Z_s) \mathbb{X}_{st}^{(2)} + o(t-s), \quad 0 \leq s \leq t \leq T, \quad (5.2)$$

where $Z: [0, T] \rightarrow \mathbb{R}^k$ is an unknown path.

We define the remainders $Z^{[3]}$ and $Z^{[2]}$ by

$$Z_{st}^{[3]} = \delta Z_{st} - \sigma(Z_s) \mathbb{X}_{st}^{(1)} + \sigma_2(Z_s) \mathbb{X}_{st}^{(2)}, \quad Z_{st}^{[2]} = \delta Z_{st} - \sigma(Z_s) \mathbb{X}_{st}^{(1)}, \quad (5.3)$$

so that a solution of (5.2) is a path $Z: [0, T] \rightarrow \mathbb{R}^k$ such that

$$Z_{st}^{[3]} = o(t-s).$$

We want to show that under certain hypothesis this rough difference equation is well-posed.

5.1. TAYLOR IDENTITIES FOR $\alpha \in (1/3, 1/2]$

We now introduce two identities that will be a key tool in the following.

LEMMA 5.1. (TAYLOR IDENTITIES $\alpha \in (\frac{1}{3}, \frac{1}{2}]$) *Let us fix $z_1, z_2 \in \mathbb{R}^k$ and $x \in \mathbb{R}^d$. If $\sigma: \mathbb{R}^k \rightarrow \mathbb{R}^k \otimes (\mathbb{R}^d)^*$ is C^1 , and σ_2 is defined by (5.1) we can write, for $\delta z := z_2 - z_1$,*

$$\begin{aligned} & \sigma(z_2) - \sigma(z_1) - \sigma_2(z_1)x \\ &= \nabla \sigma(z_1)(\delta z - \sigma(z_1)x) + \int_0^1 [(\nabla \sigma(z_1 + r_1 \delta z) - \nabla \sigma(z_1)) \delta z] dr_1, \end{aligned} \quad (5.4)$$

and

$$\begin{aligned} \sigma(z_2) - \sigma(z_1) - \sigma_2(z_1)x &= \int_0^1 [(\sigma_2(z_1 + r_1 \delta z) - \sigma_2(z_1))x] dr_1 \\ &+ \int_0^1 [\nabla \sigma(z_1 + r_1 \delta z)(\delta z - \sigma(z_1)x)] dr_1 \\ &- \int_0^1 \nabla \sigma(z_1 + r_1 \delta z) \left(\int_0^{r_1} [\nabla \sigma(z_1 + r_2 \delta z) \delta z x] dr_2 \right) dr_1. \end{aligned} \quad (5.5)$$

Proof. A proof can be found in [5] and is purely based on computations and on the relation

$$\sigma(z_2) - \sigma(z_1) = \int_0^1 [\nabla \sigma(z_1 + r_1 \delta z) \delta z] dr_1.$$

□

5.2. A PRIORI ESTIMATES FOR $\alpha \in (1/3, 1/2]$

We want to prove some estimates for a solution of equation (5.2), this will be useful when later we will be showing the continuity and uniqueness of solution.

We first show a result on the regularity of a solution Z . It turns out that it has the same regularity as the driving path $X \in C^\alpha$.

PROPOSITION 5.2. *Let us take Z solution of (5.2) and $\sigma \in C^1$. Then we have $Z^{[2]} \in C^{2\alpha}$ and $Z \in C^\alpha$.*

Proof. By the definition of a solution we can write

$$\delta Z_{st} = \sum_{i=1}^n \sigma_i(Z_s) \mathbb{X}_{st}^i + o(t-s),$$

so that

$$|\delta Z_{st}| \leq \sup_{0 \leq s \leq t} \{|\sigma(Z_s)| + |\sigma_2(Z_s)|\} (|\mathbb{X}_{st}^{(1)}| + |\mathbb{X}_{st}^{(2)}|) + o(t-s).$$

We want to show that the constant $C := \sup_{0 \leq s \leq t} \{|\sigma(Z_s)| + |\sigma_2(Z_s)|\}$ is finite. Since $\sigma \in C^1$, it follows that σ_2 is continuous, therefore it suffices to show that Z is bounded.

Let us fix $\bar{\delta} > 0$ such that $|o(t-s)| \leq 1$ for all $0 \leq s \leq t \leq T$ with $|t-s| \leq \bar{\delta}$. Since we can write $[0, T]$ as a finite union of intervals $[\bar{s}, \bar{t}]$ such that $\bar{t} - \bar{s} \leq \bar{\delta}$, we will focus on only one interval of this kind. By the definition of solution we can write

$$\sup_{t \in [\bar{s}, \bar{t}]} |Z_t| \leq |Z_{\bar{s}}| + |\sigma(Z_{\bar{s}})| \sup_{t \in [\bar{s}, \bar{t}]} |\mathbb{X}_{st}^{(1)}| + |\sigma_2(Z_{\bar{s}})| \sup_{t \in [\bar{s}, \bar{t}]} |\mathbb{X}_{st}^{(2)}| + 1 < \infty.$$

We proved that Z is bounded, therefore $C < \infty$. Observe now that we can write

$$|Z_{st}^{[2]}| \leq C |\mathbb{X}_{st}^{(2)}| + o(t-s).$$

Since $\mathbb{X}^{(2)} \in C^{2\alpha}$, we conclude that $Z^{[2]} \in C^{2\alpha}$ and $Z \in C^\alpha$.

□

We will now show that $Z_{st}^{[3]} \in C^{3\alpha}$ and we give an explicit bound.

The technique that we will use consists in estimating $\|Z^{[3]}\|_{3\alpha, \tau}$ by the Weighted Sewing Bound, since for Z solution we have $Z_{st}^{[3]} = o(t-s)$ and $3\alpha > 1$. So it makes sense to look for an expression of $\delta Z_{sut}^{[3]}$. We have

$$\begin{aligned} \delta Z_{sut}^{[3]} &= Z_{st}^{[3]} - Z_{su}^{[3]} - Z_{ut}^{[3]} \\ &= \delta Z_{st} - \sigma(Z_s) \mathbb{X}_{st}^{(1)} - \sigma_2(Z_s) \mathbb{X}_{st}^{(2)} - \delta Z_{su} + \sigma(Z_s) \mathbb{X}_{su}^{(1)} + \sigma_2(Z_s) \mathbb{X}_{su}^{(2)} \\ &\quad - \delta Z_{ut} + \sigma(Z_u) \mathbb{X}_{ut}^{(1)} + \sigma_2(Z_u) \mathbb{X}_{ut}^{(2)} \\ &= -\sigma(Z_s) \mathbb{X}_{ut}^{(1)} - \sigma_2(Z_s) \mathbb{X}_{st}^{(2)} + \sigma_2(Z_s) \mathbb{X}_{su}^{(2)} + \sigma(Z_u) \mathbb{X}_{ut}^{(1)} + \sigma_2(Z_u) \mathbb{X}_{ut}^{(2)} \\ &= (\sigma(Z_u) - \sigma(Z_s)) \mathbb{X}_{ut}^{(1)} - \sigma_2(Z_s) (\mathbb{X}_{st}^{(2)} - \mathbb{X}_{su}^{(2)} - \mathbb{X}_{ut}^{(2)}) + (\sigma_2(Z_u) - \sigma_2(Z_s)) \mathbb{X}_{ut}^{(2)} \\ &= (\sigma(Z_u) - \sigma(Z_s)) \mathbb{X}_{ut}^{(1)} - \sigma_2(Z_s) \delta \mathbb{X}_{sut}^{(2)} + (\sigma_2(Z_u) - \sigma_2(Z_s)) \mathbb{X}_{ut}^{(2)} \end{aligned}$$

By Chen's relation (4.2) we have $\delta \mathbb{X}_{sut}^2 = \mathbb{X}_{su}^{(1)} \otimes \mathbb{X}_{ut}^{(1)}$, so we conclude that

$$\begin{aligned} \delta Z_{sut}^{[3]} &= (\sigma(Z_u) - \sigma(Z_s) - \sigma_2(Z_s) \mathbb{X}_{ut}^{(1)}) \mathbb{X}_{ut}^{(1)} + (\sigma_2(Z_u) - \sigma_2(Z_s)) \mathbb{X}_{ut}^{(2)} \\ &=: B_{su} \mathbb{X}_{ut}^{(1)} + (\sigma_2(Z_u) - \sigma_2(Z_s)) \mathbb{X}_{ut}^{(2)}. \end{aligned} \quad (5.6)$$

THEOREM 5.3. (A PRIORI ESTIMATE $\alpha \in (1/3, 1/2]$) *Let σ and σ_2 be globally Lipschitz. For Z solution of (5.2) we have $Z^{[3]} \in C_2^{3\alpha}$, in particular, for all $\tau > 0$,*

$$\|Z^{[3]}\|_{3\alpha, \tau} \leq K_{3\alpha} c'_{\alpha, \mathbb{X}, \sigma} (\|\delta Z\|_{\alpha, \tau} + \|Z^{[2]}\|_{2\alpha, \tau}). \quad (5.7)$$

Moreover if T or τ is small enough, we have

$$\|\delta Z\|_{\alpha, \tau} + \|Z^{[2]}\|_{2\alpha, \tau} \leq 2(\sigma(Z_0) \|\mathbb{X}^{(1)}\|_{\alpha} + \sigma_2(Z_0) \|\mathbb{X}^{(2)}\|_{2\alpha}). \quad (5.8)$$

Proof. Since $3\alpha > 1$ and $Z_{st}^{[3]} = o(t-s)$ we apply the weighted Sewing Bound (2.17) to write $\|Z^{[3]}\|_{3\alpha, \tau} \leq K_{3\alpha} \|\delta Z^{[3]}\|_{3\alpha, \tau}$. By (5.6) we can write

$$\|\delta Z^{[3]}\|_{3\alpha, \tau} \leq \|B\|_{2\alpha, \tau} \|\mathbb{X}^{(1)}\|_{\alpha} + \|\delta \sigma_2(Z)\|_{\alpha, \tau} \|\mathbb{X}^{(2)}\|_{2\alpha}. \quad (5.9)$$

For the last term we observe that

$$\|\delta \sigma_2(Z)\|_{\alpha, \tau} \leq \|\nabla \sigma_2\|_{\infty} \|Z\|_{\alpha, \tau}. \quad (5.10)$$

For the first term we can use (5.5) to rewrite B_{su} from (5.6) as

$$\begin{aligned} B_{su} &= \int_0^1 [(\sigma_2(Z_s + r_1 \delta Z_{su}) - \sigma_2(Z_s)) \mathbb{X}_{su}^{(1)}] dr_1 + \int_0^1 [\nabla \sigma(Z_s + r_1 \delta Z_{su}) Z_{su}^{[2]}] dr_1 \\ &\quad - \int_0^1 \nabla \sigma(Z_s + r_1 \delta Z_{su}) \left(\int_0^{r_1} [\nabla \sigma(Z_{su} + r_2 \delta Z_{su}) \delta Z_{su} \mathbb{X}_{su}^{(1)}] dr_2 \right) dr_1. \end{aligned}$$

Now using (2.14) we get

$$\|B\|_{2\alpha, \tau} \leq (\|\nabla \sigma_2\|_{\infty} + \|\nabla \sigma\|_{\infty}^2) \|\mathbb{X}^{(1)}\|_{\alpha} \|\delta Z\|_{\alpha, \tau} + \|\nabla \sigma\|_{\infty} \|Z^{[2]}\|_{2\alpha, \tau}. \quad (5.11)$$

Plugging (5.10), (5.11) in (5.9) we get (5.7).

Now we wish to bound $\|\delta Z\|_{\alpha, \tau}$ and $\|Z^{[2]}\|_{2\alpha, \tau}$. We define $\varepsilon := (\tau \wedge T)^{\alpha}$ to ease the notation. Observe that we can write $Z_{st}^{[2]} = \sigma_2(Z_s) \mathbb{X}_{st}^{(2)} + Z_{st}^{[3]}$ and $\delta Z_{st} = \sigma(Z_s) \mathbb{X}_{st}^{(1)} + Z_{st}^{[2]}$, so that we get by (2.13) and (2.10)

$$\|Z^{[2]}\|_{2\alpha, \tau} \leq \|\sigma_2(Z)\|_{\infty, \tau} \|\mathbb{X}^{(2)}\|_{2\alpha} + \varepsilon \|Z^{[3]}\|_{3\alpha, \tau}.$$

Now by (2.11) and (5.7) we can write

$$\begin{aligned} \|Z^{[2]}\|_{2\alpha, \tau} &\leq \sigma_2(Z_0) \|\mathbb{X}^{(2)}\|_{2\alpha} + 3\varepsilon \|\nabla \sigma\|_{\infty} \|\delta Z\|_{\alpha, \tau} \\ &\quad + K_{3\alpha} c'_{\alpha, \mathbb{X}, \sigma} (\|\delta Z\|_{\alpha, \tau} + \|Z^{[2]}\|_{2\alpha, \tau}). \end{aligned} \quad (5.12)$$

Similarly

$$\begin{aligned} \|\delta Z\|_{\alpha, \tau} &\leq \|\sigma(Z)\|_{\infty, \tau} \|\mathbb{X}^{(1)}\|_{\alpha} + \varepsilon \|Z^{[2]}\|_{2\alpha, \tau} \\ &\leq \sigma(Z_0) \|\mathbb{X}^{(1)}\|_{\alpha} + 3\varepsilon \|\nabla \sigma\|_{\infty} \|\delta Z\|_{\alpha, \tau} + \varepsilon \|Z^{[2]}\|_{2\alpha, \tau}. \end{aligned} \quad (5.13)$$

For ε small enough, summing (5.12) and (5.13) we obtain

$$\|\delta Z\|_{\alpha, \tau} + \|Z^{[2]}\|_{2\alpha, \tau} \leq \sigma(Z_0) \|\mathbb{X}^{(1)}\|_{\alpha} + \sigma_2(Z_0) \|\mathbb{X}^{(2)}\|_{2\alpha} + \frac{1}{2} (\|\delta Z\|_{\alpha, \tau} + \|Z^{[2]}\|_{2\alpha, \tau}).$$

Since by Proposition (5.2) we know $\|\delta Z\|_{\alpha, \tau} + \|Z^{[2]}\|_{2\alpha, \tau} < \infty$, we proved (5.8).

□

5.3. UNIQUENESS FOR $\alpha \in (1/3, 1/2]$

We will prove uniqueness of solutions of (5.2).

THEOREM 5.4. (UNIQUENESS FOR $\alpha \in (1/3, 1/2]$) *Let us consider the setting of equation (5.2). Let us assume that σ is C^2 with locally Lipschitz Hessian $\nabla^2\sigma$. Then for every $z_0 \in \mathbb{R}^k$ there exists at most one solution Z of (5.2) such that $Z_0 = z_0$.*

Proof. *Let us take two solutions $Z, \bar{Z}$ of (5.2) and define $Y := Z - \bar{Z}$, $Y_{st}^{[2]} := Z_{st}^{[2]} - \bar{Z}_{st}^{[2]}$, $Y_{st}^{[3]} := Z_{st}^{[3]} - \bar{Z}_{st}^{[3]}$.*

We will show that for $\tau > 0$ small, we have $\|Y\|_{\infty, \tau} \leq 2|Y_0|$, which will imply $\|Y\|_{\infty, \tau} = 0$ since $Y_0 = Z_0 - \bar{Z}_0 = z_0 - z_0 = 0$. Therefore we conclude $Z = \bar{Z}$.

From (2.11) we know that

$$\|Y\|_{\infty, \tau} \leq |Y_0| + 3\tau^\alpha \|\delta Y\|_{\alpha, \tau}, \quad (5.14)$$

so that it will be enough to show that $\|\delta Y\|_{\alpha, \tau} \leq c_1 \|Y\|_{\infty, \tau}$. Indeed, by taking τ small we could argue by (5.14) that $\|Y\|_{\infty, \tau} \leq |Y_0| + \frac{1}{2} \|\delta Y\|_{\alpha, \tau}$, which concludes the proof. We will instead show that for $\tau > 0$ small enough we have

$$\|\delta Y\|_{\alpha, \tau} \leq c_2 \|Y\|_{\infty, \tau} + \tau^\alpha \|Y^{[2]}\|_{2\alpha, \tau} \quad (5.15)$$

$$\|Y^{[2]}\|_{2\alpha, \tau} \leq c_3 \|Y\|_{\infty, \tau} + \tau^\alpha \|Y^{[3]}\|_{3\alpha, \tau} \quad (5.16)$$

$$\|Y^{[3]}\|_{3\alpha, \tau} \leq c_4 \|Y\|_{\infty, \tau} + c'_4 \tau^\alpha \|Y^{[3]}\|_{3\alpha, \tau} \quad (5.17)$$

for c_i, c'_i depending on $Z, \bar{Z}, \mathbb{X}, \sigma$, but not on τ .

Then by substituting repeatedly from bottom to top we will get the desired result.

We need to prove (5.15)-(5.17). Let us first observe, using the notation of (2.8), that by hypothesis

$$C_{\psi, R} := \sup \{ |\psi(x)| : x \in \mathbb{R}^k, |x| \leq \max(\|Z\|_\infty, \|\bar{Z}\|_\infty) \} < \infty$$

for $\psi \in \{\nabla\sigma, \nabla^2\sigma, \nabla^3\sigma, \nabla\sigma_2, \nabla^2\sigma_2\}$, so that we can bound these quantities by

$$C := \max \{ C_{\psi, R} : \psi \in \{\nabla\sigma, \nabla^2\sigma, \nabla^3\sigma, \nabla\sigma_2, \nabla^2\sigma_2\} \}.$$

We can prove (5.15) and (5.16) similarly. We observe that

$$\begin{aligned} \delta Y_{st} &= (\sigma(Z_s) - \sigma(\bar{Z}_s)) \mathbb{X}_{st}^{(1)} + Y_{st}^{[2]}, \\ Y_{st}^{[2]} &= (\sigma_2(Z_s) - \sigma_2(\bar{Z}_s)) \mathbb{X}_{st}^{(2)} + Y_{st}^{[3]}, \end{aligned}$$

so that by (2.10) and (2.13)

$$\begin{aligned} \|\delta Y\|_{\alpha, \tau} &\leq \|\sigma(Z_s) - \sigma(\bar{Z}_s)\|_{\infty, \tau} \|\mathbb{X}^{(1)}\|_\alpha + \tau^\alpha \|Y^{[2]}\|_{2\alpha, \tau} \\ &\leq C \|Y\|_{\infty, \tau} \|\mathbb{X}^{(1)}\|_\alpha + \tau^\alpha \|Y^{[2]}\|_{2\alpha, \tau} \\ &\leq c_2 \|Y\|_{\infty, \tau} + \tau^\alpha \|Y^{[2]}\|_{2\alpha, \tau}, \end{aligned}$$

$$\begin{aligned} \|Y^{[2]}\|_{2\alpha, \tau} &\leq \|\sigma_2(Z_s) - \sigma_2(\bar{Z}_s)\|_{\infty, \tau} \|\mathbb{X}^{(2)}\|_{2\alpha} + \tau^\alpha \|Y^{[3]}\|_{3\alpha, \tau} \\ &\leq C \|Y\|_{\infty, \tau} \|\mathbb{X}^{(2)}\|_{2\alpha} + \tau^\alpha \|Y^{[3]}\|_{3\alpha, \tau} \\ &\leq c_3 \|Y\|_{\infty, \tau} + \tau^\alpha \|Y^{[3]}\|_{3\alpha, \tau}. \end{aligned}$$

We have proved (5.15) and (5.16). For (5.17) we can use the weighted Sewing Bound to write

$$\|Y^{[3]}\|_{3\alpha,\tau} \leq K_{3\alpha} \|\delta Y^{[3]}\|_{3\alpha,\tau}.$$

We will now prove

$$\|\delta Y^{[3]}\|_{3\alpha,\tau} \leq \tilde{c}_4 \|Y\|_{\infty,\tau} + \tilde{c}_4' \|\delta Y\|_{\alpha,\tau} + \tilde{c}_4'' \|Y^{[2]}\|_{2\alpha,\tau}. \quad (5.18)$$

Using (5.15) and (5.16) it is easy to obtain (5.17). Recalling (5.6) we can write

$$\delta Y_{sut}^{[3]} = (B_{su} - \bar{B}_{su}) \mathbb{X}_{ut}^{(1)} + (\delta \sigma_2(Z) - \delta \sigma_2(\bar{Z}))_{su} \mathbb{X}_{ut}^{(2)},$$

hence by (2.15)

$$\|\delta Y^{[3]}\|_{3\alpha,\tau} \leq \|B - \bar{B}\|_{2\alpha,\tau} \|\mathbb{X}^{(1)}\|_{\alpha} + \|\delta \sigma_2(Z) - \delta \sigma_2(\bar{Z})\|_{\alpha,\tau} \|\mathbb{X}^{(2)}\|_{2\alpha}. \quad (5.19)$$

We need to bound the RHS by linear combinations of $\|Y\|_{\infty,\tau}$, $\|\delta Y\|_{\alpha,\tau}$, $\|Y^{[2]}\|_{2\alpha,\tau}$.

Applying (2.9) we can bound

$$|\delta \sigma_2(Z)_{su} - \delta \sigma_2(\bar{Z})_{su}| \leq C |\delta Y_{su}| + C \{|Y_s| + |Y_u|\} |\delta Z_{su}|,$$

so that

$$\|\delta \sigma_2(Z) - \delta \sigma_2(\bar{Z})\|_{\alpha,\tau} \leq C \|\delta Y\|_{\alpha,\tau} + C \{2\|Y\|_{\infty,\tau}\} \|\delta Z\|_{\alpha}.$$

For $\|B - \bar{B}\|_{2\alpha,\tau}$ we observe that by (5.4)

$$B_{su} = \nabla \sigma(Z_s) Z_{su}^{[2]} + \int_0^1 \underbrace{(\nabla \sigma(Z_u + r_1 \delta Z_{su}) - \nabla \sigma(Z_u))}_{I_{su}} \delta Z_{su} dr_1$$

and similarly for $\bar{B}_{su}$, so that

$$|B_{su} - \bar{B}_{su}| \leq |\nabla \sigma(Z_s) Z_{su}^{[2]} - \nabla \sigma(\bar{Z}_s) \bar{Z}_{su}^{[2]}| + \int_0^1 |I_{su} \delta Z_{su} - \bar{I}_{su} \delta \bar{Z}_{su}| dr_1.$$

From now on we will use repeatedly the estimate $|ab - a'b'| \leq |a - a'| |b| + |a'| |b - b'|$.

We get

$$\begin{aligned} |\nabla \sigma(Z_s) Z_{su}^{[2]} - \nabla \sigma(\bar{Z}_s) \bar{Z}_{su}^{[2]}| &\leq |\nabla \sigma(Z_s) - \nabla \sigma(\bar{Z}_s)| |\bar{Z}_{su}^{[2]}| + |\nabla \sigma(\bar{Z}_s)| |Z_{su}^{[2]} - \bar{Z}_{su}^{[2]}| \\ &\leq C |Y_s| |Z_{su}^{[2]}| + C |Y_{su}^{[2]}|, \end{aligned}$$

so that

$$\|\nabla \sigma(Z) Z^{[2]} - \nabla \sigma(\bar{Z}) \bar{Z}^{[2]}\|_{2\alpha,\tau} \leq C \|Y\|_{\infty,\tau} \|Z^{[2]}\|_{2\alpha,\tau} + C \|Y^{[2]}\|_{2\alpha,\tau}.$$

We can also write, noting that $|\bar{I}_{su}| \leq C r_1 |\delta \bar{Z}_{su}|$,

$$\begin{aligned} |I_{su} \delta Z_{su} - \bar{I}_{su} \delta \bar{Z}_{su}| &\leq |I_{su} - \bar{I}_{su}| |\delta Z_{su}| + |\bar{I}_{su}| |\delta Z_{su} - \delta \bar{Z}_{su}| \\ &\leq |I_{su} - \bar{I}_{su}| |\delta Z_{su}| + C r_1 |\delta \bar{Z}_{su}| |\delta Y_{su}|. \end{aligned} \quad (5.20)$$

Now by (2.9) we estimate, for $0 \leq r_1 \leq 1$,

$$\begin{aligned} |I_{su} - \bar{I}_{su}| &= |(\nabla \sigma(Z_u + r_1 \delta Z_{su}) - \nabla \sigma(Z_u)) - (\nabla \sigma(Z_u + r_1 \delta \bar{Z}_{su}) - \nabla \sigma(\bar{Z}_u))| \\ &\leq C r |\delta Y_{su}| + C \{|Y_u + r_1 \delta Y_{ut}| + |Y_u|\} r_1 |\delta Z_{ut}|. \end{aligned}$$

Plugging the last bound into (5.20) and using (2.13) and (2.14), we get

$$\|I\delta Z - \bar{I}\delta\bar{Z}\|_{2\alpha,\tau} \leq (C\|\delta Y\|_{\alpha,\tau} + 2C\|Y\|_{\infty,\tau}\|\delta Z\|_{\alpha,\tau})\|\delta Z\|_{\alpha} + C\|\delta Z\|_{\alpha}\|\delta Y\|_{\alpha,\tau}.$$

We proved that $\|B - \bar{B}\|_{2\alpha,\tau}$ and $\|\delta\sigma_2(Z) - \delta\sigma_2(\bar{Z})\|_{\alpha,\tau}$ so that by (5.19) we can obtain (5.18), which completes the proof. $\square$

5.4. CONTINUITY OF SOLUTION FOR $\alpha \in (1/3, 1/2]$

We denote by $\mathcal{R}_{\alpha,d}$ the space of d -dimensional rough paths $\mathbb{X} = (\mathbb{X}^{(1)}, \mathbb{X}^{(2)})$ endowed with the norm $\|\mathbb{X}^{(1)}\|_{\alpha} + \|\mathbb{X}^{(2)}\|_{2\alpha}$. We consider the solution map

$$\begin{aligned} \Phi: \mathbb{R}^d \times \mathcal{R}_{\alpha,d} &\longrightarrow \mathcal{C}^{\alpha} \\ (Z_0, \mathbb{X}) &\longmapsto Z \end{aligned} \quad (5.21)$$

where Z is the unique solution of (5.2) for $t \in [0, T]$ starting from Z_0 .

THEOREM 5.5. (CONTINUITY OF SOLUTION MAP FOR $\alpha \in (1/3, 1/2]$) *Let σ, σ_2 be such that*

$$\|\nabla\sigma\|_{\infty}, \|\nabla^2\sigma\|_{\infty}, \|\nabla^3\sigma\|_{\infty} < \infty, \quad \|\nabla\sigma_2\|_{\infty}, \|\nabla^2\sigma_2\|_{\infty} < \infty.$$

Then, for any $T > 0$ and $\alpha \in (\frac{1}{2}, \frac{1}{3}]$, the solution map (5.21) is locally Lipschitz.

More explicitly if we consider the starting points $Z_0, \bar{Z}_0 \in \mathbb{R}^d$ and rough paths $\mathbb{X}, \bar{\mathbb{X}} \in \mathcal{R}_{\alpha,d}$, and we assume

$$\max \{ \|\nabla\sigma\|_{\infty}, \|\nabla^2\sigma\|_{\infty}, \|\nabla^3\sigma\|_{\infty}, \|\nabla\sigma_2\|_{\infty}, \|\nabla^2\sigma_2\|_{\infty} \} \leq D, \quad (5.22)$$

$$\max \{ |\sigma(Z_0)|, |\sigma_2(Z_0)|, |\sigma(\bar{Z}_0)|, |\sigma_2(\bar{Z}_0)| \} \leq D_0, \quad (5.23)$$

$$\max \{ \|\mathbb{X}^{(1)}\|_{\alpha}, \|\mathbb{X}^{(2)}\|_{2\alpha}, \|\bar{\mathbb{X}}^{(1)}\|_{\alpha}, \|\bar{\mathbb{X}}^{(2)}\|_{2\alpha} \} \leq M, \quad (5.24)$$

then, for a suitable $\tau > 0$ small enough, the corresponding solutions $Z, \bar{Z}$ of (5.2) satisfy

$$\begin{aligned} \|Y\|_{\infty,\tau} + \|\delta Y\|_{\alpha,\tau} + \|Y^{[2]}\|_{2\alpha,\tau} \\ \leq C_M |Y_0| + CD_0 (\|\mathbb{X}^{(1)} - \bar{\mathbb{X}}^{(1)}\|_{\alpha} + \|\mathbb{X}^{(2)} - \bar{\mathbb{X}}^{(2)}\|_{2\alpha}), \end{aligned} \quad (5.25)$$

where we defined $Y := Z - \bar{Z}$, $Y^{[2]} := Z^{[2]} - \bar{Z}^{[2]}$ and C is a constant depending on M .

Proof. *Let us fix some notation and bounds we will use throughout the proof. We use the notations*

$$\begin{aligned} \varepsilon &:= (\tau \wedge T)^{\alpha}, \\ c_M &:= (\|\nabla\sigma\|_{\infty} + \|\nabla\sigma_2\|_{\infty})M \leq 2DM. \end{aligned}$$

By the a priori estimate (5.8) for $\varepsilon = (\tau \wedge T)^{\alpha}$ small enough we can write

$$\|\delta Z\|_{\alpha,\tau} + \|Z^{[2]}\|_{2\alpha,\tau} \leq 4D_0M, \quad (5.26)$$

so that we have

$$\max \{ \|\delta\sigma(Z)\|_{\alpha,\tau}, \|\delta\sigma_2(Z)\|_{\alpha,\tau} \} \leq \max \{ \|\nabla\sigma\|_{\infty}, \|\nabla\sigma_2\|_{\infty} \} \|\delta Z\|_{\alpha,\tau} \leq 4D_0c_M \quad (5.27)$$

and by (2.12)

$$\max \{ \|\sigma(Z)\|_{\infty, \tau}, \|\sigma_2(Z)\|_{\infty, \tau} \} \leq D_0 + 3\varepsilon 4D_0 c_M \leq 2D_0. \quad (5.28)$$

We will also use multiple times the estimate

$$|ab - a'b'| \leq |a - a'| |b| + |a'| |b - b'|. \quad (5.29)$$

We want to show that for ε small enough

$$\|\delta Y\|_{\alpha, \tau} \leq c_M \|Y\|_{\infty, \tau} + 2D_0 \|\mathbb{X}^{(1)} - \bar{\mathbb{X}}^{(1)}\|_{\alpha} + \varepsilon \|Y^{[2]}\|_{2\alpha, \tau}, \quad (5.30)$$

$$\|Y^{[2]}\|_{2\alpha, \tau} \leq c_M \|Y\|_{\infty, \tau} + 2D_0 \|\mathbb{X}^{(2)} - \bar{\mathbb{X}}^{(2)}\|_{2\alpha} + \varepsilon \|Y^{[3]}\|_{3\alpha, \tau}, \quad (5.31)$$

and also that

$$\varepsilon \|Y^{[3]}\|_{3\alpha, \tau} \leq \|Y\|_{\infty, \tau} + D_0 (\|\mathbb{X}^{(1)} - \bar{\mathbb{X}}^{(1)}\|_{\alpha} + \|\mathbb{X}^{(2)} - \bar{\mathbb{X}}^{(2)}\|_{2\alpha}) + \|\delta Y\|_{\alpha, \tau} + \frac{1}{4} \|Y^{[2]}\|_{2\alpha, \tau} \quad (5.32)$$

Now we can deduce (5.25). Plugging (5.32) into (5.31) we get $\|Y^{[2]}\|_{2\alpha, \tau} \leq (\dots) + \frac{1}{4} \|Y^{[2]}\|_{2\alpha, \tau}$, which implies $\|Y^{[2]}\|_{2\alpha, \tau} \leq \frac{4}{3} (\dots)$. Now again plugging this expression into (5.30), we obtain $\|\delta Y\|_{\alpha, \tau} \leq C'_M \|Y\|_{\infty, \tau} + C'D_0 (\|\mathbb{X}^{(1)} - \bar{\mathbb{X}}^{(1)}\|_{\alpha} + \|\mathbb{X}^{(2)} - \bar{\mathbb{X}}^{(2)}\|_{2\alpha})$ for some constants C'_M and C' . Now looking back at $\|Y^{[2]}\|_{2\alpha, \tau}$ we get, for some constants C''_M and C'' ,

$$\|Y^{[2]}\|_{2\alpha, \tau} \leq C''_M \|Y\|_{\infty, \tau} + C'' D_0 (\|\mathbb{X}^{(1)} - \bar{\mathbb{X}}^{(1)}\|_{\alpha} + \|\mathbb{X}^{(2)} - \bar{\mathbb{X}}^{(2)}\|_{2\alpha}),$$

so that putting all together

$$\begin{aligned} \|Y\|_{\infty, \tau} + \|\delta Y\|_{\alpha, \tau} + \|Y^{[2]}\|_{2\alpha, \tau} &\leq C'''_M \|Y\|_{\infty, \tau} \\ &\quad + C''' D_0 (\|\mathbb{X}^{(1)} - \bar{\mathbb{X}}^{(1)}\|_{\alpha} + \|\mathbb{X}^{(2)} - \bar{\mathbb{X}}^{(2)}\|_{2\alpha}), \end{aligned}$$

for constants C'''_M and C''' .

Finally, since $\|Y\|_{\infty, \tau} \leq |Y_0| + 3\varepsilon \|\delta Y\|_{\alpha, \tau}$, we conclude that for ε small

$$\|Y\|_{\infty, \tau} + \|\delta Y\|_{\alpha, \tau} + \|Y^{[2]}\|_{2\alpha, \tau} \leq C_M |Y_0| + C D_0 (\|\mathbb{X}^{(1)} - \bar{\mathbb{X}}^{(1)}\|_{\alpha} + \|\mathbb{X}^{(2)} - \bar{\mathbb{X}}^{(2)}\|_{2\alpha}).$$

It remains to prove (5.30)-(5.32).

Observe that $\delta Y_{st} = \delta Z_{st} - \delta \bar{Z}_{st} = \sigma(Z_s) \mathbb{X}_{st}^{(1)} - \sigma(\bar{Z}_s) \bar{\mathbb{X}}_{st}^{(1)} + Y_{st}^{[2]}$, so that by (2.13), (5.23), (5.24) and (5.29) we have

$$\begin{aligned} \|\delta Y\|_{\alpha, \tau} &\leq \|\sigma(Z)\|_{\infty, \tau} \|\mathbb{X}^{(1)} - \bar{\mathbb{X}}^{(1)}\|_{\alpha} + \|\sigma(Z) - \sigma(\bar{Z})\|_{\infty, \tau} \|\bar{\mathbb{X}}^{(1)}\|_{\alpha} + \|Y^{[2]}\|_{\alpha, \tau} \\ &\leq 2D_0 \|\mathbb{X}^{(1)} - \bar{\mathbb{X}}^{(1)}\|_{\alpha} + \|\sigma(Z) - \sigma(\bar{Z})\|_{\infty, \tau} M + \varepsilon \|Y^{[2]}\|_{2\alpha, \tau} \\ &\leq 2D_0 \|\mathbb{X}^{(1)} - \bar{\mathbb{X}}^{(1)}\|_{\alpha} + c_M \|Y\|_{\infty, \tau} + \varepsilon \|Y^{[2]}\|_{2\alpha, \tau}, \end{aligned}$$

where in the second line we also used (5.28) and (2.10). We proved (5.30).

Note that we also have $Y_{st}^{[2]} = \sigma_2(Z_s) \mathbb{X}_{st}^{(2)} - \sigma_2(\bar{Z}_s) \bar{\mathbb{X}}_{st}^{(2)}$, so that similarly

$$\begin{aligned} \|Y^{[2]}\|_{2\alpha, \tau} &\leq \|\sigma_2(Z)\|_{\infty, \tau} \|\mathbb{X}^{(2)} - \bar{\mathbb{X}}^{(2)}\|_{2\alpha} + \|\sigma_2(Z) - \sigma_2(\bar{Z})\|_{\infty, \tau} \|\bar{\mathbb{X}}^{(2)}\|_{2\alpha} + \|Y^{[3]}\|_{2\alpha, \tau} \\ &\leq 2D_0 \|\mathbb{X}^{(2)} - \bar{\mathbb{X}}^{(2)}\|_{2\alpha} + c_M \|Y\|_{\infty, \tau} + \varepsilon \|Y^{[3]}\|_{3\alpha, \tau}, \end{aligned}$$

which is exactly (5.31).

Finally we focus on (5.32). Since $Y_{st}^{[3]} = o(t-s)$ by the definition of solution, by the weighted Sewing Bound (2.17) we can write

$$\|Y^{[3]}\|_{3\alpha, \tau} \leq K_{3\alpha} \|\delta Y^{[3]}\|_{3\alpha, \tau}.$$

By (5.6) we have $\delta Z_{sut}^{[3]} = B_{su} \mathbb{X}_{ut}^{(1)} + (\sigma_2(Z_u) - \sigma_2(Z_s)) \mathbb{X}_{ut}^{(2)}$, and analogously for $\delta \bar{Z}^{[3]}$, so that by (5.29) and (2.15) we can write

$$\begin{aligned} \|\delta Y^{[3]}\|_{3\alpha, \tau} &\leq \|B\|_{2\alpha, \tau} \|\mathbb{X}^{(1)} - \bar{\mathbb{X}}^{(1)}\|_{\alpha} + \|B - \bar{B}\|_{2\alpha, \tau} \|\bar{\mathbb{X}}^{(1)}\|_{\alpha, \tau} \\ &\quad + \|\delta \sigma_2(Z)\|_{\alpha, \tau} \|\mathbb{X}^{(2)} - \bar{\mathbb{X}}^{(2)}\|_{2\alpha} + \|\delta \sigma_2(Z) - \delta \sigma_2(\bar{Z})\|_{\alpha, \tau} \|\bar{\mathbb{X}}^{(2)}\|_{2\alpha}. \end{aligned} \quad (5.33)$$

We will estimate the for terms in the RHS.

By (5.27), for ε small enough,

$$\begin{aligned} \varepsilon K_{3\alpha} \|\delta \sigma_2(Z)\|_{\alpha, \tau} \|\mathbb{X}^{(2)} - \bar{\mathbb{X}}^{(2)}\|_{2\alpha} &\leq \varepsilon K_{3\alpha} 4D_0 c_M \|\mathbb{X}^{(1)} - \bar{\mathbb{X}}^{(1)}\|_{\alpha} \\ &\leq D_0 \|\mathbb{X}^{(1)} - \bar{\mathbb{X}}^{(1)}\|_{\alpha}. \end{aligned} \quad (5.34)$$

By (2.8) we have

$$|\delta \sigma_2(Z)_{st} - \delta \sigma_2(\bar{Z})_{st}| \leq \|\nabla \sigma_2\|_{\infty} |\delta Y_{st}| + \|\nabla^2 \sigma_2\|_{\infty} \{|Y_s| + |Y_t|\} |\delta Z_{st}|,$$

so that by (2.16) and $\bar{\tau}$ small enough to apply the a priori estimate we get

$$\begin{aligned} \|\delta \sigma_2(Z) - \delta \sigma_2(\bar{Z})\|_{\alpha, \tau} &\leq \|\nabla \sigma_2\|_{\infty} \|\delta Y\|_{\alpha, \tau} + \|\nabla^2 \sigma_2\|_{\infty} 2\|Y\|_{\infty, \tau} e^{\frac{T}{\bar{\tau}}} \|\delta Z\|_{\alpha, \bar{\tau}} \\ &\leq D \|\delta Y\|_{\alpha, \tau} + 2D \|Y\|_{\infty, \tau} e^{\frac{T}{\bar{\tau}}} \|\delta Z\|_{\alpha, \bar{\tau}}. \end{aligned}$$

By (5.24) and (5.26) we have, for ε small enough,

$$\begin{aligned} \varepsilon K_{3\alpha} \|\delta \sigma_2(Z) - \delta \sigma_2(\bar{Z})\|_{\alpha, \tau} \|\bar{\mathbb{X}}^{(2)}\|_{2\alpha} &\leq \varepsilon K_{3\alpha} \left(D M \|\delta Y\|_{\alpha, \tau} + 8D D_0 M^2 e^{\frac{T}{\bar{\tau}}} \|Y\|_{\infty, \tau} \right) \\ &\leq \frac{1}{2} (\|Y\|_{\infty, \tau} + \|\delta Y\|_{\alpha, \tau}). \end{aligned} \quad (5.35)$$

Now by (5.4) we have

$$B_{st} = \underbrace{\nabla \sigma(Z_s) Z_{st}^{[2]}}_{E_{st}} + \underbrace{\int_0^1 [(\nabla \sigma(Z_s + r_1 \delta Z_{st}) - \nabla \sigma(Z_s)) \delta Z_{st}] dr_1}_{F_{st}}$$

and similarly for $\bar{B}$. Therefore by (5.26), for ε small enough, we get

$$\begin{aligned} \varepsilon K_{3\alpha} \|B\|_{2\alpha, \tau} \|\mathbb{X}^{(1)} - \bar{\mathbb{X}}^{(1)}\|_{\alpha} &\leq \varepsilon K_{3\alpha} (\|\nabla \sigma\|_{\infty} \|Z^{[2]}\|_{2\alpha, \tau} + \|\nabla^2 \sigma\|_{\infty} \|\delta Z\|_{\alpha, \tau}^2) \|\mathbb{X}^{(1)} - \bar{\mathbb{X}}^{(1)}\|_{\alpha} \\ &\leq \varepsilon K_{3\alpha} (D 4D_0 M + D(4D_0 M)^2) \|\mathbb{X}^{(1)} - \bar{\mathbb{X}}^{(1)}\|_{\alpha} \\ &\leq D_0 \|\mathbb{X}^{(1)} - \bar{\mathbb{X}}^{(1)}\|_{\alpha}. \end{aligned} \quad (5.36)$$

Finally we observe that

$$E_{st} - \bar{E}_{st} = (\nabla \sigma(Z_s) - \nabla \sigma(\bar{Z}_s)) Z_{st}^{[2]} + \nabla \sigma(\bar{Z}_s) (Z_{st}^{[2]} - \bar{Z}_{st}^{[2]}),$$

so that by (2.16) and $\bar{\tau}$ small enough to apply the a priori estimate (5.26) we get

$$\begin{aligned} \|E - \bar{E}\|_{2\alpha, \tau} &\leq \|\nabla \sigma(Z) - \nabla \sigma(\bar{Z})\|_{\infty, \tau} e^{\frac{T}{\bar{\tau}}} \|Z^{[2]}\|_{2\alpha, \bar{\tau}} + \|\nabla \sigma\|_{\infty} \|Y^{[2]}\|_{2\alpha, \tau} \\ &\leq D \|Y\|_{\infty, \tau} e^{\frac{T}{\bar{\tau}}} 4D_0 M + D \|Y^{[2]}\|_{2\alpha, \tau}. \end{aligned} \quad (5.37)$$

Then we consider $F_{st} - \bar{F}_{st}$. By (2.4), for $0 \leq r_1 \leq 1$ we can estimate

$$\begin{aligned} &|(\nabla \sigma(Z_s + r_1 \delta Z_{st}) - \nabla \sigma(Z_s)) - (\nabla \sigma(\bar{Z}_s + r_1 \delta \bar{Z}_{st}) - \nabla \sigma(\bar{Z}_s))| |\delta Z_{st}| \\ &\leq \|\nabla^2 \sigma\|_{\infty} |\delta Y_{st}| |\delta Z_{st}| + \|\nabla^3 \sigma\|_{\infty} \max_{0 \leq u \leq 1} \{(1-u)|Y_s| + u|Y_t|\} |\delta Z_{st}|^2, \end{aligned}$$

and also

$$|\nabla \sigma(Z_s + r_1 \delta Z_{st}) - \nabla \sigma(Z_s)| |\delta Z_{st} - \delta \bar{Z}_{st}| \leq \|\nabla^2 \sigma\|_{\infty} |\delta Z_{st}| |\delta Y_{st}|,$$

so that applying (2.16) twice for $\bar{\tau}$ small enough to apply the a priori estimate (5.26) we get

$$\begin{aligned} \|F - \bar{F}\|_{2\alpha, \tau} &\leq 2\|\nabla^2 \sigma\|_{\infty} \|\delta Y\|_{\alpha, \tau} e^{\frac{T}{\bar{\tau}}} \|\delta Z\|_{\alpha, \bar{\tau}} + \|\nabla^3 \sigma\|_{\infty} \|Y\|_{\infty, \tau} e^{\frac{T}{\bar{\tau}}} \|\delta Z\|_{\alpha, \bar{\tau}}^2 \\ &\leq e^{\frac{T}{\bar{\tau}}} \{8D_0 M D \|\delta Y\|_{\alpha, \tau} + (4D_0 M)^2 D \|Y\|_{\infty, \tau}\}. \end{aligned} \quad (5.38)$$

Putting together (5.37) and (5.38) we get, for ε small enough,

$$\begin{aligned} \varepsilon K_{3\alpha} \|B - \bar{B}\|_{2\alpha, \tau} \|\bar{\mathbb{X}}^{(1)}\|_{\alpha, \tau} &\leq \|E - \bar{E}\|_{2\alpha, \tau} + \|F - \bar{F}\|_{2\alpha, \tau} \\ &\leq \frac{1}{2} (\|Y\|_{\infty, \tau} + \|\delta Y\|_{\alpha, \tau}) + \frac{1}{4} \|Y^{[2]}\|_{2\alpha, \tau}. \end{aligned} \quad (5.39)$$

In view of (5.33), summing the estimates (5.34), (5.35), (5.36) and (5.39) we finally obtain (5.32). The proof is complete. $\square$

5.5. EXISTENCE FOR $\alpha \in (1/3, 1/2]$

We will first prove local existence. We assume that σ, σ_2 are C^1 and globally Lipschitz. Let us utilize a discretization scheme. We set $t_i := \frac{i}{n}$, $i \geq 0$, and for given $y_0 \in \mathbb{R}^k$ we set

$$y_{t_{i+1}} = y_{t_i} + \sigma(y_{t_i}) \mathbb{X}_{t_i t_{i+1}}^{(1)} + \sigma_2(y_{t_i}) \mathbb{X}_{t_i t_{i+1}}^{(2)}, \quad i \geq 0. \quad (5.40)$$

We also set, for $\mathbb{T} := \{t_i : t_i \leq T\}$:

$$\begin{aligned} D &:= \max \{1, \|\nabla \sigma\|_{\infty}, \|\nabla \sigma_2\|_{\infty}\} \\ M &\geq \max \{\|\mathbb{X}^{(1)}\|_{\alpha} + \|\mathbb{X}^{(2)}\|_{2\alpha}\} \\ \delta y_{t_i t_j} &:= y_{t_j} - y_{t_i} \\ \|\delta y\|_{\alpha}^{\mathbb{T}} &:= \sup_{0 < j < i < nT} \frac{|y_{t_j} - y_{t_i}|}{|t_j - t_i|^{\alpha}} \\ A_{t_i t_j} &:= \sigma(y_{t_i}) \mathbb{X}_{t_i t_j}^{(1)} + \sigma_2(y_{t_i}) \mathbb{X}_{t_i t_j}^{(2)} \\ H_{t_i t_j} &:= \delta y_{t_i t_j} - \sigma(y_{t_i}) \mathbb{X}_{t_i t_j}^{(1)}. \end{aligned}$$

We have the following estimate:

LEMMA 5.6. *It exists $T_{M,D,\alpha} > 0$ such that, for all $T \in (0, T_{M,D,\alpha})$ and $\mathbb{X} = (\mathbb{X}^{(1)}, \mathbb{X}^{(2)}) \in \mathcal{R}_{\alpha,d}$ such that $M \geq \max \{ \|\mathbb{X}^{(1)}\|_\alpha + \|\mathbb{X}^{(2)}\|_{2\alpha} \}$, we have*

$$\|\delta y\|_\alpha^\mathbb{T} \leq 5M(|\sigma(y_0)| + |\sigma_2(y_0)|) \quad (5.41)$$

$$\|\delta y - A\|_{3\alpha}^\mathbb{T} \lesssim_{M,D,\alpha} (|\sigma(y_0)| + |\sigma_2(y_0)|) \quad (5.42)$$

Proof. Set $R_{t_it_j} = \delta y_{t_it_j} - A_{t_it_j}$. By definition $R_{t_it_{i+1}} = 0$, so that we can use the Discrete Sewing Bound on $\mathbb{T}$ so that:

$$\|R\|_{3\alpha}^\mathbb{T} \leq K_{3\alpha} \|\delta R\|_{3\alpha}^\mathbb{T}, \quad K_{3\alpha} = 2^{3\alpha} \sum_{n \geq 1} \frac{1}{n^{3\alpha}}.$$

Since $\delta R = -\delta A$, by (5.6) we can write

$$\delta R_{t_it_j t_k} = \underbrace{-(\sigma(y_{t_j}) - \sigma(y_{t_i}) - \sigma_2(y_{t_j}) \mathbb{X}_{t_it_j}^{(1)}) \mathbb{X}_{t_j t_k}^{(1)}}_{:= B_{ij}^{(2)}} - \underbrace{(\sigma_2(y_{t_j}) - \sigma(y_{t_j})) \mathbb{X}_{t_j t_k}^{(2)}}_{:= B_{ij}^{(1)}}.$$

Therefore we get

$$\|\delta R\|_{3\alpha}^\mathbb{T} \leq M(\|B^{(2)}\|_{2\alpha}^\mathbb{T} + \|B^{(1)}\|_\alpha^\mathbb{T})$$

We now need to estimate the right hand side. By (5.5) we are able to write:

$$\begin{aligned} B_{t_it_j}^{(2)} &= \int_0^1 \nabla \sigma(y_{t_i} + r_1 \delta y_{t_it_j}) H_{t_it_j} dr_1 + \int_0^1 (\sigma_2(y_{t_i} + r_1 \delta y_{t_it_j}) - \sigma_2(y_{t_i})) \mathbb{X}_{t_it_j}^{(1)} dr_1 \\ &\quad - \int_0^1 \nabla \sigma(y_{t_i} + r_1 \delta y_{t_it_j}) (\sigma(y_{t_i} + r_1 \delta y_{t_it_j}) - \sigma(y_{t_i})) \mathbb{X}_{t_it_j}^{(1)} dr_1 \\ &:= C_{ij} + F_{ij} + E_{ij}. \end{aligned}$$

We write for F and E :

$$\|F\|_{2\alpha}^\mathbb{T} \leq \|\nabla \sigma_2\|_\infty \|\delta y\|_\alpha^\mathbb{T} \|\mathbb{X}^{(1)}\|_\alpha \leq DM \|\delta y\|_\alpha^\mathbb{T},$$

$$\|E\|_{2\alpha}^\mathbb{T} \leq \|\nabla \sigma\|_\infty^2 \|\delta y\|_\alpha^\mathbb{T} \|\mathbb{X}^{(1)}\|_\alpha \leq D^2 M \|\delta y\|_\alpha^\mathbb{T}.$$

Now we observe that

$$\begin{aligned} |H_{t_it_j}| &\leq |R_{t_it_j}| + |\sigma_2(y_{t_i}) \mathbb{X}_{t_it_j}^{(2)}| \\ &\leq [T^\alpha \|R\|_{3\alpha}^\mathbb{T} + (|\sigma_2(y_0)| + T^\alpha \|\nabla \sigma_2\|_\infty \|\delta y\|_\alpha^\mathbb{T}) \|\mathbb{X}^{(2)}\|_{2\alpha}] |t_j - t_i|^{2\alpha} \\ &\leq [T^\alpha \|R\|_{3\alpha}^\mathbb{T} + M |\sigma_2(y_0)| + T^\alpha DM \|\delta y\|_\alpha^\mathbb{T}] |t_j - t_i|^{2\alpha} \end{aligned}$$

Then for C we write

$$\|C\|_{2\alpha}^\mathbb{T} \leq D \|H\|_{2\alpha}^\mathbb{T} \leq D(T^\alpha \|R\|_{3\alpha}^\mathbb{T} + M |\sigma_2(y_0)| + T^\alpha DM \|\delta y\|_\alpha^\mathbb{T})$$

We finally have

$$\begin{aligned} \|B^{(2)}\|_{2\alpha}^\mathbb{T} &\leq \|C\|_{2\alpha}^\mathbb{T} + \|F\|_{2\alpha}^\mathbb{T} + \|E\|_{2\alpha}^\mathbb{T} \\ &\leq D(M |\sigma_2(y_0)| + T^\alpha \|R\|_{3\alpha}^\mathbb{T} + DM(2 + T^\alpha) \|\delta y\|_\alpha^\mathbb{T}). \end{aligned} \quad (5.43)$$

For $B^{(1)}$ we have

$$\|B^{(1)}\|_\alpha^\mathbb{T} \leq D \|\delta y\|_\alpha^\mathbb{T}.$$

Putting together (5.43) and (5.43) we obtain

$$\|R\|_{3\alpha}^{\mathbb{T}} \leq K_{3\alpha} D M (M |\sigma_2(y_0)| + T^\alpha \|R\|_{3\alpha}^{\mathbb{T}} + [1 + D M (2 + T^\alpha)] \|\delta y\|_{\alpha}^{\mathbb{T}})$$

If $K_{3\alpha} D M T^\alpha \leq \frac{1}{2}$, then

$$\|R\|_{3\alpha}^{\mathbb{T}} \leq \underbrace{2K_{3\alpha} D M (M |\sigma_2(y_0)| + [1 + D M (2 + T^\alpha)] \|\delta y\|_{\alpha}^{\mathbb{T}})}_{:=L(y)}. \quad (5.44)$$

So we have by (5.44) and (2.10)

$$\begin{aligned} \|\delta y\|_{\alpha}^{\mathbb{T}} &\leq \|R\|_{\alpha}^{\mathbb{T}} + \|A\|_{\alpha}^{\mathbb{T}} \\ &\leq T^{2\alpha} L(y) + (|\sigma(y_0)| + |\sigma_2(y_0)| + 2D T^\alpha \|\delta y\|_{\alpha}^{\mathbb{T}}) M. \end{aligned}$$

If $2D M T^\alpha \leq \frac{1}{2}$ we get

$$\|\delta y\|_{\alpha}^{\mathbb{T}} \leq 2T^{2\alpha} L(y) + 2M(|\sigma(y_0)| + |\sigma_2(y_0)|).$$

Furthermore, looking at $L(y)$, if $2K_{3\alpha} D M [1 + D M (2 + T^\alpha)] T^{2\alpha} \leq \frac{1}{2}$ we obtain

$$\|\delta y\|_{\alpha}^{\mathbb{T}} \leq 5M(|\sigma(y_0)| + |\sigma_2(y_0)|)$$

and

$$\begin{aligned} L(y) &\leq 2K_{3\alpha} D M \{M |\sigma_2(y_0)| + [1 + D M (2 + T^\alpha)] 5M(|\sigma(y_0)| + |\sigma_2(y_0)|)\} \\ &:= S. \end{aligned} \quad (5.45)$$

We conclude by (5.44) and (5.45) that

$$\|\delta y - A\|_{3\alpha}^{\mathbb{T}} = \|R\|_{3\alpha}^{\mathbb{T}} \leq S \simeq_{M,D,\alpha} (|\sigma(y_0)| + |\sigma_2(y_0)|).$$

□

THEOREM 5.7. (LOCAL EXISTENCE FOR $\alpha \in (\frac{1}{2}, \frac{1}{3}]$) Suppose σ, σ_2 are C^1 with $\|\nabla \sigma\|_{\infty} + \|\nabla \sigma_2\|_{\infty} < +\infty$. There exists $T_{M,D,\alpha} > 0$ such that, for all $T \in (0, T_{M,D,\alpha})$ and $\mathbb{X} = (\mathbb{X}^{(1)}, \mathbb{X}^{(2)}) \in \mathcal{R}_{\alpha,d}$ such that $M \geq \max\{\|\mathbb{X}^{(1)}\|_{\alpha} + \|\mathbb{X}^{(2)}\|_{2\alpha}\}$, there exists a solution Z of (5.2) on $[0, T]$ such that $Z_0 = z_0$ and

$$\|Z\|_{\alpha} \leq 15M(|\sigma(z_0)| + |\sigma_2(z_0)|). \quad (5.46)$$

Proof. Let us define, in the same way of (5.40), the function Z^n on the partition $\mathbb{T}_n := \{t_i: t_i = \frac{i}{n} \leq T\}$ by

$$Z_{t_{i+1}}^n = Z_{t_i}^n + \sigma(Z_{t_i}^n) \mathbb{X}_{t_i t_{i+1}}^{(1)} + \sigma_2(Z_{t_i}^n) \mathbb{X}_{t_i t_{i+1}}^{(2)}.$$

We can extend Z^n by linear interpolation to a continuous function on $[0, T]$. Given $t_i \leq s < t \leq t_{i+1}$, since $\delta Z_{st}^n = \frac{t-s}{t_{i+1}-t_i} \delta Z_{t_i t_{i+1}}^n$, we can bound for $\alpha \in (0, 1]$

$$\frac{|\delta Z_{st}^n|}{(t-s)^{\alpha}} = \left(\frac{t-s}{t_{i+1}-t_i} \right)^{1-\alpha} \frac{|\delta Z_{t_i t_{i+1}}^n|}{(t_{i+1}-t_i)^{\alpha}} \leq \frac{|\delta Z_{t_i t_{i+1}}^n|}{(t_{i+1}-t_i)^{\alpha}}.$$

It follows by the triangular inequality that

$$\|\delta Z^n\|_\alpha \leq 3\|\delta Z^n\|_\alpha^{\mathbb{T}_n}.$$

Now by (5.41) we can write

$$\|\delta Z^n\|_\alpha \leq 15M(|\sigma(z_0)| + |\sigma_2(z_0)|).$$

Let us now focus on Z^n for $n=2^m$, $m \in \mathbb{N}$. The family of functions $(Z^{2^m})_{m \in \mathbb{N}}$ is equicontinuous by (5.46) and equibounded, since $Z^{2^m} = z_0$ for all $m \in \mathbb{N}$, hence by the Arzelà-Ascoli theorem (refer for example to [20],[3]) it is compact in $C([0, T], \mathbb{R}^k)$. Let us denote by $Z: [0, T] \rightarrow \mathbb{R}^k$ a limit point. By (5.42) we can write

$$|\delta Z_{st}^n - \sigma(Z_s^n) \mathbb{X}_{st}^{(1)} - \sigma_2(Z_s^n) \mathbb{X}_{st}^{(2)}| \lesssim_{M,D,\alpha} c(z_0)(t-s)^{3\alpha} \quad \forall s < t \in \mathbb{T}_n, \quad (5.47)$$

where $c(z_0) = (|\sigma(y_0)| + |\sigma_2(y_0)|)$. Letting $n=2^m \rightarrow \infty$ we see that (5.47) holds also with Z in place of Z^n and $\mathbb{T} := \bigcup_{m \in \mathbb{N}} \mathbb{T}_{2^m}$ in place of $\mathbb{T}_n$:

$$|\delta Z_{st} - \sigma(Z_s) \mathbb{X}_{st}^{(1)} - \sigma_2(Z_s) \mathbb{X}_{st}^{(2)}| \lesssim_{M,D,\alpha} c(z_0)(t-s)^{3\alpha} \quad \forall s < t \in \mathbb{T}.$$

Since $\mathbb{T}$ is dense in $[0, T]$ and Z is continuous, this bound extends to all $0 \leq s \leq t \leq T$, so that Z is a solution of (5.2). $\square$

THEOREM 5.8. (GLOBAL EXISTENCE FOR $\alpha \in (\frac{1}{2}, \frac{1}{3}]$) Suppose σ, σ_2 are C^1 with $\|\nabla \sigma\|_\infty + \|\nabla \sigma_2\|_\infty < +\infty$, then for every $z_0 \in \mathbb{R}^k$ and $T > 0$ there is a unique solution $(Z_t)_{t \in [0, T]}$ of (5.2) such that $Z_0 = z_0$.

Proof. Let $\Lambda \subseteq [0, T]$ be the set of all $u \in [0, T]$ such that there is a solution $(Z_t)_{t \in [0, u]}$ to (5.2) such that $Z_0 = z_0$. By Theorem 5.7 Λ is not empty.

We can show that Λ is an open subset of $[0, T]$. Indeed, given $u \in \Lambda$ and a solution $(Z'_t)_{t \in [0, u]}$, by Theorem 5.7 for h small enough there exists a solution $(Z''_t)_{t \in [u, u+h]}$ with starting point $Z''_u = Z'_u$, so that by a standard pasting procedure we obtain a solution $(Z_t)_{t \in [0, u+h]}$, so that $u+h \in \Lambda$, which implies Λ is open.

We will show now that Λ is also closed in $[0, T]$, which will imply $\Lambda = [0, T]$, thus concluding the proof.

Given $(u_n)_n \subset \Lambda$ increasing, we associate solutions $Z^n = (Z_t^n)_{t \in [0, u_n]}$. Then defining $u := \lim_{n \rightarrow \infty} u_n \in [0, T]$ we will show $u \in \Lambda$. As in Theorem 5.7 we extend Z^n by linear interpolation to a continuous function on $[0, T]$. By the a priori estimate (5.8), if we fix $\tau > 0$ small enough we have $\|\delta Z^n\|_{\alpha, \tau} \leq 2(\sigma(Z_0)\|\mathbb{X}^{(1)}\|_\alpha + \sigma_2(Z_0)\|\mathbb{X}^{(2)}\|_{2\alpha})$ for every $n \in \mathbb{N}$, which shows that the family of functions $(Z^n)_{n \in \mathbb{N}}$ is equicontinuous. Since $Z^n = z_0$ for every $n \in \mathbb{N}$ it is also equibounded. Therefore by the Arzelà-Ascoli theorem it is compact and it exists a convergent subsequence $Z^{n_k} \rightarrow Z$ in $C([0, T], \mathbb{R}^k)$.

We now only need to show that Z is indeed a solution of (5.2). Using again the a priori estimate (5.8) in (5.7) we get

$$|\delta Z_{st}^n - \sigma(Z_s^n) \mathbb{X}_{st}^{(1)} - \sigma_2(Z_s^n) \mathbb{X}_{st}^{(2)}| \leq c(z_0)(t-s)^{3\alpha} \quad \text{for all } 0 \leq s < t \leq u_n,$$

where $c(z_0) = K_{3\alpha} c'_{\alpha, \mathbb{X}, \sigma} 2(\sigma(Z_0)\|\mathbb{X}^{(1)}\|_\alpha + \sigma_2(Z_0)\|\mathbb{X}^{(2)}\|_{2\alpha})$.

Fix now $0 \leq s < t < u$ and note that $u < u_n$ for large $n \in \mathbb{N}$, so that letting $n \rightarrow \infty$ we get

$$|\delta Z_{st} - \sigma(Z_s)\mathbb{X}_{st}^{(1)} - \sigma_2(Z_s)\mathbb{X}_{st}^{(2)}| \leq c(z_0)(t-s)^{3\alpha} \quad \text{for all } 0 \leq s < t < u.$$

By continuity this holds also for $t = u$, so we conclude that $Z = (Z_t)_{t \in [0, u]}$ is a solution of (5.2) with $Z_0 = z_0$.

□

CHAPTER 6

GENERAL ROUGH EQUATION

In this final chapter we present some original results that have as goal the extension of the theory seen for $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ to the general case $\alpha \in (\frac{1}{n+1}, \frac{1}{n}]$ for $n \in \mathbb{N}$. The main difficulty of this new setting consists in finding estimates that are similar to the ones presented in Chapter 5. Indeed, we don't have a straightforward adaptation of the Taylor identities (5.4), (5.5) and to get something similar it will be fundamental to first use an easy and understandable representation of σ_i which we defined in (3.11) and then to use properties of geometric rough paths.

6.1. GENERAL ROUGH EQUATION

Let us fix a time horizon $T > 0$ and the dimensions $d, k \in \mathbb{N}$. We also fix a path $X: [0, T] \rightarrow \mathbb{R}^d$ such that $X \in \mathcal{C}^\alpha$ for $\alpha \in (\frac{1}{n+1}, \frac{1}{n}]$, an α -rough path $\mathbb{X} = (\mathbb{X}^{(1)}, \dots, \mathbb{X}^{(n)})$ over X and a function $\sigma: \mathbb{R}^k \rightarrow \mathbb{R}^k \otimes (\mathbb{R}^d)^*$ of class C^{n-1} . Since $\sigma \in C^{n-1}$ we are able to define σ_i for $i \in \{1, \dots, n\}$ as in (3.11).

Motivated by the discussion of Chapter 3 and 4 we consider the rough difference equation

$$\delta Z_{st} = \sum_{i=1}^n \sigma_i(Z_s) \mathbb{X}_{st}^{(i)} + o(t-s), \quad 0 \leq s \leq t \leq T, \quad (6.1)$$

where $Z: [0, T] \rightarrow \mathbb{R}^k$ is an unknown path.

We define the remainders $Z^{[i]}$ for $i \in \{2, \dots, n+1\}$ by

$$Z_{st}^{[i]} = \delta Z_{st} - \sum_{j=1}^{i-1} \sigma_j(Z_s) \mathbb{X}_{st}^{(j)}, \quad (6.2)$$

so that a solution of (5.2) is a path $Z: [0, T] \rightarrow \mathbb{R}^k$ such that

$$Z_{st}^{[n+1]} = o(t-s).$$

The final goal is to show that under certain hypothesis this rough difference equation is well-posed. In this thesis we will be able to show that the solution is unique under certain conditions.

6.2. σ_n AND TREES

We call a rooted tree τ a graph with a designated vertex, called root, such that there is a unique path from the root to any other vertex in the tree. $|\tau|$ indicates the tree order, i.e. the number of vertices. It is well known, for example from [4], that σ_n has a representation as a sum of trees

$$\sigma_n = \sum_{|\tau|=n} \alpha(\tau) \tau,$$

where α is the *Connes-Moscovici coefficient*

$$\alpha(\tau) := \frac{|\tau|!}{\tau! \sigma(\tau)}$$

and $\tau!$ is defined by $\bullet! := 1$ and

$$([\tau_1 \cdots \tau_n])! := (|\tau_1| + \cdots + |\tau_n| + 1) \tau_1! \cdots \tau_n!,$$

while $\sigma(\tau)$ is defined by $\sigma(\bullet) := 1$ and

$$\sigma([\tau_1^{n_1} \cdots \tau_k^{n_k}]) := \prod_{i=1}^k n_i! (\sigma(\tau_i))^{n_i},$$

where in the last formula $\tau_1, \dots, \tau_k$ are pairwise distinct.

It was also shown in [6] that in the tree representation a number M of edges coming out from a node indicates a multiplication for the M -th derivative of σ and every terminal node represents a multiplication for σ . This implies for example that

$$\nabla \sigma \cdot \sigma_n = \sigma_n \curvearrowright \bullet, \tag{6.3}$$

where by $\curvearrowright$ we indicate the grafting operation.

Moreover in tree form we have the following formula:

$$\sigma_n = \bullet \curvearrowright \sigma_{n-1}, \tag{6.4}$$

which means that σ_n is a forest consisting of the trees obtained by attaching a new edge and vertex to every node of σ_{n-1} .

6.3. TAYLOR'S IDENTITIES

We know from (6.4) and (6.3) that for every $n \in \mathbb{N}$, $\sigma_{n+1} = \bullet \curvearrowright \sigma_n$ and $\nabla \sigma \cdot \sigma_n = \sigma_n \curvearrowright \bullet$.

Computing σ_n for the first values of n we get different trees that have a different number of branches coming out of the root. This in our notation means a multiplication for a derivative of σ . We will say that all the trees in the expression of σ_n that have M branches going out the root are in the “level” M of σ_n .

This way we obtain a recursion on every level. Indeed, to pass from σ_n to σ_{n+1} we graft $\bullet$ into σ_n and this means that the only trees of level M of σ_{n+1} are obtained by grafting $\bullet$ into the root of the level $M-1$ of σ_n and the level M of σ_n . we can therefore say that the level M of σ_{n+1} is equal to the level $M-1$ of σ_n plus $\bullet$ grafted into the level M of σ_n .

More explicitly, looking at the first cases a pattern is clear:

$$\begin{aligned} \sigma_2 &= \begin{array}{c} \bullet \\ | \\ \bullet \end{array} & \sigma_3 &= \begin{array}{c} \bullet & \bullet \\ & \diagdown \quad \diagup \\ & \bullet \end{array} + \begin{array}{c} \bullet \\ | \\ \bullet \\ | \\ \bullet \end{array} \\ \sigma_4 &= \begin{array}{c} \bullet & \bullet & \bullet \\ & \diagdown \quad | \quad \diagup \\ & \bullet \end{array} + 3 \begin{array}{c} \bullet & \bullet \\ & \diagdown \quad \diagup \\ & \bullet \end{array} + \begin{array}{c} \bullet & \bullet \\ & \diagdown \quad \diagup \\ & \bullet \\ | \\ \bullet \end{array} + \begin{array}{c} \bullet \\ | \\ \bullet \\ | \\ \bullet \\ | \\ \bullet \end{array} \end{aligned}$$

We note that, with $[\dots]$ being the operation of attaching trees to a common root we can write:

$$\sigma_3 = [\sigma, \sigma] + [\sigma_2],$$

$$\sigma_4 = [\sigma, \sigma, \sigma] + 3[\sigma, \sigma_2] + [\sigma_3].$$

Iterating we can guess a formula.

We define for $k \in \mathbb{N}$, $l, m \in \mathbb{N}^k$

$$C(k, l, m) = \frac{1}{m_k! \dots m_1!} \cdot \frac{(l_k m_k + \dots + l_1 m_1)!}{(l_k!)^{m_k} \dots (l_1!)^{m_1}}.$$

PROPOSITION 6.1. For $n \geq 2$

$$\sigma_n = \sum_{k=1}^{n-1} \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n-1}} C(k, l, m) [\sigma_{l_k}^{m_k} \dots \sigma_{l_1}^{m_1}],$$

where the sum is over $l_1, \dots, l_k, m_1, \dots, m_k \geq 1$.

Proof.

We proceed by induction. For $n=2$ we have $\sigma_2 = [\sigma_1]$ so that the formula is true.

We suppose that the relation holds for σ_n . We have:

$$\begin{aligned} \sigma_{n+1} &= \bullet \curvearrowright \sigma_n \\ &= \sum_{k=1}^{n-1} \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n-1}} C(k, l, m) \bullet \curvearrowright [\sigma_{l_k}^{m_k} \dots \sigma_{l_1}^{m_1}] \end{aligned}$$

where we used the linearity of the grafting operation. We note that in this formula we have

$$C(k, l, m) = \frac{1}{m_k! \dots m_1!} \cdot \frac{(n-1)!}{(l_k!)^{m_k} \dots (l_1!)^{m_1}}$$

by the constraint $\sum_i m_i l_i = n - 1$.

Now we observe that if we have a tree $\tau = [\tau_1 \dots \tau_k]$ then $\bullet \curvearrowright \tau$ is obtained by grafting $\bullet$ to the root of τ , to τ_1 , to τ_2 and so on. We get:

$$\bullet \curvearrowright [\tau_1 \dots \tau_k] = [\tau_1 \dots \tau_k \bullet] + [(\bullet \curvearrowright \tau_1) \dots \tau_k] + \dots + [\tau_1 \dots (\bullet \curvearrowright \tau_k)]$$

and

$$\bullet \curvearrowright [\tau_1^{m_1} \dots \tau_k^{m_k}] = [\tau_1^{m_1} \dots \tau_k^{m_k} \bullet] + m_1 [(\bullet \curvearrowright \tau_1) \tau_1^{m_1-1} \dots \tau_k^{m_k}] + \dots + m_k [\tau_1^{m_1} \dots \tau_k^{m_k-1} (\bullet \curvearrowright \tau_k)].$$

We can therefore write

$$\sigma_{n+1} = \sum_{k=1}^{n-1} \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n-1}} C(k, l, m) \left([\sigma_{l_k}^{m_k} \dots \sigma_{l_1}^{m_1} \sigma_1] + \sum_{j=1}^k m_j [\sigma_{l_k}^{m_k} \dots \sigma_{l_{j+1}} \sigma_{l_j}^{m_j-1} \dots \sigma_{l_1}^{m_1}] \right).$$

Now we have

$$\begin{aligned} & \sum_{k=1}^{n-1} \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n-1}} C(k, l, m) [\sigma_{l_k}^{m_k} \dots \sigma_{l_1}^{m_1} \sigma_1] = \\ &= \sum_{k=1}^{n-1} \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n-1}} (\mathbb{1}_{(l_1=1)} + \mathbb{1}_{(l_1>1)}) C(k, l, m) [\sigma_{l_k}^{m_k} \dots \sigma_{l_1}^{m_1} \sigma_1] \\ &= \sum_{k=1}^{n-1} \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n}} \mathbb{1}_{(l_1=1, m_1>1)} \frac{m_1}{n} C(k, l, m) [\sigma_{l_k}^{m_k} \dots \sigma_{l_1}^{m_1}] \\ & \quad + \sum_{k=1}^{n-1} \sum_{\substack{l_1 < \dots < l_{k+1} \\ \sum_i m_i l_i = n}} \mathbb{1}_{(l_1=1, m_1=1)} \frac{1}{n} C(k+1, (l_{k+1}, \dots, l_1), (m_{k+1}, \dots, m_1)) [\sigma_{l_{k+1}}^{m_{k+1}} \dots \sigma_{l_1}^{m_1}] \\ &= \sum_{k=1}^n \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n}} \mathbb{1}_{(l_1=1)} \frac{m_1 l_1}{n} C(k, l, m) [\sigma_{l_k}^{m_k} \dots \sigma_{l_1}^{m_1}]. \end{aligned}$$

For $j \in \{1, \dots, k\}$ we use the fact that

$$1 = \mathbb{1}_{(j=k)} + \mathbb{1}_{(j < k, l_{j+1} > l_j + 1)} + \mathbb{1}_{(j < k, l_{j+1} = l_j + 1)}$$

and we obtain

$$\begin{aligned} & m_j C(k, l, m) = \\ &= \mathbb{1}_{(j=k)} \frac{l_k + 1}{n} C(k+1, (l_k+1, l_k, \dots, l_1), (1, m_k-1, \dots, m_1)) \\ & \quad + \mathbb{1}_{(j < k, l_{j+1} > l_j + 1)} \frac{l_j + 1}{n} C(k+1, (l_k, \dots, l_j+1, l_j, \dots, l_1), (m_k, \dots, 1, m_j-1, \dots, m_1)) \\ & \quad + \mathbb{1}_{(j < k, l_{j+1} = l_j + 1)} \frac{m_{j+1} + 1}{n} l_{j+1} C(k, l, (m_k, \dots, m_{j+1}+1, m_j-1, \dots, m_1)) \end{aligned}$$

and moreover

$$\begin{aligned} & C(k+1, (l_k, \dots, l_j+1, l_j, \dots, l_1), (m_k, \dots, 1, m_j-1, \dots, m_1)) = \\ &= \mathbb{1}_{(m_j > 1)} C(k+1, (l_k, \dots, l_j+1, l_j, \dots, l_1), (m_k, \dots, 1, m_j-1, \dots, m_1)) + \\ &+ \mathbb{1}_{(m_j = 1)} C(k, (l_k, \dots, l_j+1, l_{j-1}, \dots, l_1), (m_k, \dots, 1, m_{j-1}, \dots, m_1)), \end{aligned}$$

$$\begin{aligned} & C(k, l, (m_k, \dots, m_{j+1}+1, m_j-1, \dots, m_1)) = \mathbb{1}_{(m_j > 1)} C(k, l, (m_k, \dots, m_{j+1}+1, m_j-1, \dots, \\ & m_1)) \\ &+ \mathbb{1}_{(m_j = 1)} C(k-1, (l_k, \dots, l_{j+1}, l_{j-1}, \dots, l_1), (m_k, \dots, m_{j+1}+1, m_{j-1}, \dots, m_1)). \end{aligned}$$

Now (denoting $l_0 := 0$) we obtain for $j = k$ or $l_{j+1} > l_j + 1, m_j > 1$

$$\begin{aligned} & \sum_{k=1}^{n-1} \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n-1}} C(k, l, m) \sum_{j=1}^k \mathbb{1}_{(j=k \text{ or } l_{j+1} > l_j + 1)} \mathbb{1}_{(m_j > 1)} m_j [\sigma_{l_k}^{m_k} \dots \sigma_{l_{j+1}} \sigma_{l_j}^{m_j-1} \dots \sigma_{l_1}^{m_1}] = \\ &= \sum_{k=1}^{n-1} \sum_{\substack{l_1 < \dots < l_{k+1} \\ \sum_i m_i l_i = n}} C(k+1, l, m) [\sigma_{l_{k+1}}^{m_{k+1}} \dots \sigma_{l_1}^{m_1}] \sum_{j=1}^k \mathbb{1}_{(l_{j+1} = l_j + 1, m_{j+1} = 1)} \frac{m_{j+1} l_{j+1}}{n} = \\ &= \sum_{k=2}^n \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n}} C(k, l, m) [\sigma_{l_k}^{m_k} \dots \sigma_{l_1}^{m_1}] \sum_{j=2}^k \mathbb{1}_{(l_j = l_{j-1} + 1, m_j = 1)} \frac{m_j l_j}{n}, \end{aligned}$$

and for $j = k$ or $l_{j+1} > l_j + 1, m_j = 1$

$$\begin{aligned} & \sum_{k=1}^{n-1} \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n-1}} C(k, l, m) \sum_{j=1}^k \mathbb{1}_{(j=k \text{ or } l_{j+1} > l_j + 1)} \mathbb{1}_{(m_j = 1)} m_j [\sigma_{l_k}^{m_k} \dots \sigma_{l_{j+1}} \sigma_{l_j}^{m_j-1} \dots \sigma_{l_1}^{m_1}] = \\ &= \sum_{k=2}^n \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n}} C(k, l, m) [\sigma_{l_k}^{m_k} \dots \sigma_{l_1}^{m_1}] \sum_{j=1}^k \mathbb{1}_{(l_j > l_{j-1} + 1, m_j = 1)} \frac{m_j l_j}{n}. \end{aligned}$$

Next, for $l_{j+1} = l_j + 1, m_j > 1$

$$\begin{aligned} & \sum_{k=1}^{n-1} \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n-1}} C(k, l, m) \sum_{j=1}^{k-1} \mathbb{1}_{(l_{j+1} = l_j + 1, m_j > 1)} m_j [\sigma_{l_k}^{m_k} \dots \sigma_{l_{j+1}} \sigma_{l_j}^{m_j-1} \dots \sigma_{l_1}^{m_1}] = \\ &= \sum_{k=2}^{n-1} \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n}} C(k, l, m) [\sigma_{l_k}^{m_k} \dots \sigma_{l_1}^{m_1}] \sum_{j=1}^{k-1} \mathbb{1}_{(l_{j+1} = l_j + 1, m_{j+1} > 1)} \frac{m_{j+1} l_{j+1}}{n} = \\ &= \sum_{k=2}^n \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n}} C(k, l, m) [\sigma_{l_k}^{m_k} \dots \sigma_{l_1}^{m_1}] \sum_{j=2}^k \mathbb{1}_{(l_j = l_{j-1} + 1, m_j > 1)} \frac{m_j l_j}{n}, \end{aligned}$$

and for $l_{j+1} = l_j + 1, m_j = 1$

$$\begin{aligned} & \sum_{k=1}^{n-1} \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n-1}} C(k, l, m) \sum_{j=1}^{k-1} \mathbb{1}_{(l_{j+1}=l_j+1, m_j=1)} m_j [\sigma_{l_k}^{m_k} \dots \sigma_{l_{j+1}} \sigma_{l_j}^{m_j-1} \dots \sigma_{l_1}^{m_1}] = \\ &= \sum_{k=2}^n \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n}} C(k, l, m) [\sigma_{l_k}^{m_k} \dots \sigma_{l_1}^{m_1}] \sum_{j=1}^k \mathbb{1}_{(l_j > l_{j-1}+1, m_j > 1)} \frac{m_j l_j}{n}. \end{aligned}$$

Therefore

$$\begin{aligned} & \sum_{k=1}^{n-1} \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n-1}} C(k, l, m) \sum_{j=1}^k m_j [\sigma_{l_k}^{m_k} \dots \sigma_{l_{j+1}} \sigma_{l_j}^{m_j-1} \dots \sigma_{l_1}^{m_1}] \\ &= \sum_{k=1}^n \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n}} C(k, l, m) [\sigma_{l_k}^{m_k} \dots \sigma_{l_1}^{m_1}] \left(\mathbb{1}_{(l_1 > 1)} \frac{m_1 l_1}{n} + \sum_{j=2}^k \frac{m_j l_j}{n} \right). \end{aligned}$$

We therefore conclude:

$$\begin{aligned} \sigma_{n+1} &= \sum_{k=1}^n \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n}} C(k, l, m) [\sigma_{l_k}^{m_k} \dots \sigma_{l_1}^{m_1}] \sum_{j=1}^k \frac{m_j l_j}{n} \\ &= \sum_{k=1}^n \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n}} C(k, l, m) [\sigma_{l_k}^{m_k} \dots \sigma_{l_1}^{m_1}]. \end{aligned}$$

The proof is complete. $\square$

Note that Proposition 6.1 holds in principle only on the tree structure. Now we want to give a similar formula that holds for the differential function σ_n .

PROPOSITION 6.2. *For $n \geq 2$, suppose $m_k + \dots + m_1 = M$, $A = \{a_1, \dots, a_M\}$, $I = \{i_{j_1}, \dots, i_{j_n-1}\}$, and for $r \in [1, \dots, k]$, $s \in [1, \dots, m_r]$ I_r^s represents the indexes relative to the s -th σ_{l_r} and suppose that every I_r^s is ordered.*

Then, if we call $\mathcal{C}(I)$ the combinations of the indexes I in the subsets I_r^s , maintaining order in each of the I_r^s , we have:

$$\sigma_n = \sum_{k=1}^{n-1} \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n-1}} \sum_{A_M} \partial_{A_M}^M \sigma_u^v \sum_{\mathcal{C}(I)} (\sigma_{l_k})_{I_k^1}^{a_1} \dots (\sigma_{l_k})_{I_k^{m_k}}^{a_{m_k}} \dots (\sigma_{l_1})_{I_1^1}^{a_{\sum m_i+1}} \dots (\sigma_{l_1})_{I_1^{m_1}}^{a_M},$$

where the sum is over $l_1, \dots, l_k, m_1, \dots, m_k \geq 1$.

Proof.

We proceed by induction. For $n=2$ we have $\sigma_2 = \partial_a \sigma_u^v (\sigma_1)_{i_1}^a$, so that the formula is true.

We suppose that the relation holds for σ_n . We have:

$$\begin{aligned}\sigma_{n+1} &= \bullet \curvearrowright \sigma_n \\ &= \sum_{k=1}^{n-1} \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n-1}} \sum_{A_M} \partial_{A_M}^M \sigma_u^v \sum_{\mathcal{C}(I)} \bullet \curvearrowright \left\{ (\sigma_{l_k})_{I_k^1}^{a_1} \dots (\sigma_{l_k})_{I_k^{m_k}}^{a_{m_k}} \dots (\sigma_{l_1})_{I_1^1}^{a_{\sum m_i+1}} \dots (\sigma_{l_1})_{I_1^{m_1}}^{a_M} \right\},\end{aligned}$$

where we used the linearity of the grafting operation and the inductive hypothesis. We note that the grafting operation has two different actions, it can transform a σ_{l_j} in σ_{l_j+1} or add a σ_1 at the end. In both cases we have a new index i_n that needs to be assigned to σ_{l_j+1} or σ_1 .

We have:

$$\begin{aligned}\sigma_{n+1} &= \\ &= \sum_{k=1}^{n-1} \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n-1}} \sum_{A_{M+1}} \partial_{A_{M+1}}^{M+1} \sigma_u^v \{ \\ &\quad \sum_{\mathcal{C}(I)} (\sigma_{l_k})_{I_k^1}^{a_1} \dots (\sigma_{l_k})_{I_k^{m_k}}^{a_{m_k}} \dots (\sigma_{l_1})_{I_1^1}^{a_{\sum m_i+1}} \dots (\sigma_{l_1})_{I_1^{m_1}}^{a_M} (\sigma_1)_{i_n}^{a_{M+1}} \\ &\quad + \sum_{p=1}^k \mathbb{1}_{(l_p+1 \neq l_{p+1})} \sum_{\mathcal{C}(I)} \sum_{r=1}^{m_p} (\sigma_{l_k})_{I_k^1}^{a_1} \dots (\sigma_{l_k})_{I_k^{m_k}}^{a_{m_k}} \dots (\sigma_{l_p+1})_{\{I_p^r, i_n\}}^{a_r} \dots (\sigma_{l_1})_{I_1^1}^{a_{\sum m_i+1}} \dots (\sigma_{l_1})_{I_1^{m_1}}^{a_{M+1}} \} \\ &\quad + \sum_{k=1}^{n-1} \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n-1}} \sum_{A_M} \partial_{A_M}^M \sigma_u^v \{ \\ &\quad \sum_{p=1}^k \mathbb{1}_{(l_p+1 = l_{p+1})} \sum_{\mathcal{C}(I)} \sum_{r=1}^{m_p} (\sigma_{l_k})_{I_k^1}^{a_1} \dots (\sigma_{l_k})_{I_k^{m_k}}^{a_{m_k}} \dots (\sigma_{l_p+1})_{\{I_p^r, i_n\}}^{a_r} \dots (\sigma_{l_1})_{I_1^1}^{a_{\sum m_i+1}} \dots (\sigma_{l_1})_{I_1^{m_1}}^{a_M} \}.\end{aligned}$$

Now we can isolate two cases, when the lowest term is σ_1 and when it isn't. We get for the first case:

$$\begin{aligned}&\sum_{k=1}^{n-1} \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n-1}} \left\{ \sum_{A_{M+1}} \partial_{A_{M+1}}^{M+1} \sigma_u^v \left[\sum_{\mathcal{C}(I)} (\sigma_{l_k})_{I_k^1}^{a_1} \dots (\sigma_{l_k})_{I_k^{m_k}}^{a_{m_k}} \dots (\sigma_{l_1})_{I_1^1}^{a_{\sum m_i+1}} \dots (\sigma_{l_1})_{I_1^{m_1}}^{a_M} (\sigma_1)_{i_n}^{a_{M+1}} + \right. \right. \\ &\quad \mathbb{1}_{(l_1=1)} \sum_{p=1}^k \mathbb{1}_{(l_p+1 \neq l_{p+1})} \\ &\quad \sum_{\mathcal{C}(I)} \sum_{r=1}^{m_p} (\sigma_{l_k})_{I_k^1}^{a_1} \dots (\sigma_{l_k})_{I_k^{m_k}}^{a_{m_k}} \dots (\sigma_{l_p+1})_{\{I_p^r, i_n\}}^{a_r} \dots (\sigma_1)_{I_1^1}^{a_{\sum m_i+1}} \dots (\sigma_1)_{I_1^{m_1}}^{a_{M+1}} \left. \right] + \\ &\quad \mathbb{1}_{(l_1=1)} \sum_{A_M} \partial_{A_M}^M \sigma_u^v \sum_{p=1}^k \mathbb{1}_{(l_p+1 = l_{p+1})} \\ &\quad \sum_{\mathcal{C}(I)} \sum_{r=1}^{m_p} (\sigma_{l_k})_{I_k^1}^{a_1} \dots (\sigma_{l_k})_{I_k^{m_k}}^{a_{m_k}} \dots (\sigma_{l_p+1})_{\{I_p^r, i_n\}}^{a_r} \dots (\sigma_1)_{I_1^1}^{a_{\sum m_i+1}} \dots (\sigma_1)_{I_1^{m_1}}^{a_M} \left. \right\}.\end{aligned}$$

We observe that renaming the l_i 's and the m_i 's we have all the tuples $l_1 < \dots < l_k$ with $1 \leq k \leq n$, $l_1 = 1$, such that $\sum_i m_i l_i = n$ and that we also have all the possible positions of the new index i_n . We can therefore rewrite the last expression as:

$$\sum_{k=1}^n \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n}} \mathbb{1}_{(l_1=1)} \sum_{A_M} \partial_{A_M}^M \sigma_u^v \sum_{\mathcal{C}(\{I, i_{j_n}\})} (\sigma_{l_k})_{I_k^1}^{a_1} \dots (\sigma_{l_k})_{I_k^{m_k}}^{a_{m_k}} \dots (\sigma_{l_1})_{I_1^1}^{a_{\sum m_i+1}} \dots (\sigma_{l_1})_{I_1^{m_1}}^{a_M}.$$

For the other case we do an analogous procedure:

$$\begin{aligned} & \sum_{k=1}^{n-1} \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n-1}} \left\{ \sum_{A_{M+1}} \partial_{A_{M+1}}^{M+1} \sigma_u^v \left[\right. \right. \\ & \mathbb{1}_{(l_1 > 1)} \sum_{p=1}^k \mathbb{1}_{(l_{p+1} \neq l_{p+1})} \sum_{\mathcal{C}(I)} \sum_{r=1}^{m_p} (\sigma_{l_k})_{I_k^1}^{a_1} \dots (\sigma_{l_k})_{I_k^{m_k}}^{a_{m_k}} \dots (\sigma_{l_{p+1}})_{\{I_p^r, i_n\}}^{a_r} \dots (\sigma_{l_1})_{I_1^1}^{a_{\sum m_i+1}} \dots (\sigma_{l_1})_{I_1^{m_1}}^{a_{M+1}} \left. \right] \\ & + \sum_{A_M} \partial_{A_M}^M \sigma_u^v \left[\right. \\ & \mathbb{1}_{(l_1 > 1)} \sum_{p=1}^k \mathbb{1}_{(l_{p+1} = l_{p+1})} \sum_{\mathcal{C}(I)} \sum_{r=1}^{m_p} (\sigma_{l_k})_{I_k^1}^{a_1} \dots (\sigma_{l_k})_{I_k^{m_k}}^{a_{m_k}} \dots (\sigma_{l_{p+1}})_{\{I_p^r, i_n\}}^{a_r} \dots (\sigma_{l_1})_{I_1^1}^{a_{\sum m_i+1}} \dots (\sigma_{l_1})_{I_1^{m_1}}^{a_M} \left. \right] \left. \right\} \\ & = \sum_{k=1}^n \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n}} \mathbb{1}_{(l_1 > 1)} \sum_{A_M} \partial_{A_M}^M \sigma_u^v \sum_{\mathcal{C}(\{I, i_{j_n}\})} (\sigma_{l_k})_{I_k^1}^{a_1} \dots (\sigma_{l_k})_{I_k^{m_k}}^{a_{m_k}} \dots (\sigma_{l_1})_{I_1^1}^{a_{\sum m_i+1}} \dots (\sigma_{l_1})_{I_1^{m_1}}^{a_M}. \end{aligned}$$

Summing the two cases we get:

$$\sigma_{n+1} = \sum_{k=1}^n \sum_{\substack{l_1 < \dots < l_k \\ \sum_i m_i l_i = n}} \sum_{A_M} \partial_{A_M}^M \sigma_u^v \sum_{\mathcal{C}(\{I, i_{j_n}\})} (\sigma_{l_k})_{I_k^1}^{a_1} \dots (\sigma_{l_k})_{I_k^{m_k}}^{a_{m_k}} \dots (\sigma_{l_1})_{I_1^1}^{a_{\sum m_i+1}} \dots (\sigma_{l_1})_{I_1^{m_1}}^{a_M}.$$

The proof is complete. $\square$

We note that we have a single σ_{l_1} only when $k=1$ and $m_1=1$, and in that case $l_1=n-1$, so that we get a term which is equal to $\nabla \sigma \cdot \sigma_{n-1}$.

Now we want to be able to use this representation to split the geometric rough path of order n in rough paths of lower order, in other words we would like to prove that $\sigma_{n+1} \mathbb{X}^{(n)} = \sum_{k=1}^n \frac{\nabla^k \sigma}{k!} \left(\sum_{\substack{(l_1, \dots, l_k) \\ \sum l_i = n}} \prod_{i=1}^k \sigma_{l_i} \mathbb{X}^{(l_i)} \right)$. Here's an example for the specific case $n=4$, $k=2$, $l_1=l_2=2$.

Example 6.3.

As an example we show how the product $\frac{\nabla^2 \sigma}{2}(\sigma_2 \mathbb{X}^{(2)} \otimes \sigma_2 \mathbb{X}^{(2)})$ can be recovered from one of the elements of the tree form of σ_5 . Note that below, 3 is the Connes-Moscovici coefficient of the tree.

$$\begin{aligned}
3 \quad \begin{array}{c} \bullet \quad \bullet \\ | \quad | \\ \bullet \quad \bullet \\ \diagdown \quad \diagup \\ \bullet \end{array} &= \begin{array}{c} i_2 \quad \bullet \quad i_4 \\ | \quad \quad | \\ i_1 \quad \bullet \quad i_3 \\ \diagdown \quad \diagup \\ u \end{array} + \begin{array}{c} i_3 \quad \bullet \quad i_4 \\ | \quad \quad | \\ i_1 \quad \bullet \quad i_2 \\ \diagdown \quad \diagup \\ u \end{array} + \begin{array}{c} i_4 \quad \bullet \quad i_3 \\ | \quad \quad | \\ i_1 \quad \bullet \quad i_2 \\ \diagdown \quad \diagup \\ u \end{array} = \\
&= \sum_{a,b,c,d} \partial_{ab}^2 \sigma_u^v \{ \partial_c \sigma_{i_1}^a \sigma_{i_2}^c \partial_d \sigma_{i_3}^b \sigma_{i_4}^d + \partial_c \sigma_{i_1}^a \sigma_{i_3}^c \partial_d \sigma_{i_2}^b \sigma_{i_4}^d + \partial_c \sigma_{i_1}^a \sigma_{i_4}^c \partial_d \sigma_{i_2}^b \sigma_{i_3}^d \} \mathbb{X}^{i_4 i_3 i_2 i_1} \\
&= \frac{1}{2} \sum_{a,b,c,d} \partial_{ab}^2 \sigma_u^v \{ \partial_c \sigma_{i_1}^a \sigma_{i_2}^c \partial_d \sigma_{i_3}^b \sigma_{i_4}^d + \partial_c \sigma_{i_1}^a \sigma_{i_3}^c \partial_d \sigma_{i_2}^b \sigma_{i_4}^d + \partial_c \sigma_{i_1}^a \sigma_{i_4}^c \partial_d \sigma_{i_2}^b \sigma_{i_3}^d \\
&\quad + \partial_c \sigma_{i_3}^a \sigma_{i_4}^c \partial_d \sigma_{i_1}^b \sigma_{i_2}^d + \partial_c \sigma_{i_2}^a \sigma_{i_4}^c \partial_d \sigma_{i_1}^b \sigma_{i_3}^d + \partial_c \sigma_{i_2}^a \sigma_{i_3}^c \partial_d \sigma_{i_1}^b \sigma_{i_4}^d \} \mathbb{X}^{i_4 i_3 i_2 i_1} \\
&= \frac{1}{2} \sum_{a,b} \partial_{ab}^2 \sigma_u^v \{ (\sigma_2)_{i_1 i_2}^a (\sigma_2)_{i_3 i_4}^b + (\sigma_2)_{i_1 i_3}^a (\sigma_2)_{i_2 i_4}^b + (\sigma_2)_{i_1 i_4}^a (\sigma_2)_{i_2 i_3}^b + (\sigma_2)_{i_3 i_4}^a (\sigma_2)_{i_1 i_2}^b \\
&\quad + (\sigma_2)_{i_2 i_4}^a (\sigma_2)_{i_1 i_3}^b + (\sigma_2)_{i_2 i_3}^a (\sigma_2)_{i_1 i_4}^b \} \mathbb{X}^{i_4 i_3 i_2 i_1} \\
&= \frac{1}{2} \sum_{a,b} \partial_{ab}^2 \sigma_u^v \{ (\sigma_2)_{i_1 i_2}^a (\sigma_2)_{i_3 i_4}^b \} (\mathbb{X}^{i_4 i_3 i_2 i_1} + \mathbb{X}^{i_4 i_2 i_3 i_1} + \mathbb{X}^{i_2 i_4 i_3 i_1} + \mathbb{X}^{i_4 i_2 i_1 i_3} + \mathbb{X}^{i_2 i_4 i_1 i_3} \\
&\quad + \mathbb{X}^{i_2 i_1 i_4 i_3}) \\
&= \frac{1}{2} \sum_{a,b} \partial_{ab}^2 \sigma_u^v \{ (\sigma_2)_{i_1 i_2}^a (\sigma_2)_{i_3 i_4}^b \} \mathbb{X}^{i_4 i_3 \sqcup i_2 i_1} \\
&= \frac{1}{2} \sum_{a,b} \partial_{ab}^2 \sigma_u^v \{ (\sigma_2)_{i_1 i_2}^a \mathbb{X}^{i_2 i_1} (\sigma_2)_{i_3 i_4}^b \mathbb{X}^{i_4 i_3} \},
\end{aligned}$$

where in the second line we used the symmetry of a,b and of c,d.

We want to formalize the passage of the example in which we change the name of the indexes and we recover the shuffle product on the components of $\mathbb{X}$. This trick can be summed up in the following proposition.

PROPOSITION 6.4.

Fix k . Fix n, m such that $n, m < k$, $n + m = k$. Fix some indexes $i_1 < i_2 < \dots < i_k$. Now consider the permutations $i_{l_1} \dots i_{l_k}$ such that $i_{l_1} < \dots < i_{l_n}$, $i_{l_{n+1}} < \dots < i_{l_{m+n}} = i_k$, which we call $P(i_{l_1} \dots i_{l_k})$. For each permutation $i_{l_1} \dots i_{l_k}$ we define h_j as the position of the index i_j in $i_{l_1} \dots i_{l_k}$.

Then

$$\sum_{P(i_{l_1} \dots i_{l_k})} i_{h_k} \dots i_{h_1} = (i_k \dots i_{n+1}) \sqcup (i_n \dots i_1). \quad (6.5)$$

Proof.

We will proceed by induction on k .

For $k=2$, necessarily $n=m=1$ and we have

$$\sum_{P(i_1 i_2)} i_{h_2} i_{h_1} = i_1 i_2 + i_2 i_1 = i_2 \sqcup i_1.$$

Assume that the relation is true for k and we want to prove it for $k+1$.

Notice that the index i_1 , being the smallest, can only be in position 1 or $n+1$, in other words we have $i_{l_1} = 1$ or $i_{l_{n+1}} = 1$, otherwise the condition $i_{l_1} < \dots < i_{l_n}$, $i_{l_{n+1}} < \dots < i_{l_{m+n}} = i_{l_{k+1}}$ would not be satisfied. Therefore we have $h_1 = i_1$ or $h_1 = i_{n+1}$. So we can differentiate between this two cases:

$$\begin{aligned} \sum_{P(i_1 \dots i_{l_{k+1}})} i_{h_{k+1}} \dots i_{h_1} &= \left(\sum_{P(i_2 \dots i_{l_{k+1}})} i_{h_{k+1}} \dots i_{h_2} \right) i_{n+1} + \left(\sum_{P(i_2 \dots i_{l_{k+1}})} i_{h_{k+1}} \dots i_{h_2} \right) i_1 \\ &= \left(\sum_{P(i_1 \dots i_{l_n} i_{l_{n+2}} \dots i_{l_{k+1}})} i_{h_{k+1}} \dots i_{h_2} \right) i_{n+1} + \left(\sum_{P(i_2 \dots i_{l_{k+1}})} i_{h_{k+1}} \dots i_{h_2} \right) i_1, \end{aligned}$$

where in the first term we renamed the indexes.

We now observe that in the first element we sum over the permutations of distinct indexes $i_{l_1}, \dots, i_{l_n}, i_{l_{n+2}}, \dots, i_{l_{k+1}} \in \{i_1, \dots, i_n, i_{n+2}, \dots, i_{k+1}\}$ such that $i_{l_1} < \dots < i_{l_n}$, $i_{l_{n+2}} < \dots < i_{l_{k+1}}$, and in the second over $i_{l_2}, \dots, i_{l_{k+1}} \in \{i_2, \dots, i_{k+1}\}$ such that $i_{l_2} < \dots < i_{l_n}$, $i_{l_{n+1}} < \dots < i_{l_{k+1}}$ and we can use the inductive hypothesis since we have k indexes.

We obtain

$$\begin{aligned} \sum_{P(i_1 \dots i_{l_{k+1}})} i_{h_{k+1}} \dots i_{h_1} &= [(i_{k+1} i_k \dots i_{n+2}) \sqcup (i_n \dots i_1)] i_{n+1} \\ &\quad + [(i_{k+1} i_k \dots i_{n+1}) \sqcup (i_n \dots i_2)] i_1. \end{aligned}$$

Now if we name $v = i_{k+1} i_k \dots i_{n+2}$, $w = i_n \dots i_2$, $a = i_{n+1}$, $b = i_1$ we can write:

$$\begin{aligned} \sum_{P(i_1 \dots i_{l_{k+1}})} i_{h_{k+1}} \dots i_{h_1} &= [(i_{k+1} i_k \dots i_{n+2}) \sqcup (i_n \dots i_2)] i_1 \\ &\quad + [(i_{k+1} i_k \dots i_{n+2}) (i_{n+1}) \sqcup (i_n \dots i_2)] i_1 \\ &= (v \sqcup w b) a + (v a \sqcup w) b \\ &= (v a) \sqcup (w b) \\ &= (i_{k+1} \dots i_{n+1}) \sqcup (i_n \dots i_1), \end{aligned}$$

where we used the definition of the shuffle product.

This proves what we wanted. □

Note that we can extend this computation to multiple shuffle products, just by iteration of the product with just two elements.

We now will show how to generalize Example 6.3 to a generic multiplication of $\sigma_{l_k}, \dots, \sigma_{l_1}$. The idea is to use Proposition 6.4.

Suppose $m_k + \dots + m_1 = M$, $A = \{a_1, \dots, a_M\}$, $I = \{i_{j_1}, \dots, i_{j_n}\}$, and for $r \in [1, \dots, k]$, $s \in [1, \dots, m_r]$ I_r^s represents the indexes relative to the s -th σ_{l_r} and suppose that every I_r^s is ordered. We suppose $l_k m_k + \dots + l_1 m_1 = n$ and we call $\mathcal{C}(I)$ the combinations of the indexes I in the subsets I_r^s , maintaining order in each of the I_r^s and $P(I)$ the combinations of the indexes I in the subsets I_r^s , maintaining order in each of the I_r^s . Suppose that $\mathbb{X}^{(n)}$ is the n -th element of a geometric rough path $\mathbb{X} = (\mathbb{X}^{(1)}, \dots, \mathbb{X}^{(n)})$. We have:

$$\begin{aligned} & \left\{ \sum_{A_M} \partial_{A_M}^M \sigma_u^v \sum_{\mathcal{C}(I)} (\sigma_{l_k})_{I_k^1}^{a_1} \dots (\sigma_{l_k})_{I_k^{m_k}}^{a_{m_k}} \dots (\sigma_{l_1})_{I_1^1}^{a_{\sum m_i+1}} \dots (\sigma_{l_1})_{I_1^{m_1}}^{a_M} \right\} \mathbb{X}^{i_1 \dots i_n} \\ &= \left\{ \frac{1}{m_k! \dots m_1!} \sum_{A_M} \partial_{A_M}^M \sigma_u^v \sum_{P(I)} (\sigma_{l_k})_{I_k^1}^{a_1} \dots (\sigma_{l_k})_{I_k^{m_k}}^{a_{m_k}} \dots (\sigma_{l_1})_{I_1^1}^{a_{\sum m_i+1}} \dots (\sigma_{l_1})_{I_1^{m_1}}^{a_M} \right\} \mathbb{X}^{i_1 \dots i_n} \\ &= \left\{ \frac{1}{m_k! \dots m_1!} \sum_{A_M} \partial_{A_M}^M \sigma_u^v (\sigma_{l_k})_{I_k^1}^{a_1} \dots (\sigma_{l_k})_{I_k^{m_k}}^{a_{m_k}} \dots (\sigma_{l_1})_{I_1^1}^{a_{\sum m_i+1}} \dots (\sigma_{l_1})_{I_1^{m_1}}^{a_M} \right\} \sum_{P(I)} \mathbb{X}^{i_{h_n} \dots i_{h_1}}, \end{aligned}$$

where in the last line we changed the names of the indexes and where h_s is the position of i_s in the permutation. Now we can apply (6.5) and get:

$$\begin{aligned} & \left\{ \frac{1}{m_k! \dots m_1!} \sum_{A_M} \partial_{A_M}^M \sigma_u^v (\sigma_{l_k})_{I_k^1}^{a_1} \dots (\sigma_{l_k})_{I_k^{m_k}}^{a_{m_k}} \dots (\sigma_{l_1})_{I_1^1}^{a_{\sum m_i+1}} \dots (\sigma_{l_1})_{I_1^{m_1}}^{a_M} \right\} \sum_{P(I)} \mathbb{X}^{i_{h_n} \dots i_{h_1}} \\ &= \left\{ \frac{1}{m_k! \dots m_1!} \sum_{A_M} \partial_{A_M}^M \sigma_u^v (\sigma_{l_k})_{I_k^1}^{a_1} \dots (\sigma_{l_k})_{I_k^{m_k}}^{a_{m_k}} \dots (\sigma_{l_1})_{I_1^1}^{a_{\sum m_i+1}} \dots (\sigma_{l_1})_{I_1^{m_1}}^{a_M} \right\} \sum_{P(I)} \mathbb{X}^{i_n \dots i_{n-l_1} \sqcup \dots \sqcup i_k \dots i_1}. \end{aligned}$$

We conclude by the geometricity of the rough path that

$$\begin{aligned} & \left\{ \sum_{A_M} \partial_{A_M}^M \sigma_u^v \sum_{\mathcal{C}(I)} (\sigma_{l_k})_{I_k^1}^{a_1} \dots (\sigma_{l_k})_{I_k^{m_k}}^{a_{m_k}} \dots (\sigma_{l_1})_{I_1^1}^{a_{\sum m_i+1}} \dots (\sigma_{l_1})_{I_1^{m_1}}^{a_M} \right\} \mathbb{X}^{i_1 \dots i_n} = \\ & \frac{1}{m_k! \dots m_1!} \sum_{A_M} \partial_{A_M}^M \sigma_u^v \left[\left(\prod_{i_n-l_1, \dots, i_n}^{m_k} (\sigma_{l_k})_{i_n-l_1, \dots, i_n} \mathbb{X}^{i_n \dots i_{n-l_1}} \right) \otimes \dots \otimes \left(\prod_{i_1, \dots, i_k}^{m_1} (\sigma_{l_1})_{i_1, \dots, i_k} \mathbb{X}^{i_k \dots i_1} \right) \right], \end{aligned}$$

which corresponds in symbolic form to

$$\frac{1}{m_k! \dots m_1!} \nabla^M \sigma \left[\left(\prod_{i_n-l_1, \dots, i_n}^{m_k} \sigma_{l_k} \mathbb{X}^{(l_k)} \right) \otimes \dots \otimes \left(\prod_{i_1, \dots, i_k}^{m_1} \sigma_{l_1} \mathbb{X}^{(l_1)} \right) \right].$$

Moreover by symmetry of the shuffle product we can write:

$$\begin{aligned} & \frac{1}{m_k! \dots m_1!} \nabla^M \sigma \left[\left(\prod_{i_n-l_1, \dots, i_n}^{m_k} \sigma_{l_k} \mathbb{X}^{(l_k)} \right) \otimes \dots \otimes \left(\prod_{i_1, \dots, i_k}^{m_1} \sigma_{l_1} \mathbb{X}^{(l_1)} \right) \right] \\ &= \frac{1}{m_k! \dots m_1!} \cdot \frac{m_k! \dots m_1!}{M!} \nabla^M \sigma \sum_{P_{m_k, \dots, m_1}(I)} (\sigma_{l_{j_n}} \mathbb{X}^{(l_{j_n})} \otimes \dots \otimes \sigma_{l_{j_1}} \mathbb{X}^{(l_{j_1})}) \\ &= \frac{1}{M!} \nabla^M \sigma \sum_{P_{m_k, \dots, m_1}(I)} (\sigma_{l_{j_n}} \mathbb{X}^{(l_{j_n})} \otimes \dots \otimes \sigma_{l_{j_1}} \mathbb{X}^{(l_{j_1})}), \end{aligned}$$

where $P_{m_k, \dots, m_1}$ represents the permutation of the multiset $m_k, \dots, m_1$ or in other words the anagrams of a word with N letters where the same letter appears $m_k, \dots, m_1$ times. We finally get:

$$\left\{ \sum_{A_M} \partial_{A_M}^M \sigma_u^v \sum_{\mathcal{C}(I)} (\sigma_{l_k})_{I_k^1}^{a_1} \dots (\sigma_{l_k})_{I_k^{m_k}}^{a_{m_k}} \dots (\sigma_{l_1})_{I_1^1}^{a_{\sum m_i+1}} \dots (\sigma_{l_1})_{I_1^{m_1}}^{a_M} \right\} \mathbb{X}^{i_1 \dots i_n} \\ = \frac{1}{M!} \sum_{P_{m_k, \dots, m_1}(l_{j_n}, \dots, l_{j_1})} \sigma_{l_{j_n}} \mathbb{X}^{(l_{j_n})} \otimes \dots \otimes \sigma_{l_{j_1}} \mathbb{X}^{(l_{j_1})},$$

which is the desired result.

Using the last result and Proposition 6.2 we have the following theorem.

THEOREM 6.5. *Suppose that $\mathbb{X} = (\mathbb{X}^{(1)}, \dots, \mathbb{X}^{(n)})$ is a geometric rough path and σ_i is defined as in (3.11). We have*

$$(\sigma_{n+1} - \nabla \sigma \cdot \sigma_n) \mathbb{X}^{(n)} = \sum_{k=2}^n \frac{\nabla^k \sigma}{k!} \left(\sum_{\substack{(l_1, \dots, l_k) \\ \sum l_i = n}} \prod_{i=1}^k \sigma_{l_i} \mathbb{X}^{(l_i)} \right). \quad (6.6)$$

Remark 6.6. Observe that we can generalize Theorem 6.6. For some $m, n \in \mathbb{N}$ we can write

$$(\sigma_{m+n} - \nabla \sigma_m \cdot \sigma_n) \mathbb{X}^{(n)} = \sum_{k=2}^n \frac{\nabla^k \sigma_m}{k!} \left(\sum_{\substack{(l_1, \dots, l_k) \\ \sum l_i = n}} \prod_{i=1}^k \sigma_{l_i} \mathbb{X}^{(l_i)} \right). \quad (6.7)$$

This might seem not completely trivial, but, without entering in the details, by the recursive nature of σ_i we can repeat the same computations we did before but now substituting $\nabla \sigma_m$ to $\nabla \sigma$. To visualize better this fact, observe that we can consider a completely analogous tree structure where this time a number M of edges coming out from the root indicates a multiplication for the M -th derivative of σ_m .

We now have the instruments necessary to state some formulas that have the goal of reproducing the Taylor identities (5.4), (5.5).

PROPOSITION 6.7. *Let us take a geometric rough path $\mathbb{X} = (\mathbb{X}^{(1)}, \dots, \mathbb{X}^{(n)})$ and σ_i defined in (3.11). We have*

$$\sigma(z_2) - \sigma(z_1) - \sum_{i=2}^{n+1} \sigma_i(z_1) \mathbb{X}_{12}^{(i-1)} \\ = \int_0^1 \int_0^{r_1} \dots \int_0^{r_{n-1}} [\nabla^n \sigma(z_1 + r_n \delta z_{12}) - \nabla^n \sigma(z_1)] \delta z_{12} \otimes \dots \otimes \delta z_{12} dr_n \dots dr_1 \\ + \sum_{M=1}^{n+1} \frac{\nabla^M \sigma(z_1)}{M!} R_M^{[n+1]}, \quad (6.8)$$

where $R_{M,12}^{[n+1]} := R_M$ and

$$R_1^{[k]} = z_{12}^{[k]},$$

$$R_M^{[k]} = \underbrace{z_{12}^{[k-M+1]} \otimes \delta z_{12} \otimes \dots \otimes \delta z_{12}}_{M \text{ terms}} + \sigma(z_1) \mathbb{X}_{12}^{(1)} \otimes R_{M-1}^{[k-1]} + \dots + \sigma_{k-M}(z_1) \mathbb{X}_{12}^{(k-M)} \otimes R_{M-1}^{[M]}.$$

Proof. Notice that by simple computations we have

$$\begin{aligned} & \sigma(z_2) - \sigma(z_1) - \sum_{i=2}^{n+1} \sigma_i(z_1) \mathbb{X}_{12}^{(i-1)} \\ &= \int_0^1 \int_0^{r_1} \dots \int_0^{r_{n-1}} [\nabla^n \sigma(z_1 + r_n \delta z_{12}) - \nabla^n \sigma(z_1)] \delta z_{12} \otimes \dots \otimes \delta z_{12} dr_n \dots dr_1 \\ &+ \nabla \sigma(z_1) z_{12}^{[n+1]} + \sum_{k=2}^n (\nabla \sigma(z_1) \cdot \sigma_k(z_1) - \sigma_{k+1}(z_1)) \mathbb{X}_{12}^{(k)} + \sum_{k=2}^n \frac{\nabla^k \sigma(z_1)}{k!} \left(\prod_{i=1}^k \delta z_{12} \right) \end{aligned}$$

To ease the notation we will write δz instead of δz_{12} and $\mathbb{X}$ instead of $\mathbb{X}_{12}$.

Now by (6.6) we write

$$(\sigma_{k+1}(z_1) - \nabla \sigma(z_1) \cdot \sigma_k(z_1)) \mathbb{X}^{(k)} = \sum_{m=2}^k \frac{\nabla^m \sigma(z_1)}{m!} \left(\sum_{\substack{(l_1, \dots, l_m) \\ \sum l_i = k}} \prod_{i=1}^m \sigma_{l_i}(z_1) \mathbb{X}^{(l_i)} \right),$$

so that

$$\begin{aligned} & \sum_{k=2}^n ((\nabla \sigma(z_1) \cdot \sigma_k(z_1) - \sigma_{k+1}(z_1)) \mathbb{X}^{(k)}) + \sum_{k=2}^n \frac{\nabla^k \sigma(z_1)}{k!} \left(\prod_{i=1}^k \delta z \right) \\ &= \left(\sum_{k=2}^n \frac{\nabla^k \sigma(z_1)}{k!} \prod_{i=1}^k \delta z \right) - \left(\sum_{k=2}^n \sum_{m=2}^k \frac{\nabla^m \sigma(z_1)}{m!} \sum_{\substack{(l_1, \dots, l_m) \\ \sum l_i = k}} \prod_{i=1}^m \sigma_{l_i}(z_1) \mathbb{X}^{(l_i)} \right). \end{aligned}$$

Let us look at the terms corresponding to $\frac{\nabla^M \sigma(z_1)}{M!}$, for a generic $M \geq 2$. We want to prove

$$\prod_{i=1}^M \delta z - \sum_{k=M}^n \sum_{\substack{(l_1, \dots, l_M) \\ \sum l_i = k}} \prod_{i=1}^M \sigma_{l_i}(z_1) \mathbb{X}^{(l_i)} = R_M^{[n+1]}.$$

We will use induction on M . We see that for $M=2$ and for n generic we have

$$\begin{aligned} & \delta z \otimes \delta z - \sum_{k=2}^n \sum_{\substack{(l_1, l_2) \\ \sum l_i = k}} \sigma_{l_1}(z_1) \mathbb{X}^{(l_1)} \otimes \sigma_{l_2}(z_1) \mathbb{X}^{(l_2)} \\ &= \delta z \otimes \delta z - \sum_{i=1}^{n-1} \sigma_i(z_1) \mathbb{X}^{(i)} \otimes \delta z + \sum_{i=1}^{n+1-M} \sigma_i(z_1) \mathbb{X}^{(i)} \otimes \delta z \\ & \quad - \sum_{k=2}^n \sigma(z_1) \mathbb{X}^{(1)} \otimes \sigma_{k-1}(z_1) \mathbb{X}^{(k-1)} - \dots - \end{aligned}$$

$$\begin{aligned}
& - \sum_{k=n}^n \sigma_{(n-1)}(z_1) \mathbb{X}^{(n-1)} \otimes \sigma_{k-(n-1)}(z_1) \mathbb{X}^{(k-(n-1))} \\
& = z_{12}^{[n]} \otimes \delta z + \sigma(z_1) \mathbb{X}^{(1)} \otimes \left(\delta z - \sum_{j=1}^{n-1} \sigma_j(z_1) \mathbb{X}^{(j)} \right) + \cdots + \\
& \quad + \sigma_{(n-1)}(z_1) \mathbb{X}^{(n-1)} \otimes (\delta z - \sigma(z_1) \mathbb{X}^{(1)}) \\
& = z_{12}^{[n]} \otimes \delta z + \sigma(z_1) \mathbb{X}^{(1)} \otimes z_{12}^{[n]} + \cdots + \sigma_{(n-1)}(z_1) \mathbb{X}^{(n-1)} \otimes z_{12}^{[2]} \\
& = z_{12}^{[n]} \otimes \delta z + \sigma(z_1) \mathbb{X}^{(1)} \otimes R_1^{[n]} + \cdots + \sigma_{(n-1)}(z_1) \mathbb{X}^{(n-1)} \otimes R_1^{[2]} \\
& = R_2^{[n]}.
\end{aligned}$$

We now suppose that

$$\prod_{i=1}^{M'} \delta z - \sum_{k=M}^{n'} \sum_{\substack{(l_1, \dots, l_{M'}) \\ \sum l_i = k}} \prod_{i=1}^{M'} \sigma_{l_i}(z_1) \mathbb{X}^{(l_i)} = R_{M'}^{[n'+1]},$$

for $M' = M - 1$ and n' generic.

We have:

$$\begin{aligned}
& \prod_{i=1}^M \delta z - \sum_{k=M}^n \sum_{\substack{(l_1, \dots, l_M) \\ \sum l_i = k}} \prod_{i=1}^M \sigma_{l_i}(z_1) \mathbb{X}^{(l_i)} \\
& = \prod_{i=1}^M \delta z - \sum_{\substack{(l_1, \dots, l_M) \\ \sum l_i = n}} \prod_{i=1}^M \sigma_{l_i}(z_1) \mathbb{X}^{(l_i)} - \cdots - \sum_{\substack{(l_1, \dots, l_M) \\ \sum l_i = M}} \prod_{i=1}^M \sigma_{l_i}(z_1) \mathbb{X}^{(l_i)} \\
& = \prod_{i=1}^M \delta z - \sum_{i=1}^{n+1-M} \sigma_i(z_1) \mathbb{X}^{(i)} \otimes \delta z \otimes \cdots \otimes \delta z + \sum_{i=1}^{n+1-M} \sigma_i(z_1) \mathbb{X}^{(i)} \otimes \delta z \otimes \cdots \otimes \delta z \\
& \quad - \sum_{k=M}^n \sum_{\substack{(l_2, \dots, l_M) \\ \sum_{i=2}^M l_i = k-1}} \sigma(z_1) \mathbb{X}^{(1)} \otimes \prod_{i=2}^M \sigma_{l_i}(z_1) \mathbb{X}^{(l_i)} - \cdots - \\
& \quad - \sum_{k=n}^n \sum_{\substack{(l_2, \dots, l_M) \\ \sum_{i=2}^M l_i = k-(n+1-M)}} \sigma_{n+1-M}(z_1) \mathbb{X}^{(n+1-M)} \otimes \prod_{i=2}^M \sigma_{l_i}(z_1) \mathbb{X}^{(l_i)} \\
& = z_{12}^{[n+1-M+1]} \otimes \prod_{i=1}^{M-1} \delta z \\
& \quad + \sigma(z_1) \mathbb{X}^{(1)} \otimes \left(\prod_{i=1}^{M-1} \delta z - \sum_{j=M-1}^{n-1} \sum_{\substack{(l_1, \dots, l_{M-1}) \\ \sum l_i = j}} \prod_{i=1}^{M-1} \sigma_{l_i}(z_1) \mathbb{X}^{(l_i)} \right) + \cdots + \\
& \quad + \sigma_{n+1-M}(z_1) \mathbb{X}^{(n+1-M)} \otimes \left(\prod_{i=1}^{M-1} \delta z - \sum_{j=M-1}^{M-1} \sum_{\substack{(l_1, \dots, l_{M-1}) \\ \sum l_i = j}} \prod_{i=1}^{M-1} \sigma_{l_i}(z_1) \mathbb{X}^{(l_i)} \right)
\end{aligned}$$

$$\begin{aligned}
&= z_{12}^{[n+1-M+1]} \otimes \prod_{i=1}^{M-1} \delta z \\
&\quad + \sigma(z_1) \mathbb{X}^{(1)} \otimes R_{M-1}^{[n]} + \cdots + \sigma_{n+1-M}(z_1) \mathbb{X}^{(n+1-M)} \otimes R_{M-1}^{[M]} \\
&= R_M^{[n+1]}.
\end{aligned}$$

Since we fixed generic M , this holds for every M , thus proving the desired result. $\square$

Note 6.8.

Note that in the formula above every rest $R_M^{[k]}$ is a tensor product of M terms and is of class $\mathcal{C}^{k\alpha}$, therefore every term in the right hand side of the formula is of class $\mathcal{C}^{(n+1)\alpha}$.

Remark 6.9. We observe that Proposition (6.7) can be generalized to expressions with form $\sigma_k(z_2) - \sigma_k(z_1) - \sum_{j=k+1}^{n+1} \sigma_j(z_1) \mathbb{X}_{12}^{(j-k)}$ for all $k \in \{1, \dots, n-1\}$. Indeed, we can write, calling $m := n - k$,

$$\begin{aligned}
&\sigma_k(z_2) - \sigma_k(z_1) - \sum_{j=k+1}^{n+1} \sigma_j(z_1) \mathbb{X}_{12}^{(j-k)} \\
&= \int_0^1 \int_0^{r_1} \cdots \int_0^{r_{m-1}} [\nabla^m \sigma_k(z_1 + r_m \delta z_{12}) - \nabla^m \sigma_k(z_1)] \delta z_{12} \otimes \cdots \otimes \delta z_{12} dr_m \dots dr_1 \\
&\quad + \nabla \sigma_k(z_1) z_{12}^{[n+1-k]} + \sum_{j=1}^{n-k} \nabla \sigma_k(z_1) \cdot \sigma_j(z_1) \mathbb{X}_{12}^{(j)} \\
&\quad + \sum_{h=2}^m \frac{\nabla^h \sigma_k(z_1)}{h!} \left(\prod_{i=1}^h \delta z_{12} \right) - \sum_{j=k+1}^n \sigma_j(z_1) \mathbb{X}_{12}^{(j-k)} \\
&= \int_0^1 \int_0^{r_1} \cdots \int_0^{r_{m-1}} [\nabla^m \sigma_k(z_1 + r_m \delta z_{12}) - \nabla^m \sigma_k(z_1)] \delta z_{12} \otimes \cdots \otimes \delta z_{12} dr_m \dots dr_1 \\
&\quad + \nabla \sigma_k(z_1) z_{12}^{[n+1-k]} + \sum_{j=2}^{n-k} \nabla \sigma_k(z_1) \cdot \sigma_j(z_1) \mathbb{X}_{12}^{(j)} \\
&\quad + \sum_{h=2}^{n-k} \frac{\nabla^h \sigma_k(z_1)}{h!} \left(\prod_{i=1}^h \delta z_{12} \right) \\
&= \int_0^1 \int_0^{r_1} \cdots \int_0^{r_{m-1}} [\nabla^m \sigma_k(z_1 + r_m \delta z_{12}) - \nabla^m \sigma_k(z_1)] \delta z_{12} \otimes \cdots \otimes \delta z_{12} dr_m \dots dr_1 \\
&\quad + \nabla \sigma_k(z_1) z_{12}^{[n+1-k]} + \sum_{h=2}^{n-k} \frac{\nabla^h \sigma_k(z_1)}{h!} \left(\prod_{i=1}^h \delta z_{12} \right) \\
&\quad + \sum_{j=1}^{n-k} (\nabla \sigma_k(z_1) \cdot \sigma_j(z_1) - \sigma_{k+j}(z_1)) \mathbb{X}_{12}^{(j)}.
\end{aligned}$$

We can now use (6.7) to write

$$(\sigma_{k+j} - \nabla \sigma_k \cdot \sigma_j) \mathbb{X}^{(j)} = \sum_{M=2}^j \frac{\nabla^M \sigma_k}{M!} \left(\sum_{\substack{(l_1, \dots, l_M) \\ \sum l_i = j}} \prod_{i=1}^M \sigma_{l_i} \mathbb{X}^{(l_i)} \right),$$

which permits us to get analogously to Proposition 6.7:

$$\begin{aligned} \sigma_k(z_2) - \sigma_k(z_1) - \sum_{j=k+1}^{n+1} \sigma_j(z_1) \mathbb{X}_{12}^{(j-k)} \\ = \int_0^1 \int_0^{r_1} \dots \int_0^{r_{m-1}} [\nabla^m \sigma_k(z_1 + r_m \delta z_{12}) - \nabla^m \sigma_k(z_1)] \delta z_{12} \otimes \dots \otimes \delta z_{12} dr_m \dots dr_1 \\ + \sum_{M=1}^{m+1} \frac{\nabla^M \sigma_k(z_1)}{M!} R_M^{[m+1]}, \end{aligned} \quad (6.9)$$

where again $m := n - k$. Note that in this general formula every term in the right hand side is of class $\mathcal{C}^{(n-k+1)\alpha}$.

With similar computations we can find another identity, that is more compact and has the characteristic of being linear in the remainders $z^{[i]}$. We will use the notation $z^{[1]} := \delta z$.

PROPOSITION 6.10. *We have*

$$\begin{aligned} \sigma(z_2) - \sigma(z_1) - \sum_{i=2}^{n+1} \sigma(z_1) \mathbb{X}^{(i-1)} \\ = \sum_{k=1}^n \sum_{\substack{(l_1, \dots, l_k) \\ \sum l_i = n+1}} \int_0^1 \int_0^{r_1} \dots \int_0^{r_{k-1}} \nabla^k \sigma(z_1 + r \delta z_{12}) z_{12}^{[l_1]} \otimes \prod_{j=2}^k \sigma_{l_j}(z_1) \mathbb{X}^{(l_j)} dr_k \dots dr_1. \end{aligned} \quad (6.10)$$

Proof. For convenience we will write $\delta z := \delta z_{12}$, $z^{[i]} := z_{12}^{[i]}$ and $\mathbb{X} := \mathbb{X}_{12}$. We proceed by induction. For $n = 1$ we have:

$$\begin{aligned} \sigma(z_2) - \sigma(z_1) - \sigma_2(z_1) \mathbb{X}^{(1)} &= \int_0^1 \nabla \sigma(z_1 + r_1 \delta z) \delta z dr_1 - \sigma_2(z_1) \mathbb{X}^{(1)} \\ &= \int_0^1 \nabla \sigma(z_1 + r_1 \delta z) \delta z dr_1 - \nabla \sigma(z_1) \sigma(z_1) \mathbb{X}^{(1)} \\ &= \int_0^1 \nabla \sigma(z_1 + r_1 \delta z) \delta z dr_1 - \int_0^1 \nabla \sigma(z_1 + r_1 \delta z) \sigma(z_1) \mathbb{X}^{(1)} dr_1 \\ &\quad + \int_0^1 \nabla \sigma(z_1 + r_1 \delta z) \sigma(z_1) \mathbb{X}^{(1)} dr_1 - \nabla \sigma(z_1) \sigma(z_1) \mathbb{X}^{(1)} \\ &= \int_0^1 \nabla \sigma(z_1 + r_1 \delta z) z^{[2]} dr_1 \\ &\quad + \int_0^1 [\nabla \sigma(z_1 + r_1 \delta z) - \nabla \sigma(z_1)] \sigma(z_1) \mathbb{X}^{(1)} dr_1 \end{aligned}$$

$$\begin{aligned}
&= \int_0^1 \nabla \sigma(z_1 + r_1 \delta z) z^{[2]} dr_1 \\
&\quad + \int_0^1 \int_0^{r_1} \nabla^2 \sigma(z_1 + r_2 \delta z) \delta z \otimes \sigma(z_1) \mathbb{X}^{(1)} dr_2 dr_1,
\end{aligned}$$

which corresponds exactly to (6.10).

We now suppose the relation is true for n and we prove it for $n+1$. We have:

$$\begin{aligned}
&\sigma(z_2) - \sigma(z_1) - \sum_{i=2}^n \sigma(z_1) \mathbb{X}^{(i-1)} \\
&= \sum_{k=1}^n \sum_{\substack{(l_1, \dots, l_k) \\ \sum l_i = n}} \int_0^1 \int_0^{r_1} \dots \int_0^{r_{k-1}} \nabla^k \sigma(z_1 + r \delta z) z^{[l_1]} \otimes \prod_{j=2}^k \sigma_{l_j}(z_1) \mathbb{X}^{(l_j)} dr_k \dots dr_1 \\
&\quad - \sigma_{n+1}(z_1) \mathbb{X}^{(n)}.
\end{aligned}$$

By Theorem 6.6 we can write

$$\sigma_{n+1}(z_1) \mathbb{X}^{(n)} = - \sum_{k=1}^n \frac{\nabla^k \sigma(z_1)}{k!} \left(\sum_{\substack{(l_1, \dots, l_k) \\ \sum l_i = n}} \sigma_{l_1}(z_1) \mathbb{X}^{(l_1)} \otimes \dots \otimes \sigma_{l_k}(z_1) \mathbb{X}^{(l_k)} \right),$$

so that we obtain

$$\begin{aligned}
&\sigma(z_2) - \sigma(z_1) - \sum_{i=2}^n \sigma(z_1) \mathbb{X}^{(i-1)} \\
&= \sum_{k=1}^n \sum_{\substack{(l_1, \dots, l_k) \\ \sum l_i = n}} \int_0^1 \int_0^{r_1} \dots \int_0^{r_{k-1}} \nabla^k \sigma(z_1 + r \delta z) z^{[l_1]} \otimes \prod_{j=2}^k \sigma_{l_j}(z_1) \mathbb{X}^{(l_j)} dr_k \dots dr_1 \\
&\quad - \sum_{k=1}^n \frac{\nabla^k \sigma(z_1)}{k!} \left(\sum_{\substack{(l_1, \dots, l_k) \\ \sum l_i = n}} \sigma_{l_1}(z_1) \mathbb{X}^{(l_1)} \otimes \dots \otimes \sigma_{l_k}(z_1) \mathbb{X}^{(l_k)} \right) \\
&= \sum_{k=1}^n \sum_{\substack{(l_1, \dots, l_k) \\ \sum l_i = n}} \int_0^1 \int_0^{r_1} \dots \int_0^{r_{k-1}} \nabla^k \sigma(z_1 + r \delta z) z^{[l_1]} \otimes \prod_{j=2}^k \sigma_{l_j}(z_1) \mathbb{X}^{(l_j)} dr_k \dots dr_1 \\
&\quad - \sum_{k=1}^n \int_0^1 \dots \int_0^{r_{k-1}} \nabla^k \sigma(z_1 + r_k \delta z) \sum_{\substack{(l_1, \dots, l_k) \\ \sum l_i = n}} \sigma_{l_1}(z_1) \mathbb{X}^{(l_1)} \otimes \dots \otimes \sigma_{l_k}(z_1) \mathbb{X}^{(l_k)} dr_1 \dots dr_k \\
&\quad + \sum_{k=1}^n \int_0^1 \dots \int_0^{r_{k-1}} \nabla^k \sigma(z_1 + r_k \delta z) \sum_{\substack{(l_1, \dots, l_k) \\ \sum l_i = n}} \sigma_{l_1}(z_1) \mathbb{X}^{(l_1)} \otimes \dots \otimes \sigma_{l_k}(z_1) \mathbb{X}^{(l_k)} dr_1 \dots dr_k \\
&\quad - \sum_{k=1}^n \frac{\nabla^k \sigma(z_1)}{k!} \left(\sum_{\substack{(l_1, \dots, l_k) \\ \sum l_i = n}} \sigma_{l_1}(z_1) \mathbb{X}^{(l_1)} \otimes \dots \otimes \sigma_{l_k}(z_1) \mathbb{X}^{(l_k)} \right).
\end{aligned}$$

We now look at the last two terms. We have

$$\begin{aligned}
& \sum_{k=1}^n \int_0^1 \dots \int_0^{r_{k-1}} \nabla^k \sigma(z_1 + r_k \delta z) \sum_{\substack{(l_1, \dots, l_k) \\ \sum_i l_i = n}} \sigma_{l_1}(z_1) \mathbb{X}^{(l_1)} \otimes \dots \otimes \sigma_{l_k}(z_1) \mathbb{X}^{(l_k)} dr_1 \dots dr_k \\
& - \sum_{k=1}^n \frac{\nabla^k \sigma(z_1)}{k!} \left(\sum_{\substack{(l_1, \dots, l_k) \\ \sum_i l_i = n}} \sigma_{l_1}(z_1) \mathbb{X}^{(l_1)} \otimes \dots \otimes \sigma_{l_k}(z_1) \mathbb{X}^{(l_k)} \right) \\
& = \sum_{k=1}^n \int_0^1 \dots \int_0^{r_{k-1}} \nabla^k \sigma(z_1 + r_k \delta z) \sum_{\substack{(l_1, \dots, l_k) \\ \sum_i l_i = n}} \sigma_{l_1}(z_1) \mathbb{X}^{(l_1)} \otimes \dots \otimes \sigma_{l_k}(z_1) \mathbb{X}^{(l_k)} dr_1 \dots dr_k \\
& - \sum_{k=1}^n \int_0^1 \dots \int_0^{r_{k-1}} \nabla^k \sigma(z_1) \sum_{\substack{(l_1, \dots, l_k) \\ \sum_i l_i = n}} \sigma_{l_1}(z_1) \mathbb{X}^{(l_1)} \otimes \dots \otimes \sigma_{l_k}(z_1) \mathbb{X}^{(l_k)},
\end{aligned}$$

which is equal to

$$\sum_{k=1}^n \int_0^1 \dots \int_0^{r_k} \nabla^{k+1} \sigma(z_1 + r_{k+1} \delta z) \delta z \otimes \sum_{\substack{(l_1, \dots, l_k) \\ \sum_i l_i = n}} \sigma_{l_1}(z_1) \mathbb{X}^{(l_1)} \otimes \dots \otimes \sigma_{l_k}(z_1) \mathbb{X}^{(l_k)} dr_1 \dots dr_{k+1}$$

which we can reformulate as

$$\mathbb{1}_{(l_1=1)} \sum_{k=2}^n \sum_{\substack{(l_1, \dots, l_k) \\ \sum_i l_i = n+1}} \int_0^1 \int_0^{r_1} \dots \int_0^{r_{k-1}} \nabla^k \sigma(z_1 + r \delta z) z^{[l_1]} \otimes \prod_{j=2}^k \sigma_{l_j}(z_1) \mathbb{X}^{(l_j)} dr_k \dots dr_1.$$

Looking at the first two terms, for a generic k we have:

$$\begin{aligned}
& \sum_{\substack{(l_1, \dots, l_k) \\ \sum_i l_i = n}} \int_0^1 \int_0^{r_1} \dots \int_0^{r_{k-1}} \nabla^k \sigma(z_1 + r \delta z) z^{[l_1]} \otimes \prod_{j=2}^k \sigma_{l_j}(z_1) \mathbb{X}^{(l_j)} dr_k \dots dr_1 \\
& - \int_0^1 \dots \int_0^{r_{k-1}} \nabla^k \sigma(z_1 + r_k \delta z) \sum_{\substack{(l_1, \dots, l_k) \\ \sum_i l_i = n}} \sigma_{l_1}(z_1) \mathbb{X}^{(l_1)} \otimes \dots \otimes \sigma_{l_k}(z_1) \mathbb{X}^{(l_k)} dr_1 \dots dr_k.
\end{aligned}$$

Now we write:

$$\begin{aligned}
& \sum_{\substack{(l_1, \dots, l_k) \\ \sum_i l_i = n}} z^{[l_1]} \otimes \prod_{j=2}^k \sigma_{l_j}(z_1) \mathbb{X}^{(l_j)} - \sum_{\substack{(l_1, \dots, l_k) \\ \sum_i l_i = n}} \sigma_{l_1}(z_1) \mathbb{X}^{(l_1)} \otimes \dots \otimes \sigma_{l_k}(z_1) \mathbb{X}^{(l_k)} \\
& = \sum_{\substack{(l_2, \dots, l_k) \\ \sum_i l_i = n}} z^{[n - \sum_{i=2}^k l_i]} \otimes \prod_{j=2}^k \sigma_{l_j}(z_1) \mathbb{X}^{(l_j)} \\
& - \sum_{\substack{(l_2, \dots, l_k) \\ \sum_i l_i = n}} \sigma_{n - \sum_{i=2}^k l_i}(z_1) \mathbb{X}^{(n - \sum_{i=2}^k l_i)} \otimes \prod_{j=2}^k \sigma_{l_j}(z_1) \mathbb{X}^{(l_j)}
\end{aligned}$$

$$\begin{aligned}
&= \sum_{\substack{(l_2, \dots, l_k) \\ \sum l_i = n}} \left(z^{[n - \sum_{i=2}^k l_i]} - \sigma_{n - \sum_{i=2}^k l_i}(z_1) \mathbb{X}^{(n - \sum_{i=2}^k l_i)} \right) \otimes \prod_{j=2}^k \sigma_{l_j}(z_1) \mathbb{X}^{(l_j)} \\
&= \sum_{\substack{(l_2, \dots, l_k) \\ \sum l_i = n}} z^{[n+1 - \sum_{i=2}^k l_i]} \otimes \prod_{j=2}^k \sigma_{l_j}(z_1) \mathbb{X}^{(l_j)} \\
&= \mathbb{1}_{(l_1 > 1)} \sum_{\substack{(l_1, \dots, l_k) \\ \sum l_i = n+1}} z^{[l_1]} \otimes \prod_{j=2}^k \sigma_{l_j}(z_1) \mathbb{X}^{(l_j)},
\end{aligned}$$

so that the first two terms become

$$\mathbb{1}_{(l_1 > 1)} \sum_{k=1}^n \sum_{\substack{(l_1, \dots, l_k) \\ \sum l_i = n+1}} \int_0^1 \int_0^{r_1} \dots \int_0^{r_{k-1}} \nabla^k \sigma(z_1 + r \delta z) z^{[l_1]} \otimes \prod_{j=2}^k \sigma_{l_j}(z_1) \mathbb{X}^{(l_j)} dr_k \dots dr_1.$$

Putting everything together we get

$$\begin{aligned}
&\sigma(z_2) - \sigma(z_1) - \sum_{i=2}^n \sigma(z_1) \mathbb{X}^{(i-1)} \\
&= \mathbb{1}_{(l_1=1)} \sum_{k=2}^n \sum_{\substack{(l_1, \dots, l_k) \\ \sum l_i = n+1}} \int_0^1 \int_0^{r_1} \dots \int_0^{r_{k-1}} \nabla^k \sigma(z_1 + r \delta z) z^{[l_1]} \otimes \prod_{j=2}^k \sigma_{l_j}(z_1) \mathbb{X}^{(l_j)} dr_k \dots dr_1 \\
&+ \mathbb{1}_{(l_1 > 1)} \sum_{k=1}^n \sum_{\substack{(l_1, \dots, l_k) \\ \sum l_i = n+1}} \int_0^1 \int_0^{r_1} \dots \int_0^{r_{k-1}} \nabla^k \sigma(z_1 + r \delta z) z^{[l_1]} \otimes \prod_{j=2}^k \sigma_{l_j}(z_1) \mathbb{X}^{(l_j)} dr_k \dots dr_1 \\
&= \sum_{k=1}^n \sum_{\substack{(l_1, \dots, l_k) \\ \sum l_i = n+1}} \int_0^1 \int_0^{r_1} \dots \int_0^{r_{k-1}} \nabla^k \sigma(z_1 + r \delta z) z^{[l_1]} \otimes \prod_{j=2}^k \sigma_{l_j}(z_1) \mathbb{X}^{(l_j)} dr_k \dots dr_1,
\end{aligned}$$

which is exactly what we wanted. $\square$

6.4. A PRIORI ESTIMATES

PROPOSITION 6.11. *Let us take Z solution of (6.1) and $\sigma \in C^{n-1}$. Then we have $Z^{[i]} \in C^{i\alpha}$ and $Z \in C^\alpha$.*

Proof. By the definition of a solution we can write

$$\delta Z_{st} = \sum_{i=1}^n \sigma_i(Z_s) \mathbb{X}_{st}^i + o(t-s),$$

so that

$$|\delta Z_{st}| \leq \sup_{0 \leq s \leq t} \{ |\sigma(Z_s)| + \dots + |\sigma_n(Z_s)| \} \sum_{i=1}^n |\mathbb{X}_{st}^{(i)}| + o(t-s).$$

We want to show that the constant $C := \sup_{0 \leq s \leq t} \{ |\sigma(Z_s)| + \dots + |\sigma_n(Z_s)| \}$ is finite. Since $\sigma \in C^{n-1}$, it follows that σ_i is continuous for every $i \in \{1, \dots, n\}$, therefore it suffices to show that Z is bounded.

Let us fix $\bar{\delta} > 0$ such that $|o(t-s)| \leq 1$ for all $0 \leq s \leq t \leq T$ with $|t-s| \leq \bar{\delta}$. Since we can write $[0, T]$ as a finite union of intervals $[\bar{s}, \bar{t}]$ such that $\bar{t} - \bar{s} \leq \bar{\delta}$, we will focus on only one interval of this kind. By the definition of solution we can write

$$\sup_{t \in [\bar{s}, \bar{t}]} |Z_t| \leq |Z_{\bar{s}}| + \sum_{i=1}^n |\sigma_i(Z_{\bar{s}})| \sup_{t \in [\bar{s}, \bar{t}]} |\mathbb{X}_{st}^{(i)}| + 1 < \infty.$$

We proved that Z is bounded, therefore $C < \infty$. Observe now that we can write

$$|Z_{st}^{[2]}| \leq C \sum_{i=2}^n |\mathbb{X}_{st}^{(i)}| + o(t-s), \dots, |Z_{st}^{[n]}| \leq C |\mathbb{X}_{st}^{(n)}| + o(t-s).$$

Now since $\mathbb{X}^{(i)} \in C^{i\alpha}$, we conclude that $Z^{[i]} \in C^{i\alpha}$ and $Z \in C^\alpha$. $\square$

We now record a computation that will be useful later. We will find a formula for $\delta Z_{sut}^{[n+1]}$. We write

$$\begin{aligned} \delta Z_{sut}^{[n+1]} &= Z_{st}^{[n+1]} - Z_{su}^{[n+1]} - Z_{ut}^{[n+1]} \\ &= - \sum_{i=1}^n \sigma_i(Z_s) \mathbb{X}_{st}^{(i)} + \sum_{i=1}^n \sigma_i(Z_s) \mathbb{X}_{su}^{(i)} + \sum_{i=1}^n \sigma_i(Z_u) \mathbb{X}_{ut}^{(i)} \end{aligned}$$

Now by using Chen's relation we get:

$$\begin{aligned} \delta Z_{sut}^{[n+1]} &= - \sum_{i=1}^n \sigma_i(Z_s) \sum_{j=0}^i \mathbb{X}_{su}^{(i-j)} \otimes \mathbb{X}_{ut}^{(j)} \\ &\quad + \sum_{i=1}^n \sigma_i(Z_s) \mathbb{X}_{su}^{(i)} + \sum_{i=1}^n \sigma_i(Z_u) \mathbb{X}_{ut}^{(i)} \\ &= - \sum_{i=1}^n \sigma_i(Z_s) \mathbb{X}_{ut}^{(i)} - \sum_{i=1}^n \sigma_i(Z_s) \mathbb{X}_{su}^{(i)} \\ &\quad - \sum_{i=1}^n \sigma_i(Z_s) \sum_{j=1}^{i-1} \mathbb{X}_{su}^{(i-j)} \otimes \mathbb{X}_{ut}^{(j)} \\ &\quad + \sum_{i=1}^n \sigma_i(Z_s) \mathbb{X}_{su}^{(i)} + \sum_{i=1}^n \sigma_i(Z_u) \mathbb{X}_{ut}^{(i)} \\ &= \sum_{i=1}^n (\sigma_i(Z_u) - \sigma_i(Z_s)) \mathbb{X}_{ut}^{(i)} - \sum_{i=1}^n \sigma_i(Z_s) \sum_{j=1}^{i-1} \mathbb{X}_{su}^{(i-j)} \otimes \mathbb{X}_{ut}^{(j)} \\ &= \sum_{i=1}^n (\sigma_i(Z_u) - \sigma_i(Z_s)) \mathbb{X}_{ut}^{(i)} - \sum_{i=1}^n \sum_{j=i+1}^n \sigma_j(Z_s) \mathbb{X}_{su}^{(j-i)} \otimes \mathbb{X}_{ut}^{(i)} \\ &= \sum_{i=1}^n \left(\sigma_i(Z_u) - \sigma_i(Z_s) - \sum_{j=i+1}^n \sigma_j(Z_s) \mathbb{X}_{su}^{(j-i)} \right) \otimes \mathbb{X}_{ut}^{(i)} \\ &=: \sum_{i=1}^n B_{su}^{(n+1-i)} \mathbb{X}_{ut}^{(i)}. \end{aligned} \tag{6.11}$$

6.5. UNIQUENESS

THEOREM 6.12. (UNIQUENESS) *Let X be a path in C^α , $\alpha \in (\frac{1}{n+1}, \frac{1}{n}]$, let $\mathbb{X} = (\mathbb{X}^{(1)}, \dots, \mathbb{X}^{(n)})$ be an α -rough path over X , let σ be of class C^n with $\nabla^n \sigma$ locally Lipschitz. Then, for every $z_0 \in \mathbb{R}^k$, the rough difference equation (6.1) admits a unique solution Z such that $Z_0 = z_0$.*

Proof. Define $Y := Z - \bar{Z}$. We want to show that for $\tau > 0$ small we have $\|Y\|_{\infty, \tau} \leq 2|Y_0|$. This will imply, since $Z_0 = \bar{Z}_0$, that $\|Y\|_{\infty, \tau} = 0$, so that $Z = \bar{Z}$.

We will show:

$$\begin{aligned} \|\delta Y\|_{\alpha, \tau} &\leq c_1 \|Y\|_{\infty, \tau} + \tau^\alpha \|Y^{[2]}\|_{2\alpha, \tau}, \\ \|Y^{[2]}\|_{2\alpha, \tau} &\leq c_2 \|Y\|_{\infty, \tau} + \tau^\alpha \|Y^{[3]}\|_{3\alpha, \tau}, \\ &\dots \\ \|Y^{[n]}\|_{n\alpha, \tau} &\leq c_n \|Y\|_{\infty, \tau} + \tau^\alpha \|Y^{[n+1]}\|_{(n+1)\alpha, \tau}, \\ \|Y^{[n+1]}\|_{(n+1)\alpha, \tau} &\leq c_{n+1} \|Y\|_{\infty, \tau} + c'_{n+1} \tau^\alpha \|Y^{[n+1]}\|_{(n+1)\alpha, \tau}, \end{aligned} \quad (6.12)$$

for constants c_i, c'_{n+1} that may depend on $Z, \bar{Z}, \mathbb{X}$, but not on τ . We can then complete the proof. Indeed, by taking $\tau > 0$ small enough we can write $\|Y^{[n+1]}\|_{(n+1)\alpha, \tau} \leq 2c_{n+1} \|Y\|_{\infty, \tau}$ and by substitution $\|Y^{[n]}\|_{n\alpha, \tau} \leq 2c_n \|Y\|_{\infty, \tau}$ and so on until we get $\|\delta Y\|_{\alpha, \tau} \leq 2c_1 \|Y\|_{\infty, \tau}$. Then we can observe that by (2.12)

$$\|Y\|_{\infty, \tau} \leq |Y_0| + 3\tau^\alpha \|\delta Y\|_{\alpha, \tau},$$

so that by taking $\tau > 0$ small we get $\|Y\|_{\infty, \tau} \leq 2|Y_0|$.

We have:

$$\begin{aligned} \delta Y_{st} &= (\sigma(Z_s) - \sigma(\bar{Z}_s)) \mathbb{X}_{st}^{(1)} + Y_{st}^{[2]}, \\ Y_{st}^{[2]} &= (\sigma_2(Z_s) - \sigma_2(\bar{Z}_s)) \mathbb{X}_{st}^{(2)} + Y_{st}^{[3]}, \\ &\dots \\ Y_{st}^{[n]} &= (\sigma_n(Z_s) - \sigma_n(\bar{Z}_s)) \mathbb{X}_{st}^{(n)} + Y_{st}^{[n+1]}. \end{aligned}$$

This implies, by (2.16) and (2.13),

$$\begin{aligned} \|\delta Y\|_{\alpha, \tau} &\leq \|\sigma(Z_s) - \sigma(\bar{Z}_s)\|_{\infty, \tau} \|\mathbb{X}^{(1)}\|_\alpha + \tau^\alpha \|Y^{[2]}\|_{2\alpha, \tau} \\ &\leq c_1' \|Y\|_{\infty, \tau} \|\mathbb{X}^{(1)}\|_\alpha + \tau^\alpha \|Y^{[2]}\|_{2\alpha, \tau} \leq c_1 \|Y\|_{\infty, \tau} + \tau^\alpha \|Y^{[2]}\|_{2\alpha, \tau}, \end{aligned} \quad (6.13)$$

and in general for $2 \leq i \leq n$:

$$\begin{aligned} \|Y^{[i]}\|_{i\alpha, \tau} &\leq \|\sigma_i(Z_s) - \sigma_i(\bar{Z}_s)\|_{\infty, \tau} \|\mathbb{X}^{(i)}\|_{i\alpha} + \tau^\alpha \|Y^{[i+1]}\|_{(i+1)\alpha, \tau} \\ &\leq c_i' \|Y\|_{\infty, \tau} \|\mathbb{X}^{(i)}\|_{i\alpha} + \tau^\alpha \|Y^{[i+1]}\|_{(i+1)\alpha, \tau} \\ &\leq c_i \|Y\|_{\infty, \tau} + \tau^\alpha \|Y^{[i+1]}\|_{(i+1)\alpha, \tau}. \end{aligned} \quad (6.14)$$

We now only need to prove (6.12):

$$\|Y^{[n+1]}\|_{(n+1)\alpha, \tau} \leq c_{n+1} \|Y\|_{\infty, \tau} + c'_{n+1} \tau^\alpha \|Y^{[n+1]}\|_{(n+1)\alpha, \tau}.$$

Since $(n+1)\alpha \geq 1$ and $Y_{st}^{[n+1]} = o(t-s)$ by the definition of solution, we can use the weighted Sewing Bound (2.17):

$$\|Y^{[n+1]}\|_{(n+1)\alpha, \tau} \leq K_{(n+1)\alpha} \|\delta Y^{[n+1]}\|_{(n+1)\alpha, \tau}.$$

In view of the previous bounds (6.13), (6.14), it suffices to prove

$$\|\delta Y^{[n+1]}\|_{(n+1)\alpha, \tau} \leq \tilde{c}_{n+1} \|Y\|_{\infty, \tau} + \tilde{c}_{n+1}^{(1)} \|\delta Y\|_{\alpha, \tau} + \tilde{c}_{n+1}^{(2)} \|Y^{[2]}\|_{2\alpha, \tau} + \dots + \tilde{c}_{n+1}^{(n)} \|Y^{[n]}\|_{n\alpha, \tau},$$

where the constants $\tilde{c}_{n+1}^{(i)}$ may depend on $Z, \bar{Z}, \mathbb{X}, \sigma$ but not τ .

In other words we want to show that $\|\delta Y^{[n+1]}\|_{(n+1)\alpha, \tau}$ can be bounded by a linear combination of $\|Y\|_{\infty, \tau}$, $\|\delta Y\|_{\alpha, \tau}$ and $\|Y^{[i]}\|_{i\alpha, \tau}$ for $i \in \{2, \dots, n\}$.

By (6.11) we can write

$$\begin{aligned} \delta Y_{sut}^{[n+1]} &= \sum_{i=1}^n (B_{su}^{(i)} - \bar{B}_{su}^{(i)}) \mathbb{X}_{ut}^{(n+1-i)} \\ &= \sum_{i=2}^n (B_{su}^{(i)} - \bar{B}_{su}^{(i)}) \mathbb{X}_{ut}^{(n+1-i)} + (\delta\sigma_n(Z) - \delta\sigma_n(\bar{Z}))_{su} \mathbb{X}_{ut}^{(n)}, \end{aligned}$$

so that by (2.15)

$$\begin{aligned} \|\delta Y^{[n+1]}\|_{(n+1)\alpha, \tau} &\leq \sum_{i=2}^n \|B^{(i)} - \bar{B}^{(i)}\|_{i\alpha, \tau} \|\mathbb{X}^{(n+1-i)}\|_{(n+1-i)\alpha} \\ &\quad + \|\delta\sigma_n(Z) - \delta\sigma_n(\bar{Z})\|_{\alpha, \tau} \|\mathbb{X}^{(n)}\|_{n\alpha}. \end{aligned} \quad (6.15)$$

We focus on the last term of (6.15). We will use the notation

$$C_k^{(h)} := \sup \{ |\nabla^h \sigma_k(x)| : x \in \mathbb{R}^k, |x| \leq (\|Z\|_{\infty} \vee \|\bar{Z}\|_{\infty}) \}. \quad (6.16)$$

We have, by (2.9),

$$\|\delta\sigma_n(Z) - \delta\sigma_n(\bar{Z})\|_{\alpha, \tau} \leq C_n^{(1)} \|\delta Y\|_{\alpha, \tau} + C_n^{(2)} \{2\|Y\|_{\infty, \tau}\} \|\delta Z\|_{\alpha}.$$

We will now bound the terms $B_{su}^{(i)} - \bar{B}_{su}^{(i)}$. We will use extensively the trivial bound $|a \otimes b - \bar{a} \otimes \bar{b}| \leq |a - \bar{a}| \otimes |b| + |\bar{a}| \otimes |b - \bar{b}|$.

We start from $B_{su}^{(n)} - \bar{B}_{su}^{(n)}$. Thanks to Proposition 6.7 we can write

$$\begin{aligned} B_{su}^{(n)} &= \sigma(Z_u) - \sigma(Z_s) - \sum_{i=2}^n \sigma_i(Z_s) \mathbb{X}_{st}^{(i-1)} \\ &= \int_0^1 \int_0^{r_1} \dots \int_0^{r_{n-2}} [\nabla^{n-1} \sigma(Z_s + r_{n-1} \delta Z_{su}) - \nabla^{n-1} \sigma(Z_s)] \delta Z_{su} \otimes \dots \otimes \delta Z_{su} dr_{n-1} \dots dr_1 \\ &\quad + \sum_{M=1}^n \frac{\nabla^M \sigma(Z_s)}{M!} R_M^{[n]} \\ &= \int_0^1 \int_0^{r_1} \dots \int_0^{r_{n-2}} [\nabla^{n-1} \sigma(Z_s + r_{n-1} \delta Z_{su}) - \nabla^{n-1} \sigma(Z_s)] \delta Z_{su} \otimes \dots \otimes \delta Z_{su} dr_{n-1} \dots dr_1 \\ &\quad + \nabla \sigma(Z_s) Z_{su}^{[n]} + \sum_{M=2}^n \frac{\nabla^M \sigma}{M!} R_M^{[n]}. \end{aligned}$$

so that, defining $F_{su} := \nabla^{n-1}\sigma(Z_s + r_{n-1}\delta Z_{su}) - \nabla^{n-1}\sigma(Z_s)$, we have

$$\begin{aligned} |B_{su}^{(n)} - \bar{B}_{su}^{(n)}| &\leq |\nabla\sigma(Z_s)Z_{su}^{[n]} - \nabla\sigma(\bar{Z}_s)\bar{Z}_{su}^{[n]}| \\ &\quad + \sum_{M=2}^n |\nabla^M\sigma(Z_s)R_M^{[n]} - \nabla^M\sigma(\bar{Z}_s)\bar{R}_M^{[n]}| \\ &+ \int_0^1 \int_0^{r_1} \dots \int_0^{r_{n-2}} |F_{su}\delta Z_{su} \otimes \dots \otimes \delta Z_{su} - \bar{F}_{su}\delta \bar{Z}_{su} \otimes \dots \otimes \delta \bar{Z}_{su}| dr_{n-1} \dots dr_1. \end{aligned} \quad (6.17)$$

For the first term we write, using (6.16),

$$\begin{aligned} |\nabla\sigma(Z_s)Z_{su}^{[n]} - \nabla\sigma(\bar{Z}_s)\bar{Z}_{su}^{[n]}| &\leq |\nabla\sigma(Z_s) - \nabla\sigma(\bar{Z}_s)||Z_{su}^{[n]}| + |\nabla\sigma(\bar{Z}_s)||Z_{su}^{[n]} - \bar{Z}_{su}^{[n]}| \\ &\leq C_1^{(2)}|Y_s||Z_{su}^{[n]}| + C_1^{(1)}|Y_{su}^{[n]}|, \end{aligned}$$

and taking the $n\alpha, \tau$ -norm we see that the RHS becomes:

$$C_1^{(2)}\|Y\|_{\infty, \tau}\|Z^{[n]}\|_{n\alpha} + C_1^{(1)}\|Y^{[n]}\|_{n\alpha, \tau}.$$

We now handle the integral term. Note that $|F_{su}| \leq C_1^{(n-1)}r_{n-1}|\delta Z_{su}|$, for $0 \leq r_{n-1} \leq 1$. We have

$$\begin{aligned} |F_{su}\delta Z_{su} \otimes \dots \otimes \delta Z_{su} - \bar{F}_{su}\delta \bar{Z}_{su} \otimes \dots \otimes \delta \bar{Z}_{su}| &\leq |F_{su} - \bar{F}_{su}||\delta Z_{su} \otimes \dots \otimes \delta Z_{su}| \\ &\quad + |F_{su}|\left|\underbrace{\delta Z_{su} \otimes \dots \otimes \delta Z_{su} - \delta \bar{Z}_{su} \otimes \dots \otimes \delta \bar{Z}_{su}}_{n-1 \text{ terms}}\right| \\ &\leq |F_{su} - \bar{F}_{su}||\delta Z_{su} \otimes \dots \otimes \delta Z_{su}| \\ &\quad + C_1^{(n-1)}r_{n-1}|\delta \bar{Z}_{su}||\delta Z_{su} \otimes \dots \otimes \delta Z_{su} - \delta \bar{Z}_{su} \otimes \dots \otimes \delta \bar{Z}_{su}|. \end{aligned} \quad (6.18)$$

Now by (2.9)

$$\begin{aligned} |F_{su} - \bar{F}_{su}| &\leq |(\nabla^{n-1}\sigma(Z_s + r_{n-1}\delta Z_{su}) - \nabla^{n-1}\sigma(Z_s)) - (\nabla^{n-1}\sigma(\bar{Z}_s + r_{n-1}\delta \bar{Z}_{su}) - \nabla^{n-1}\sigma(\bar{Z}_s))| \\ &\leq C_1^{(n-1)}r_{n-1}|\delta Y_{su}| + C_1^{(n)}\{|Y_s + r_{n-1}\delta Y_{su}| + |Y_s|\}r_{n-1}|\delta Z_{su}| \\ &\leq C_1^{(n-1)}r_{n-1}|\delta Y_{su}| + C_1^{(n)}\{(1 - r_{n-1})|Y_s| + (1 + r_{n-1})|Y_u|\}r_{n-1}|\delta Z_{su}|. \end{aligned}$$

Therefore we obtain for $0 \leq r_{n-1} \leq 1$:

$$\|F - \bar{F}\|_{\alpha, \tau} \leq C_1^{(n-1)}\|\delta Y\|_{\alpha, \tau} + 2C_1^{(n)}\|Y\|_{\infty, \tau}\|\delta Z\|_{\alpha}. \quad (6.19)$$

Let us look at $\delta Z_{su} \otimes \dots \otimes \delta Z_{su} - \delta \bar{Z}_{su} \otimes \dots \otimes \delta \bar{Z}_{su}$. We write, using $|a \otimes b - \bar{a} \otimes \bar{b}| \leq |a - \bar{a}| \otimes |b| + |\bar{a}| \otimes |b - \bar{b}|$,

$$\begin{aligned} \left| \prod_{i=1}^{n-1} \delta Z_{su} - \prod_{i=1}^{n-1} \delta \bar{Z}_{su} \right| &\leq |\delta Z_{su} - \delta \bar{Z}_{su}| \left| \prod_{i=1}^{n-2} \delta Z_{su} \right| + |\delta \bar{Z}_{su}| \left| \prod_{i=1}^{n-2} \delta Z_{su} - \prod_{i=1}^{n-2} \delta \bar{Z}_{su} \right| \\ &\leq |\delta Y_{su}| \left| \prod_{i=1}^{n-2} \delta Z_{su} \right| \\ &\quad + |\delta \bar{Z}_{su}| \left\{ |\delta Y_{su}| \left| \prod_{i=1}^{n-3} \delta Z_{su} \right| + |\delta \bar{Z}_{su}| \left| \prod_{i=1}^{n-3} \delta Z_{su} - \prod_{i=1}^{n-3} \delta \bar{Z}_{su} \right| \right\} \end{aligned}$$

Applying recursively the same bound we get

$$\left| \prod_{i=0}^{n-1} \delta Z_{su} - \prod_{i=0}^{n-1} \delta \bar{Z}_{su} \right| \leq \sum_{i=0}^{n-2} |\delta \bar{Z}_{su}|^i |\delta Y_{su}| \left| \prod_{i=0}^{n-i-2} \delta Z_{su} \right|. \quad (6.20)$$

Observe that δY_{su} appears linearly in each term on the RHS. Plugging (6.19) and (6.20) into (6.18) we get

$$\begin{aligned} \|F_{su} \delta Z_{su} \otimes \dots \otimes \delta Z_{su} - \bar{F}_{su} \delta \bar{Z}_{su} \otimes \dots \otimes \delta \bar{Z}_{su}\|_{n\alpha, \tau} \\ \leq (C_1^{(n-1)} \|\delta Y\|_{\alpha, \tau} + 2C_1^{(n)} \|Y\|_{\infty, \tau} \|\delta Z\|_{\alpha}) \|\delta Z\|_{\alpha}^{n-1} \\ + C_1^{(n-1)} \|\delta Z\|_{\alpha} \sum_{i=0}^{n-2} \|\delta \bar{Z}\|_{\alpha}^i \|\delta Y\|_{\alpha, \tau} \|\delta Z\|_{\alpha}^{n-i-2} \\ = C_1^{(n-1)} \|\delta Y\|_{\alpha, \tau} \|\delta Z\|_{\alpha}^{n-1} + 2C_1^{(n)} \|Y\|_{\infty, \tau} \|\delta Z\|_{\alpha}^n \\ + C_1^{(n-1)} \sum_{i=0}^{n-2} \|\delta \bar{Z}\|_{\alpha}^i \|\delta Y\|_{\alpha, \tau} \|\delta Z\|_{\alpha}^{n-i-1}. \end{aligned}$$

We see that the terms $\|\delta Y\|_{\alpha, \tau}$ and $\|Y\|_{\infty, \tau}$ appear linearly in the RHS.

It only remains to estimate the term $\sum_{M=2}^n |\nabla^M \sigma(Z_s) R_M^{[n]} - \nabla^M \sigma(\bar{Z}_s) \bar{R}_M^{[n]}|$. Fixing a generic M we remind that

$$R_1^{[n]} = Z_{su}^{[n]}$$

$$R_M^{[n]} = Z_{su}^{[n-M+1]} \otimes \delta Z_{su} \otimes \dots \otimes \delta Z_{su} + \sigma(Z_s) \mathbb{X}_{st}^{(1)} \otimes R_{M-1}^{[n-1]} + \dots + \sigma_{n-M}(Z_s) \mathbb{X}_{st}^{(k-M)} \otimes R_{M-1}^{[M]}$$

and we observe that $R_M^{[n]} \in \mathcal{C}^{\alpha n}$.

We can write

$$\begin{aligned} |\nabla^M \sigma(Z_s) R_M^{[n]} - \nabla^M \sigma(\bar{Z}_s) \bar{R}_M^{[n]}| &\leq |\nabla^M \sigma(Z_s) - \nabla^M \sigma(\bar{Z}_s)| |R_M^{[n]}| \\ &\quad + |\nabla^M \sigma(\bar{Z}_s)| |R_M^{[n]} - \bar{R}_M^{[n]}| \\ &\leq C_1^{(M+1)} |Y_s| |R_M^{[n]}| + C_1^{(M)} |R_M^{[n]} - \bar{R}_M^{[n]}| \end{aligned}$$

When we take the norm on the RHS we get, by (2.13),

$$C_1^{(M+1)} \|Y\|_{\infty, \tau} \|R_M^{[n]}\|_{n\alpha} + C_1^{(M)} \|R_M^{[n]} - \bar{R}_M^{[n]}\|_{n\alpha, \tau}$$

We now treat the last term. We have

$$|R_M^{[n]} - \bar{R}_M^{[n]}| \leq |Z_{su}^{[n-M+1]} \otimes \delta Z_{su} \otimes \dots \otimes \delta Z_{su} - \bar{Z}_{su}^{[n-M+1]} \otimes \delta \bar{Z}_{su} \otimes \dots \otimes \delta \bar{Z}_{su}| \quad (1)$$

$$+ |\sigma(Z_s) \mathbb{X}_{su}^{(1)} \otimes R_{M-1}^{[n-1]} - \sigma(\bar{Z}_s) \mathbb{X}_{su}^{(1)} \otimes \bar{R}_{M-1}^{[n-1]}| \quad (2)$$

$$+ \dots + |\sigma_{n-M}(Z_s) \mathbb{X}_{su}^{(n-M)} \otimes R_{M-1}^{[M]} - \sigma_{n-M}(\bar{Z}_s) \mathbb{X}_{su}^{(n-M)} \otimes \bar{R}_{M-1}^{[M]}| \quad (M)$$

We can write

$$(1) \leq |Y_{su}^{[n-M+1]}| \left| \prod_{i=0}^{M-1} \delta Z_{su} \right| + |\bar{Z}_{su}^{[n-M+1]}| \left| \prod_{i=0}^{M-1} \delta Z_{su} - \prod_{i=0}^{M-1} \delta \bar{Z}_{su} \right|$$

Now we can use (6.20) and conclude that

$$(1) \leq |Y_{su}^{[n-M+1]}| |\delta Z_{su}|^{M-1} + |\bar{Z}_{su}^{[n-M+1]}| \sum_{i=0}^{M-2} |\delta \bar{Z}_{su}|^i |\delta Y_{su}| |\delta Z_{su}|^{M-i-2}$$

It is clear that every term has order $n\alpha$ and we still have linearity in $|\delta Y_{su}|$ and $|Y_{su}^{[n-M+1]}|$.

For (2) and (M) we write

$$(2) \leq |\sigma(Z_s) - \sigma(\bar{Z}_s)| |\mathbb{X}_{su}^{(1)}| |R_{M-1}^{[n-1]}| + |\sigma(\bar{Z}_s) \mathbb{X}_{su}^{(1)}| |R_{M-1}^{[n-1]} - \bar{R}_{M-1}^{[n-1]}| \\ \leq C_1^{(1)} |Y_s| |\mathbb{X}_{su}^{(1)}| |R_{M-1}^{[n-1]}| + |\sigma(\bar{Z}_s) \mathbb{X}_{su}^{(1)}| |R_{M-1}^{[n-1]} - \bar{R}_{M-1}^{[n-1]}|,$$

$$(M) \leq |\sigma_{n-M}(Z_s) - \sigma_{n-M}(\bar{Z}_s)| |\mathbb{X}_{su}^{(n-M)}| |R_{M-1}^{[M]}| + |\sigma_{n-M}(\bar{Z}_s) \mathbb{X}_{su}^{(n-M)}| |R_{M-1}^{[M]} - \bar{R}_{M-1}^{[M]}| \\ \leq C_{n-M}^{(1)} |Y_s| |\mathbb{X}_{su}^{(n-M)}| |R_{M-1}^{[M]}| + |\sigma_{n-M}(\bar{Z}_s) \mathbb{X}_{su}^{(n-M)}| |R_{M-1}^{[M]} - \bar{R}_{M-1}^{[M]}|.$$

We can do something similar for the other terms and get:

$$|R_M^{[n]} - \bar{R}_M^{[n]}| \leq \sum_{i=1}^{n-M} C_i^{(1)} |Y_s| |\mathbb{X}_{su}^{(i)}| |R_{M-1}^{[n-i]}| \\ + \sum_{i=1}^{n-M} |\sigma_i(\bar{Z}_s) \mathbb{X}_{su}^{(i)}| |R_{M-1}^{[n-i]} - \bar{R}_{M-1}^{[n-i]}|. \quad (6.21)$$

We want to show that $|R_M^{[n]} - \bar{R}_M^{[n]}|$ can be bounded by a linear combinations of $|Y_{su}|, |\delta Y_{su}|, |\delta Y_{su}^{[i]}|$ for $i \in \{2, \dots, n\}$.

Since the base case, $|R_1^{[i]} - \bar{R}_1^{[i]}| = |\delta Z_{su}^{[i]} - \delta \bar{Z}_{su}^{[i]}| = |\delta Y_{su}^{[i]}|$ is clearly linear in $|\delta Y_{su}^{[i]}|$, by a simple recursion argument it is clear from (6.21) that also $|R_M^{[n]} - \bar{R}_M^{[n]}|$ can be bounded by a linear combinations of $|Y_{su}|, |\delta Y_{su}|, |\delta Y_{su}^{[i]}|$ for $i \in \{2, \dots, n\}$.

Observe that by using (2.13), (2.14) we can show that $\|R_M^{[n]} - \bar{R}_M^{[n]}\|_{n\alpha, \tau}$ can be bounded by a linear combinations of $\|Y\|_{\infty, \tau}, \|\delta Y\|_{\alpha, \tau}$ and $\|Y^{[i]}\|_{i\alpha, \tau}$ for $i \in \{2, \dots, n\}$.

Putting everything together, by (6.17), we have shown that $\|B^{(n)} - \bar{B}^{(n)}\|_{n\alpha, \tau}$ is bounded by a linear combinations of $\|Y\|_{\infty, \tau}, \|\delta Y\|_{\alpha, \tau}$ and $\|Y^{[i]}\|_{i\alpha, \tau}$ for $i \in \{2, \dots, n\}$.

Thanks to (6.9) in Remark 6.9, we can do a completely analogous procedure for all the other terms $\|B^{(i)} - \bar{B}^{(i)}\|_{i\alpha, \tau}$, and we can state that each of these terms is bounded by a linear combinations of $\|Y\|_{\infty, \tau}, \|\delta Y\|_{\alpha, \tau}$ and $\|Y^{[j]}\|_{j\alpha, \tau}$ for $j \in \{2, \dots, i\}$. We can therefore write that

$$\|\delta Y^{[n+1]}\|_{(n+1)\alpha, \tau} \leq \tilde{c}_{n+1} \|Y\|_{\infty, \tau} + \tilde{c}_{n+1}^{(1)} \|\delta Y\|_{\alpha, \tau} + \tilde{c}_{n+1}^{(2)} \|Y^{[2]}\|_{2\alpha, \tau} + \dots + \tilde{c}_{n+1}^{(n)} \|Y^{[n]}\|_{n\alpha, \tau},$$

which is exactly what we needed to conclude the proof. $\square$

CHAPTER 7

CONCLUSIONS

In this thesis we have explored an alternative way of solving rough difference equation, consisting in a completely analytic theory which relies solely on the properties of rough paths. In Chapter 2, following [5], we have introduced the Sewing Bound independently from the *Sewing Lemma* of [12]. Then in Chapter 3 and 4 we focused on motivating the definition of rough paths and in setting a general rough difference equation. The outcome gave sense to controlled ODEs with a really irregular driving path. In Chapter 5, in the same fashion of [5] we stated existence, uniqueness and continuity of solutions for the simplest rough case. Finally in Chapter 6 we started the extension of this results for the general case of a driving path with arbitrarily low Hölder regularity. The first step consisted in studying the algebraic properties of geometric rough paths and their relations with the recursively defined function σ_n , we first explored a tree representation introduced in [6] and then we used it to recover some identities needed for the analytic bounds. We managed to use this results to give a proof of uniqueness for a generic rough equation analogous to the simple case of [5].

In conclusion, it is clear that the next step would be proving existence and continuity of the solution in the general case. To do so, we would need a kind of a priori estimate similar to the simple case, therefore the difficulty relies in formalizing some identities similar to the ones we found that could help bounding $\delta Z^{[n+1]}$. The key could lie again in understanding the algebraic properties of geometric rough paths.

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